# Experimental apparatus for universal spin-motion control in multi-ion registers

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# Contents

# 1
# Introduction

## Contents



**Should something go here?**

## 1.1 Classical and Quantum Computing

**two paragraphs**

- **Why we want to build a quantum computer. (Feynman, Shor, Grover)**
- **What constitutes a quantum computer? (Qubits, gates, circuits)**

Quantum mechanics successfully describes the behaviour of sufficiently small things [**<empty citation>**]. The theory is useful in the description of sub-atomic particles, atoms, the molecule they compose, crystals, light, the vacuum, and many other parts of our reality. That is to say, given a choice, bet things obey quantum mechanics. From the 1980s? work began in earnest on marrying the fields of quantum mechanics and information, with the intuition that if reality is described by QM, then surely QM is the efficient language in which to “compute” reality [1]. To discuss this idea, I will first donate a paragraph to the classical computers that have gone before.

We will define “compute” as modelling some physical system with another, disparate, physical system. This could be an analogue simulacra with continuous input and outputs, or it could be some digital model that mimics the system at an arbitrary precision. Apart from the engineering practicalities of realising the device, this analogue and digital distinction does not matter. The *Computer* enters when Turing proposed a class of these systems that not only mimic one specific system, but rather can model any physical system, provided enough time and memory. This remarkable result lends *computers* their other name: the Turing machines. Suprisingly few operations are required for a device to be a Turing machine (i.e. is Turing complete), and so the question of whether a Turing machine can be constructed was never up for debate (as this was likely answered at least XX years before Turing’s birth by probably some astrolabe XX). But now comes the practicality - although a physical device is a Turing machine, it may still not be feasible for an operator to “code” some desired behaviour [2], or for a useful result to be returned in reasonable time [3]. Regardless, highly advanced and operable computers have been constructed and have successfully modelled and transformed almost all aspects of moden life. However, there still remains vast regions of computational problems that fall into this category of unfeasible on current hardware. Worse still, the scaling of resources and time required for some real-world problems suggest that classical computing will never be able to solve them.

One category of problem that is often recognised to be unfeasible on classical computers, is that of modelling quantum systems. Fundamentally this is due to the concept of superposition (**I have seen interference being the often quoted feature, check my understanding of this**) – a quantum state is generally

[1] Paraphrased, rather than the tired Feynman spiel [**<empty citation>**].

[2] play with esolangs for an hour to confirm this for yourself.

[3] factor a 256-bit number for **1e10** years to confirm this for yourself.

described by the superposition of all possible $2^n$ basis states, where $n$ is the system dimension **Check this**. Each basis state will have some amplitude and phase, and with some normalisating results in $2^n - 1$ complex numbers required to track and accurately model the generic state. This exponential scaling is not kind to a digital computer, whose own internal classical state (something about how they scale...). **Read deutsch a bit again to finalise the argument here**. Returning to the 80s?, it was thought that this detrimental scaling of quantum systems might offer a distinct advantage if we were to construct a Turing machine from quantum objects [4]. This prompted the requirements given by DiVincenzo on the components to construct such a device: Some aggregation of fundamental quantum information carrying units, most commonly two-level systems called quantum-bits (or qubits), although higher dimensional systems such as qudits [**<empty citation>**], or continuous systems such as qumodes [**<empty citation>**] are areas of active research. We further require some way to prepare these quantum objects into a known initial state. Then, a set of unitary operations (quantum gates) are used to manipulate the state and perform the desired computation. Finally, some measurement of the state is required to extract information. The performance of a physical QC is then characterised by how faithfully each of these requirements is fulfilled in the error prone world. To realise a resource scaled quantum computer, we must recognise the double-edged sword that is superposition: this large state space in which to encode information, is increasingly sensitive to environmental noise [**<empty citation>**] due to this large state space. The loss of recoverable information due to noise is called decoherence, and is the major challenge to building a practical quantum computer [**<empty citation>**].

The physical origin and coupling of noise into the quantum system is highly dependent on the hardware implementation of the computer, although the effects of noise fall into a few classes, which we shall discuss in the next chapter. This thesis

[4] As an aside, the implications of quantum mechanics on information extend further than purely mimicking other dynamic systems – using and distributing quantum objects also allows for dominant strategies in cryptography [**<empty citation>**], and some competitive (although currently esoteric) games [**<empty citation>**].

is concerned practically on the trapped ion platform, and so a specific focus will be lent to it, and we shall now spend some time outlining what it consists of.

## 1.2 Trapped Ion Platform

**two paragraphs**

- Atoms are convenient qubits.
- Description and history of experimental trapped ion systems.

An effective quantum computer requires meticulous control over a collection of simple quantum systems. A good starting point is thus selecting that simple system to build our control around. Nature provides us with a bounty of choices in this regard: Atoms. By definition it is difficult to choose a simpler object, and according to our understanding of physics, atoms of the same species are identical to one another no matter where or when you are. We may use their electronic energy structure, which atomic physicists have scrutinized for over a century, to define our quantum bits, and we can use an assortment of electromagnetic frequencies to manipulate these internal states. This is the strategy of both the “neutral-atom”, and “trapped-ion” platforms, with the latter choosing to remove or add a few electrons to more easily manipulate the atoms position in space (although we note that the neutral-atom platform has made significant progress in optical traps, and I suspect this distinction could become a moot point relatively soon).
Despite this gift of nature, there is wide interest in other “manufactured" quantum systems, such as superconducting circuits, quantum dots and, NV centers; benefitting from control over the level structure, and owing to their solid state, presumably allowing for easier integration into existing semiconductor fabrication techniques. The final platform we name is photonic qubits, arguably another of natural origin, and unique in being a "flying-qubit": photons are able to transmit information over long distances (and it is in fact quite hard to keep them still). At this time, it is unclear whether one of these platforms, or an entirely new architecture, will

be the first to reach a scaled device. Each have their own advantages and unique technical challenges, and each are in rapid active development. We shall focus almost exclusively on the trapped ion platform, however will give some discussion on common noise channels in other platforms in chapter 2
Atomic ions held in place through some combination of electric and/or magnetic fields. They are stored in ultra-high vacuum conditions, and physically distant (>10 µm) from any bulk material to limit the coupling of environmental noise to our fragile state. Lasers, radio, and microwave drives are then used to effect the electronic and motional states of the ions.

## 1.3 Scaling Trapped Ion Systems

- Why we need to scale up trapped ion systems (Steane) in register, and in depth.
- What are the technical challenges to scaling up trapped ion systems. Including motional control and hybrid systems.
- What are the new errors that appear when we scale up trapped ion systems. This is a main contribution of this thesis.

Useful circuits are proposed to be of order x qubits, x gates. increased complexity changes the relevant error processes

Errors increase with both space and time in QC, motivating hardware improvements, quantum error mitigation and correction techniques. Hardware challenges for scaling up trapped ion systems include number of qubits, addressing (control line problem), speed of operation, and motional control. Particularly, as the resource scales, new errors appear that are not present in small systems. These include spatially correlated noise, and crosstalk between qubits. This thesis focuses on the construction of a new experimental apparatus to enable the study of these errors, and the development of techniques to mitigate them.

## 1.4 Structure of this Thesis

This thesis is concerned with the construction, and characterisation of a new ion trap apparatus targetting spin and motional control of multi-ion chains; the errors that arise at this scale; and error mitigation through virtual distillation.

Chapter 2 provides motivation and theory required for the remainder of the thesis, including a background on randomised benchmarking and the relevant group theoretic results.
Then follows in chapter **??** a hardware description of apparatus constructed during my DPhil, followed by performance characterisations in chapter 4.
Chapter **??** presents a novel randomised benchmarking method for detecting and quantifying spatially correlated errors in multi-qubit devices, including experimental results on up-to an 8-ion chain.
The motional states of the ions as a computational resource are discussed in chapter **??**, with relevant device characterisation including a demonstration of squeezing, numerical simulations supporting the expected leading errors for our device, and an outlook on future experiments on hybrid spin-motion computation.
Chapter **??**, presents an experimental implementation of virtual distillation, an error mitigation technique with practical application to computation involving both spin and motional degrees of freedom.
The thesis is concluded in chapter **??** with details on the implication on this work to the field, and an outlook of future work planned on the constructed apparatus.

# 2
# Trapped-Ion Quantum Computing

**Contents**



## 2.1 Trapping Ions

### 2.1.1 Trapping Potential

From Earnshaw's theorem, $\nabla^2 V = 0$, a stable stationary point in 3D cannot be realised using only static electric potentials, $V$, as if the potential is confining in two dimensions, it will be anti-confining in the third. Therefore, to achieve stable trapping, either an oscillating electric field (Paul trap [**paul1990electromagnetic**]), or a static magnetic field (Penning trap) is used.

Will include pseudopotential relating $V_{RF}$ and $\Omega_{RF}$ to the radial mode frequencies $\omega_m$ (see Vera eqn 3.4),

$$\omega_r \propto \frac{V_{RF}}{M\Omega_{RF}}. \tag{2.1}$$

Further, will use James to give ion spacing and non-COM mode frequencies as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitati as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitation.

## 2.2 Light Matter Interactions

### 2.2.1 Rabi Flopping

Explain this is how we do single qubit gates.

### 2.2.2 Coupling to Motion

### 2.2.3 Geometric Phase Gates

## 2.3 Randomised Benchmarking

Some overview and motivation for RB. Need to cover:

- Twirling over a group
- Liouville representation of a CPTP map
- Irreps of the Clifford group
- Clifford approximates twirling over the unitary group (Haar measure)
- How Liouville representation is transformed under twirling
- The specific representation-theory tools you use later: irreps, multiplicity spaces, invariant sectors, Schur basis.

- RB protocol for single qubit Clifford group, generalisation to multi-qubit Clifford group.
- How to extract error per Clifford from decay rate

## 2.4 Noise channels

As will also be used in virtual distillation, may move to theory sections
A description of the noise channels we consider in this work. Explicit CPTP maps for these channels.

### 2.4.1 Why isn't everything white noise?

### 2.4.2 Unitary noise

Both continuous and probabilistic.

### 2.4.3 Depolarising noise

An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere. Get this from twirling over Clifford group!

### 2.4.4 Spatially correlated noise

**Common noise** An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere, but the same unitary is applied to multiple qubits. Can enforce some symmetry to this channel by twirling over the k-copy Clifford group. This is the channel we will use to model correlated noise in our experiments.

**Entangling noise** In SC ZZ entangling noise is common.

### 2.4.5 Can we make everything white noise?

Have a look at Balint paper, deep random circuits, etc.

# 3
# Apparatus

"We never experiment with just one electron or atom or (small) molecule." – Schrödinger 1952

To keep him happy we will try to load at least two.

## Contents



After 70 years of ion confinement development (since the quadrupole mass spectrometer in the 1950s [1]), and the myriad use-cases from time standards, to analytic chemistry, to quantum computing, there are naturally many systems and thesi apparatus chapters in existance today. As such, we structure this chapter in the following way: Following on from the standard requirements of an ion trap system, provided in the introduction section 2.1, we discuss the unique motivation and design specifications of our system; We then present an abstract schematic of the system in its current form, highlighting the constituent subsystems, and the optical and electrical signal paths; The remaining sections of the chapter contain the technical detail, perhaps more relevant to the practicing experimental student, and the hardware enthusiastic.

## 3.1 Specification

Ion-trap experiments are exceptionally versatile in their application, however in the vast search space of quantum-information applications, we want a system particularly tailored for three avenues of study:

1. Error channels on multi-qubit chains.
2. Continuous variable quantum computing and simulation utilising the spin and motional degrees of freedom of the confined ions.
3. Fast two-qubit entangling gates via phase stabilised optical standing waves.

These latter two research directions were explored within our group on a previous apparatus, leading to state-of-the-art two-qubit entanglement rates on the ion trap platform [2, 3], creation of highly non-classical motional states [4, 5], and a simulation of Z2 gauge theories encoded via the spin and motional degrees of freedom of a two-ion chain [6, 7]. However, hard limitations of both scaling these demonstrations to longer ion chains, and of interaction strength due to availible optical intensity, were often restrictive. To solve both these issues, a new apparatus has been designed, constructed, and characterised, motivated by these applications. Colloquilly the system is named *"FastGates"*, owing to its experimental objectives,

and we shall refer to it as such throughout this chapter.
These objectives give the following specific control requirements:

- Storage of medium length (5-10) qubit registers, section 3.2.
- Individual internal state control of each qubit, section 3.6.5.
- Stable motional modes for storing quantum information, section 3.2.1.
- Dual high numerical apperture (NA) optical access for high-intensity standing waves, section 3.4.

As the lifetime of such an apparatus extends past a single PhD generation, this thesis serves as a point-in-time description of the system, and where applicable, the planned future extensions will be highlighted as such. An abstract schematic of the current configuration is shown in figure 3.1 with blocks resembling either electronic, mechanical, or optical subsystems.

## 3.2 The Ion Trap

Recently, the micro-fabricated surface style linear Paul trap has gained popularity due to the maturity of chip fabrication technologies [9] and the potential route to scalability this offers. In the surface trap, the 3D radial and axial electrodes of a "macro" trap are effectively projected onto a 2D surface. The stable confining point of such a trap is typically designed to be on the order of $50 - 100$ µm above the chip surface. The maturity of chip fabrication has enabled the manufacturing of complex multi-zone devices with many DC electrodes, amenable to ion shuttling [10, 11], and hence furthering progress towards Quantum CCD type architectures [12]. However, this surface style geometry typically comes with two costs: the depth of the trapping potential is often greatly reduced due to anharmonicity [13], and the close proximity of the surface to the ion can be a large contributor to motional heating rates [14–17].
Our trap, provided by the *National Physical Laboratory* in the UK, brings together the advantages of chip fabrication as well as the low heating rates and high trapping

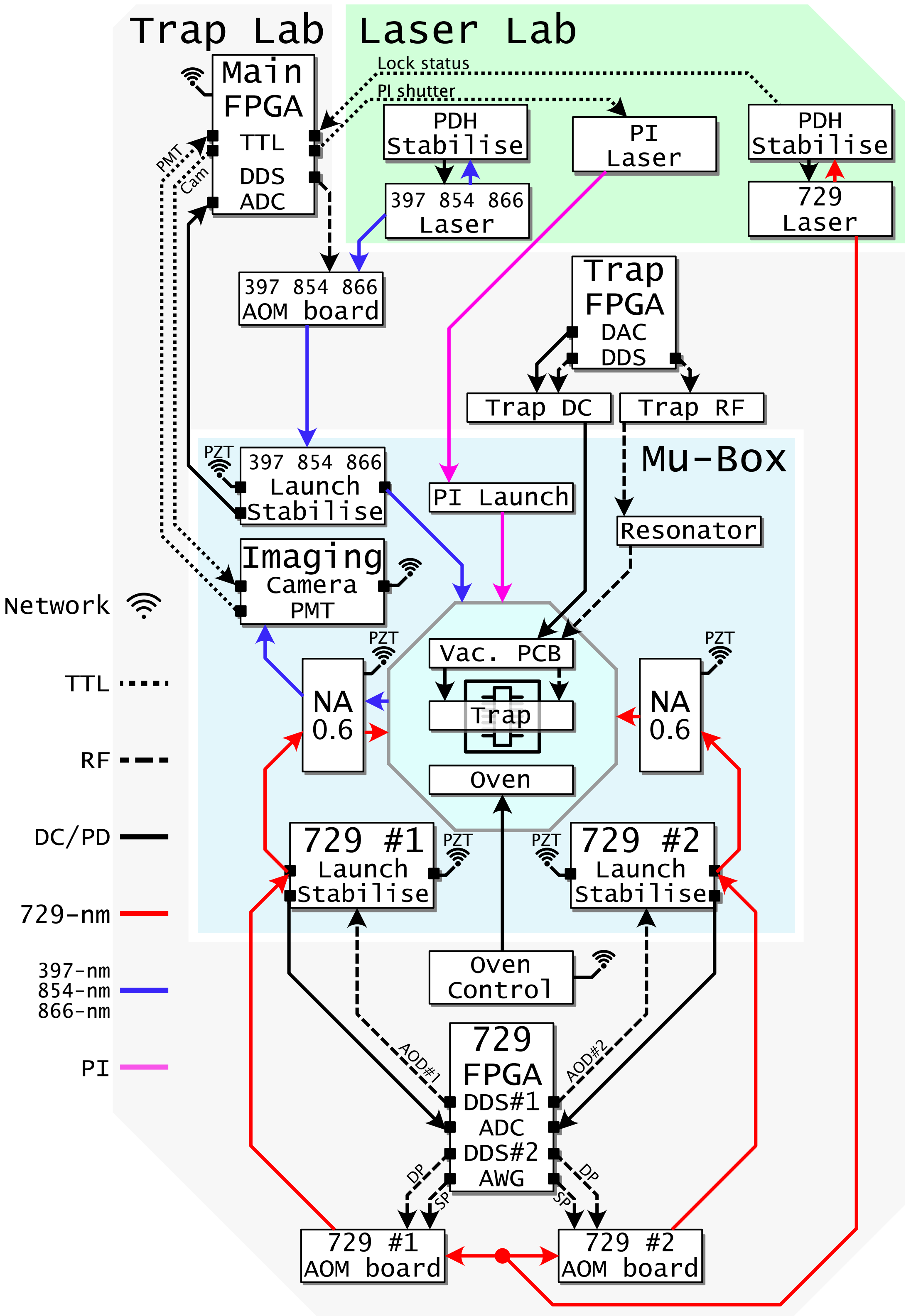



**Figure 3.1:** Block diagram schematic of the *FastGates* apparatus. Blocks resemble discrete subsystems, which shall be discussed within this chapter. Lines resemble relevant optical (free-space or in-fibre), RF, DC, and communication paths between subsystems. Block location signifies whether the device is in-vacuum, within the magnetic shielded (MuMetal) box, within the "trap-lab", or within the "laser-lab".

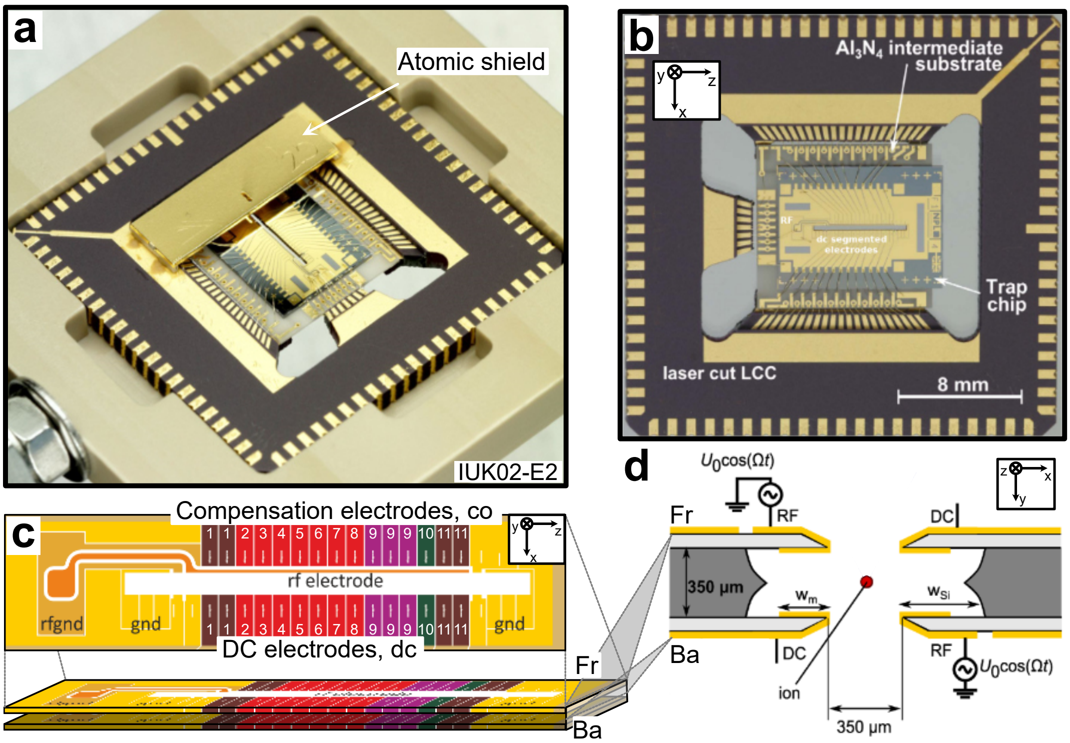



**Figure 3.2:** Schematics of the *NPL* trap chip used in this experiment. **a)** A photograph of the trap chip used in the *FastGates* experiment before being placed under vacuum. The top "atomic shield" can be seen here. **b)** A front view of the trap chip. Axial confinement is provided by a subset of the segmented DC electrodes. Radial confinement is provided by the RF rails. The trap chip and leadless chip carrier (LCC) are highlighted. **c)** A schematic of the electrode geometries and labelling for the front-face of the trap chip. A similar arrangement exists for the back-face however with the RF rail on the diagonally opposite electrode. **d)** A slice view of the trap. The red point here represents the ions, the gold are the RF and DC electrodes, and the grey is the silica substrate. It can be seen that the ion is $350 \times \sqrt{2}/2 \simeq 250\mu$m away from the nearest electrode. The numerical aperture of the ion plane due to the electrodes is $\sim 0.71$. Figures b) and d) adapted from Ref. [8], other panels adapted from private communication with the *NPL* team.

depths of a macro 3D style trap through utilising a micro-fabricated 3D trap geometry with greater ion-surface distances than typical planar traps. Details on its design and characterisations can be found in Ref. [8, 18, 19]. Figure 3.2 shows the electrode geometry of the trap and the relevant length scales. The trap package consistes of an outer leadless chip carrier (LCC) with outer pads routing DC, RF, and ground connections to an intermediate $Al_3N_4$ substrate. The connection to the front and back surfaces of the trap chip is made through vias and wire bonds to the intermediate substrate [20]. The trap chip is a monolithic double-sided silicon device with gold plated electrodes on the front (Fr) and back (Ba) surfaces. These two layers are shown schematically in the bottom panel of (*c*). The electrode schematic for the front surface is shown, and the electrode indexing convention is highlighted. Electrodes recessed from the ion due to the RF rail are named "compensation"-electrodes (co), while electrodes on the opposite side provide the strong DC potential required for axial confinement[1]. We typically trap ions centered on electrode dc5. Panel (*d*) shows a slice view of the trapping region, highlighting the ion position between the front and back surface electrodes. The dual RF rails and DC electrodes are shown in gold, and the undercut silicon bulk is highlighted in dark grey.

The next two subsections describe the source of RF and DC voltages to the trap electrodes. The physical mounting of the trap in the vacuum system is discussed in section 3.4, and the optical access to the trap is discussed in section 3.4.

### 3.2.1 Trap RF Chain

To utilise radial motional modes for low error quantum gates, we require the radio frequency (RF) field that produces the desired trapping pseudopotential to be both frequency and amplitude stable. Furthermore, as the natural time scale for two-qubit gate durations is $\tau = 2\pi/\omega_m$ (see section 2.2.3), where $\omega_m$ is the motional mode angular frequency, then greater motional mode frequency is desired for fast entanglement. From equation 2.1, it can be seen that radial motional mode frequency is proportional to RF voltage, and inversely proportional to the frequency. The *NPL* trap used in the *FastGates* apparatus is rated for a max peak-to-peak voltage of 400 Vpp on the RF electrodes. To target high motional mode frequencies, saturating this maximum voltage is desirable, however, for safe operation, typically the trap is provided a maximum RF voltage $V_{RF} = 360$ Vpp. The RF chain which provides this signal is schematically shown as:

[1]Note: Compensation electrodes co1 and co11 on both front and back surfaces are grounded on the trap package, and so are not available for use. There is a continuity fault on front dc11 and so this electrode is also not available for use.

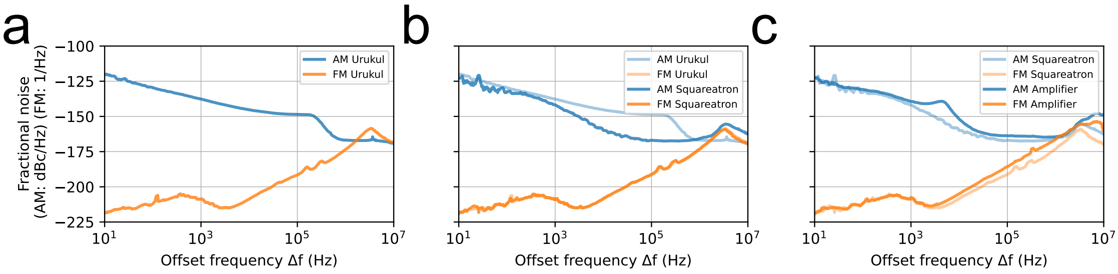



**Figure 3.3:** Characterisation of the RF chain. **a)** SSB spectra for AM and PM, given as fractional noise, measured after the *Urukul*. **b)** Measured after the *Squareatron*. **c)** Measured after the amplifier.

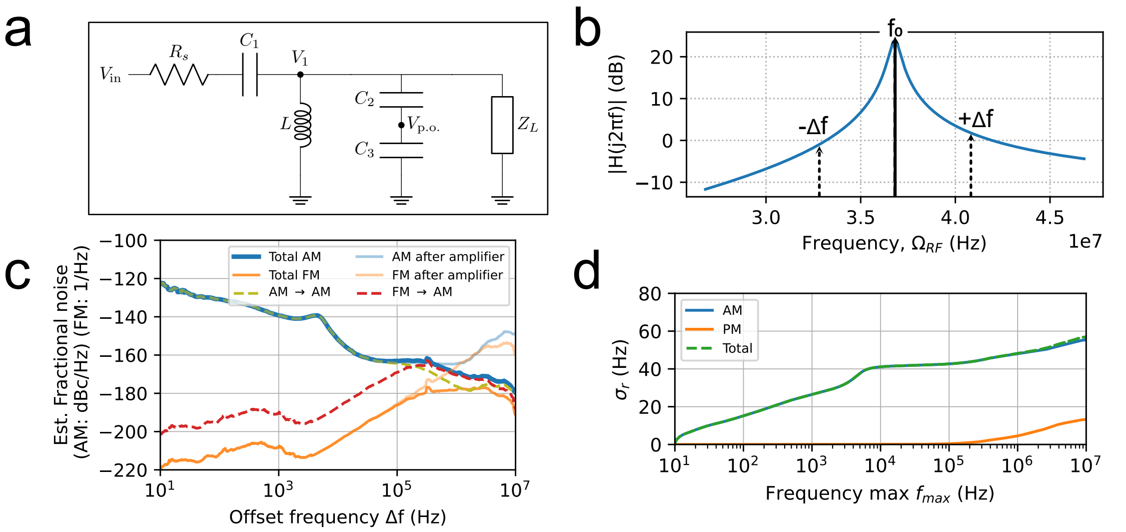
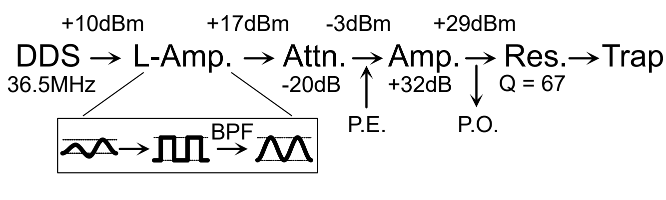



**Figure 3.4:** Expected SSB after the resonator, and estimated mode frequency noise. **a)** Resonator circuit diagram. **b)** Resonator lineshape, given by magnitude of the transfer function $|H(f)|$. **c)** Expected AM and FM fractional noise spectra after the resonator. **d)** Integrated RMS noise of radial motional mode frequency due to the RF chain.




and embedded within the schematic 3.1, beginning with the *Trap FPGA*. Each element of the chain is designed to provide a low-noise, high-voltage RF signal to the trap electrodes, and is discussed in turn below.

The frequency source is a direct-digital-synthesiser (DDS), named *Urukul* [21], part of the *Artiq Sinara* [22] hardware ecosystem. *Urukul* is used extensively in our lab for driving acousto-optic modulators (AOMs) and here used as the source drive of our trap RF. The *Urukul* is operated at maximum output power, +10 dBm, and with a frequency $\sim$ 35 MHz (set by resonator, discussed below). As can be seen in equation 2.1, amplitude and frequency instability will directly lead to radial mode instability. Expanding for small amplitude, $dV_{RF}$, and drive frequency, $d\Omega_{RF}$, errors we find the motional mode frequency, $\omega_r'$, is related to the ideal frequency, $\omega_r$, by

$$\omega_r' = \omega_r(1 + dV_{RF}/V_{RF} - d\Omega_{RF}/\Omega_{RF}), \tag{3.1}$$

where we can see both errors affect the motional mode frequency linearly. The instability of the oscillator can limit motional mode coherence time (see section 4.3.4 for characterisation with the ion). We measure noise originating from both amplitude modulation (AM) and frequency modulation (FM) across the RF chain using a phase noise analyser[2]. This was done to verify low-noise operation of the individual devices, and to estimate expected motional mode errors due to the inherent noise on the RF chain.

We quote FM as the fractional frequency stability power spectral density (PSD),

$$S_y(f) = \frac{f^2}{f_0^2} S_\phi(f), \tag{3.2}$$

where $S_\phi(f)$ is the phase noise PSD (which is the Fourier transform of the Autocorrelation function) $f$ is the offset frequency, and $f_0$ is the carrier frequency. $S_y(f)$ has units of $\left[\frac{1}{\text{Hz}}\right]$. AM noise is quoted as the single sideband (SSB) power, $S_\alpha$, in a 1 Hz bandwidth at an offset frequency, $f$, from the carrier, normalised to the total power of the carrier, and is here given in units of $\left[\frac{\text{dBc}}{\text{Hz}}\right]$.

Figure 3.3, panel ($a$) displays the *Urukul* AM and FM spectra. The *Urukul* is phase referenced to the master FPGA board (*Kasli*), which is in turn phase locked to an external 10 MHz atomic clock[3]. It can be seen that AM noise dominates up $\sim$ 1 MHz, suggesting that this will be the limiting error source for motional mode stability. The AM noise may be reduced by using a non-linear voltage clamp. We

[2] Rhode and Schwartz FSWP PNA

[3] **ToDo: Abaqus atomic clock PN**

use an “ultra-low noise limiting amplifier” named *Squareatron* developed previously in the group, and further refined by *Oxford Ionics*. *Squareatron* takes an input RF signal and converts to a stable amplitude square-wave output which is subsequently bandpassed around the desired RF frequency to recover the sinusoidal signal, but with significantly reduced AM noise. The AM and FM after the *Squareatron* is shown in panel (*b*). It can be seen that between 1 kHz and 1 MHz, AM is considerably suppressed. There appears to be an increase in AM at higher frequencies **ToDo: explore this further**. The *Squareatron* has a 17 dBm output power, which is attenuated by −20 dB before an amplification stage to reach the desired power for the trap. Highlighted in the above chain schematic is the option to input a parametric excitation signal (P.E.), which we shall discuss further in section 4.2.3. For the following measurements, the P.E. branch was removed. AM and FM introduced by the low-noise amplifier[4], can be seen in panel (*c*). Most notably an increase in AM is seen between 1 kHz and 10 kHz.

To impedance match the 50 Ω line from the amplifier to the small capacitance of the trap electrodes and vacuum feedthrough, a resonant matching circuit is used. This “resonator” has three main effects on our system:

1. The source 50 Ω is matched to the (almost) purely capacitive load.
2. The voltage step up provides the neccesarry amplitude for high motional mode frequencies.
3. The input RF is bandpassed around the desired trapping frequency.

The latter two points are practically useful, especially the bandpass effect of the resonator attenuating noise on the RF line, however, changes to the resonators Q factor and central frequency due to thermal and mechanical noise can be a significant source of low frequency motional mode instability. Opting for good passive stability, a PCB based design[5] with good thermal management was seen as

[4] Mini-Circuits ZHL-1-2W-S+

[5] This style PCB resonator has been used around the Oxford Ion Trapping Group for some time, and is based originally on a design by Tom Harty

preferable to large helical resonators, which are often used in ion trap experiments for their high Q factors. Aiming for a moderate Q factor of $\sim 70$ allows for a good voltage step up, and some bandpass filtering, without being overly sensitive to thermal and mechanical noise. The circuit diagram of the constructed resonator [6] is shown in figure 3.4 (a). The inductor, $L \sim 0.8$ µH, is hand-wound copper wire with a toroidal iron powder core[7], and the capacitors are low tempco C0G rated ceramic, $C = 5.4$ pF. The vacuum can feedthrough and trap were measured to present an $\sim 18$ pF load. The trap itself presents a $C_T = 4$ pF [20], the cable between the resonator and the vacuum feedthrough contributes $\sim 4$ pF, and the remaining presented capacitance is expected to originate from the feedthrough and in-vacuum RF line. In vacuum, the RF line is a *Kapton* coated 0.4 mm diameter single strand copper wire, which connects to an in-vacuum distribution PCB. The RF connection to the trap LCC is made by *fuzz button* vias.

Measuring the motional mode frequency of a trapped ion while varying the trap RF frequency (see section 4.3.3), yields a resonator quality factor $Q = 57.88(6)$, and central frequency 34.659(1) MHz. The effect of the resonator on the AM and FM noise can be calculated through the transfer function, $H(f)$, of the resonator circuit. Capacitors ($C2$ and $C3$) are used to provide a pick-off (p.o.) voltage which can be monitored, and if required, used in a closed feedback loop. We however have not attempted this as of yet, and shall ignore the effect of the pick-off capacitors for this analysis. The transfer function of the simplified circuit is then given by,

$$H(f) = \frac{V_{out}}{V_{in}} = \frac{Z_p}{Z_s + Z_p}, \tag{3.3}$$

where

$$\begin{aligned} Z_p &= \left( \frac{1}{j2\pi f L} + \frac{1}{Z_L} \right)^{-1}, \\ Z_s &= R_s + \frac{1}{j2\pi f C_1}. \end{aligned} \tag{3.4}$$

Given the approximate capacitance and inductance values of the resonator and trap, the analytic $|H(f)|$ is plotted in figure 3.4 panel ($b$), and shows acceptable

[6] Credit to Emily Hirsch for construction of the recent resonator.

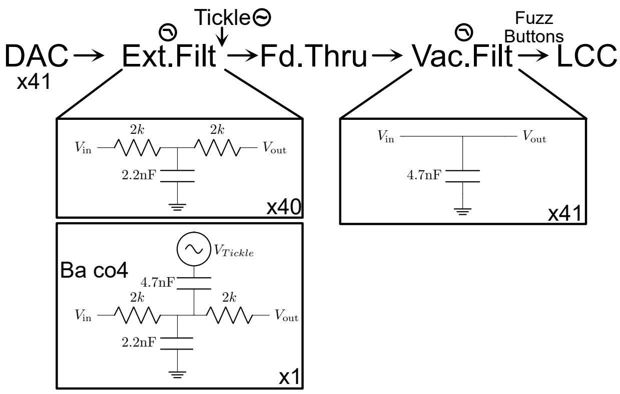
[7] Micrometals T80-6

agreement with the measured quality factor and central frequency (here $Q = 69$ and $f_0 = 36.8$MHz) with the discrepancy of quality factor expected to originate from excess resistance in the inductor not accounted for in the model. Final AM and FM spectra are calculated through considering both upper ($+\Delta f$) and lower sidebands ($-\Delta f$), as shown in panel ($b$), following reference [23]. We find that for both AM$\rightarrow$AM and FM$\rightarrow$FM (where "$\rightarrow$" represents the transform from input noise type to output noise type) the input PSD are low pass filtered with a cut-off frequency of $\sim 0.25$ MHz. We also consider the cross term, FM$\rightarrow$AM, which here we expect to be non-negligible at higher offset frequencies due to the assymmetry of the resonator lineshape. Panel ($c$) shows the expected fractional noise after the resonator, with total FM noise (orange line), total AM noise (blue line), and contributions from AM$\rightarrow$AM (dashed green), FM$\rightarrow$AM (dashed red). Integrating the AM and FM fractional noise spectra between some chosen range of offset frequencies, $f_{min}$ and $f_{max}$, gives the expected fractional RMS noise on the radial mode frequency of the ions. Assuming a mode frequency of $w_r = 3$ MHz the RMS, $\sigma_r$, is shown in figure 3.4 panel ($d$) as a function of $f_{max}$, with $f_{min} = 10$ Hz. It can be seen that the expected noise is dominated by the AM contribution, and in the frequency range considered, corresponds to a mode frequency instability of $\sigma_r = 60$ Hz. This analysis excludes any noise less than 10 Hz, and environmental noise coupled in by the resonator, which may be a significant contribution at low frequencies due to thermal and mechanical noise. The motional modes will also be subject to noise from other coupling mechanisms, such as voltage noise on the DC electrodes, or fluctuating stray electric fields, which we do not consider here.

Here we have characterised both AM and FM noise on the RF chain and estimated the expected contribution to motional mode instability. Further characterisations of the motional mode stability using the ion as a sensor is discussed in section 4.3.3.

### 3.2.2 Trap DC Voltages

Static fields are required for reshaping the trapping psuedo-potential. This is useful for setting desired mode frequency splittings, for shuttling ions, and for compensating stray electric fields. As shown in figure 3.2, the *NPL* trap has 40 independently controllable DC electrodes. A further DC bias voltage can be applied directly to the silicon bulk. DC Voltage is supplied to the LCC through the following chain:




and embedded within the schematic 3.1. Two multi-channel DACs, known as *Fastino*, part of the *Artiq Sinara* hardware ecosystem, provide the initial 41 DC channels. The Fastino outputs are filtered by a passive cascaded RC low pass filters with resistances, $R = 2k$, and capacitances $C_1 = 2.2$ nF and $C_2 = 4.7$ nF[8] yielding a $-3$ dB "cut-off" frequency of 7.4 kHz. The low pass filter circuit is shown in the above schematic, with the first stage on an external to vacuum PCB (Ext. Filt), and the final shunt capacitor on an in-vacuum PCB (Vac. Filt)[9] within close proximity of the trap chip to mitigate RF pickup on the DC lines. The feedthrough to the vacuum system has 51 pins[10] with non-DC pins used to provide ground connections. On the external filter board, one channel, corresponding to electrode *Ba co4*, has a capacitively coupled in AC signal. This originates from a DDS (*Urukul*) and is used to resonantly excite the motional modes of the ion chain for characterisation and calibration purposes (see section 4.3.3). The in-vacuum PCB is connected to the trap chip via a custom PEEK interposer with embedded *Fuzz buttons* vias[11].

[8] For in-vacuum use. 1206 NPO 4.7nF 100V, SRT 1206A472JBCB

[9] Vacuum compatible PCB using Rogers RT6002. Filter boards were designed by Mariella Minder.

[10] *Allectra* 51-way Micro-D

[11] *Custom Interconnects* Au/BeCu 0.025" diameter x 0.104" length Fuzz Button

The ion is confined in the "operation" zone, seen in figure 3.2 ($c$). The axial trapping is provided by electrodes *Ba dc5* and *Fr dc5*, where *Ba* and *Fr* correspond to the back and front plane of the trap respectively. The DAC can provide up to 10 V to these electrodes to produce an axial frequency of 1.6 MHz.

### 3.2.3 System Grounding

**Thinking of removing this section.** Poor ground design (or no design at all). RF signal and return in vacuum may be important.

## 3.3 Magnetic Field

Due to the fragility of the internal states of the ion (these are *literally* state-of-the-art magnetic field sensors), we must take great care in isolating the ion from a noisy environment. We do not have access to magnetic field insensitive transitions in $^{40}Ca^{+}$, therefore uncontrolled Zeeman shifts of the internal atomic energy levels must be suppressed through enforcing a stable external magnetic field. This magnetic field stability is required for long spin coherence times (see section 4.2.7), and for low error single- and two-qubit gates. A permanent magnet array of Samarium Cobalt[12], $Sm_2Co_{17}$, is used in a Helmholtz configuration to create a stable magnetic field of $\simeq 5$ G[13]. Samarium Cobalt was chosen for its low temperature coefficient of remenance, -0.03%/K.

The ion is shielded from unwanted external magnetic fields by a custom designed MuMetal enclosure[14], with an internal dimension of $650 \times 650 \times 650$ mm. The shielding consists of two layers of MuMetal shielding with internal shield thickness of 1.5mm and external shield thickness of 2.0mm. The factory quoted shielding factor at the center of the box is $\times 546$ for DC fields. Spin coherence time comparisons with and without magnetic shielding are shown in section 4.2.7.

[12] *Bunting Magnets* E643SM and E644SM.
[13] Designed and constructed by Sophie Howarth.
[14] *Magnetic Shields* custom design.

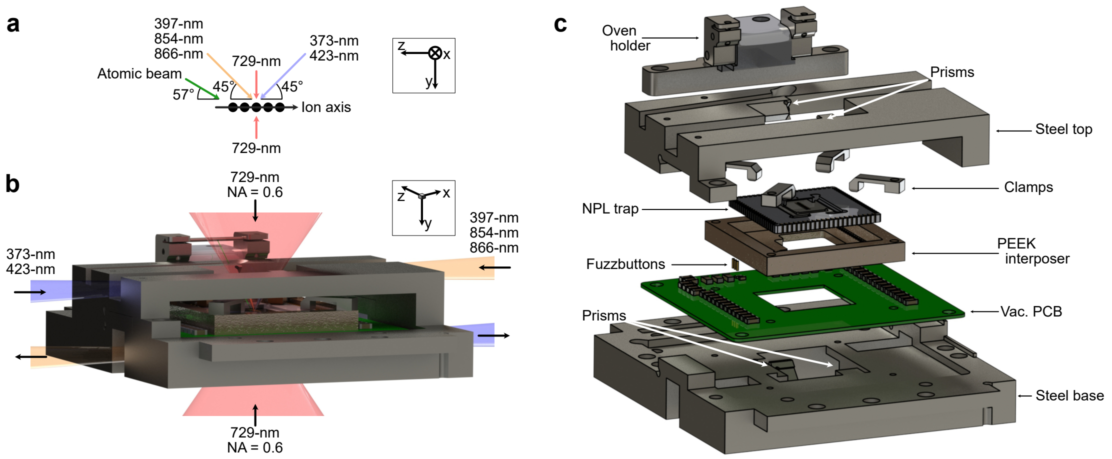



**Figure 3.5: a)** Laser and atomic beam geometries with respect to the ion chain. **b)** Compact trap housing with dual NA=0.6 access. The size of the high NA objectives restricts access for other beams, and so in-vacuum mirrors are used to redirect low-NA beams (373, 397, 423, 854, 866-nm) to the trap center. **c)** Exploded view of the trap housing with components labelled.

## 3.4 Vacuum System and Beam Geometry

Ultra High Vacuum (UHV) is required to prevent ion loss. However, UHV equipment is often bulky, and puts constraints on the access and visibility of the ion chain. Here, we describe the custom vacuum system, allowing for multiple beam access geometries including dual high numerical aperture (NA) objectives required for single addressed optical standing waves. The vacuum system was designed by Sebastian Saner, Chris Ballance, and Mariella Minder, and constructed by Sebastian Saner, Fabian Pokorny, and myself.

A residual pressure of $< 10^{-11}$ mbar is desired. For this strict requirement, care must be taken in selecting in-vacuum materials, and thorough cleaning and baking procedures must be followed. A summary of tactics that were useful in the construction of the vacuum system can be found in [24, 25].

The system consists of a 6" spherical-octagonal experimental chamber[15] connected to a spherical chamber with ion pump[16], ion gauge[17] and Titanium sublimator

[15] Kimball MCF600-SphOct-F2C8
[16] Agilent VacIon Plus 20 Pump
[17] Agilent UHV-24P Ion gauge

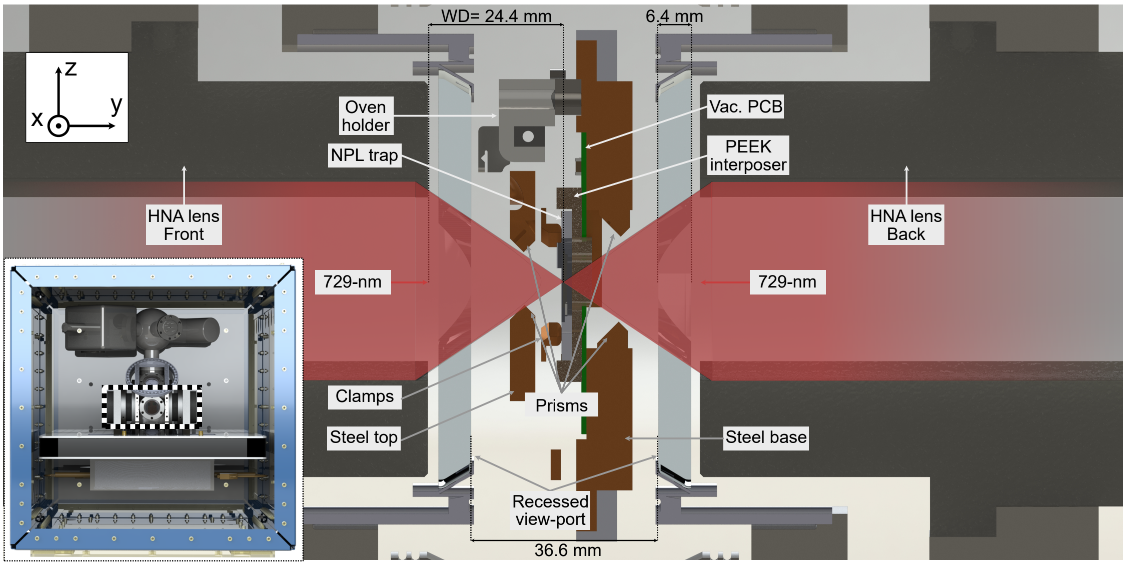



**Figure 3.6:** Cross section of the vacuum chamber and dual high NA objective sandwich. Lower right inset shows the vacuum chamber housed in the MuMetal enclosure, with dashed lines indicating the cross section region. Custom recessed CF-100 viewports are used to provide optical access for the high NA objectives (working distance = 24.4 mm).

pump (TSP)[18] attached. After the final system bake, a He leak test was performed and no external leaks were found. We do observe a gradual pressure increase that we suspect is either due to the in-vacuum optics, the optics epoxy adhesive[19], or the PCB solder. Regardless, the ion pump and TSP maintain UHV at the desired $< 10^{-11}$ mbar (the TSP is fired every 4 weeks at 41 A for 60 seconds to provide a fresh Ti layer).

Figure 3.5 (*a*) shows the beam geometries with respect to the ion chain axis. The quadrupole 729-nm beam has high NA (NA = 0.6) access perpendicular to the ion chain axis, while 373-nm, 397-nm, 423-nm, 854-nm and 866-nm beams have low NA (NA $< 0.05$) access at an angle of 45° to the ion chain axis. The atomic beam from the oven is at an angle of $\sim 57°$ to the ion chain axis. As such, there is a doppler shift of the atomic beam with respect to the 423-nm photoionisation laser, which is accounted for in the set laser frequency (given 12° angle between atomic and 423-nm beam, and 500 K mean temperature, expect +126 MHz frequency shift). Figure 3.5 (*b*) shows the custom trap housing including NA=0.6 access for the

[18] Scanwel custom housing
[19] EPO-TEK 353ND

729-nm beams, and in-vacuum mirrors (prisms) to redirect the low NA beams to the trap center. The low NA beams enter the housing via one of the four beam access channels parallel to the trap chip plane, and are reflected through the trap slit. The exiting beam is reflected by a second mirror out of the housing to prevent scattered light incident on the ion chain. Figure 3.5 ($c$) shows an exploded view of the trap housing with components labelled. The trap chip is interfaced to an in-vacuum PCB (described in section 3.2.2) via a custom PEEK interposer with embedded *Fuzzbutton* vias (one for each DC and RF channel). The trap chip is held in place by 4 steel clamp arms, which attach to a steel base plate which provides mechanical stability and houses the lower in-vacuum prisms. A top steel plate houses the upper prisms. The tube oven is mounted on two steel clamps which are electrically isolated from the base plate by a PEEK spacer. The trap housing is fastened to the vacuum can by 4 steel arms attached to alternating side ports of the octagonal chamber [20]. Figure 3.6 shows a cross section of the vacuum chamber and recessed dual high NA objectives. Two recessed CF100 viewports[21] are required to bring the custom objectives[22] to the designed working distance of 24.4 mm from the ion chain.

## 3.5 Imaging System

To image the ion chain, and readout the qubit states, fluorescent scatter of incident 397-nm light is collected. Here we briefly describe the imaging system, while section 4.2.2 describes the qubit readout procedure and fidelity. The imaging system is designed to provide high spatial resolution of the ion chain, while also providing a large field of view to accommodate long ion chains. The imaging system must also be compatible with the high NA objectives used for single addressed optical standing waves. Figure 3.7 shows the optical path schematic of the imaging system, with both a a high quantum-efficiency ($\simeq$ 70% at 397 nm) CMOS camera[23] for spatially selective qubit readout on a multi-ion chain, and a photon multiplier tube

[20]Using *Kimball Groove grabbers* P/N 53-482610
[21]*UK Atomic Energy Authority* P/N VPR100015
[22]*Photon Gear, inc.* Custom 18WD Atom Imager
[23]*Hamamatsu Orca-Fusion BT* C15440-20UP

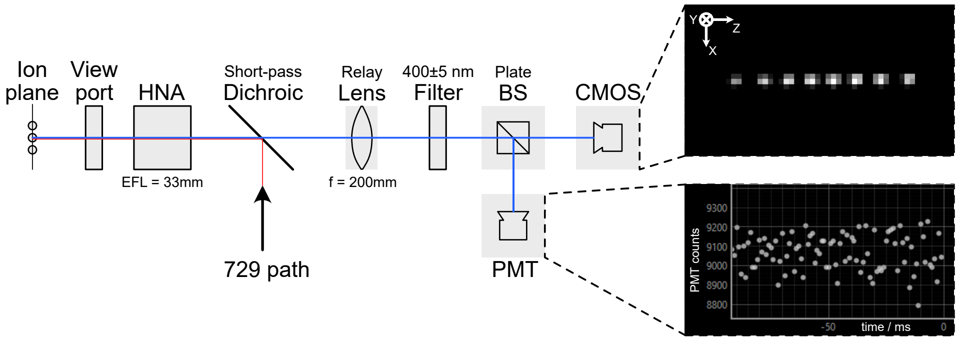



**Figure 3.7:** Optical path for fluorescent imaging of the ions using both a CMOS camera and a Photon-multiplier tube. Insets on right hand side show typical signal for: spatial resolution of eight ions in a linear chain (top), time resolution of detected photons from the ion chain (bottom).

(PMT)[24], for experiments requiring time resolved photon counting (e.g. micromotion compensation, section 4.2.3). Typical signals of both devices can be seen in right panels of figure 3.7. The optical path consists of a high numerical aperture objective[25] (effective focal length, $f$ = 33 mm), NA = 0.6, a dichroic mirror[26] (transmission: 397 nm; reflection: 729 nm), a relay lens[27] ($f$ = 200 mm), a 90 : 10 plate beamsplitter[28], and the camera and PMT.

The objective lens is custom designed to compensate for aberrations due to the recessed vacuum viewport, however, due to the large NA, accurate alignment of all degrees of freedom of the objective is critical. As such, the NA is attached to a 5-axis alignment stage[29], with z-axis focus controlled by a piezo micrometer[30] for fine adjustment of the ion focus with the MuMetal box closed (requires < μm precision).

Given the chosen lens focal lengths, we expect a magnification of ×6 from the ion plane to the imaging plane. With an expected ion separation of 5 μm, and the camera pixel pitch of 6.5 μm, we expect around 4.5 pixels between ions, which, if the system is well aligned, is more than sufficient for low cross-talk spatial readout of qubits.

[24] *Hamamatsu* Photon counting head H10682-210
[25] *Photon Gear, inc.* Custom 18WD Atom Imager
[26] ***Thorlabs* TODO**
[27] *Thorlabs* Achromat Doublet, ACT508-200-A-ML
[28] ***Thorlabs* TODO**
[29] *Newport ULTRAlign* M-562-XYZ, M-36, SM-13
[30] *Newport CONEX-TRA12CC*

## 3.6 Ca$^+$ Laser Systems

Singly-charged group 2 elements are popular in ion trap experiments due to their single outer electron resulting in Hydrogen-like energy levels. In this thesis we use $^{40}$Ca$^+$.

$^{40}$Ca$^+$has no nuclear spin giving the (relatively) simple level structure shown in figure 4.1 with the relevant laser transitions used in our apparatus are indicated. Details on the ion choice, and utility of the lasers is given in section 4.1, here we only describe the hardware and feedback control to provide this stabilised light.

The relevant wavelengths are the following: The qubit is defined by the quadrupole transition at 729-nm. Access to other excited levels, outside of our defined two level system, are crucial for ion trap quantum logic to enable qubit readout via state selective fluorescence (397-nm, and 866-nm transitions), and state preparation via optical pumping (866-nm, and 854-nm transitions). Access to two additional transitions in neutral calcium for isotope selective ionisation, 423-nm and 373-nm, are also required [26].

This section begins with a description of laser frequency and power stabilisation techniques common to multiple laser systems used in this thesis. Following this, the specific systems are described, starting with the diode lasers, which drive all but the 729-nm transition. Subsequently, the Ti:Sapph 729-nm source is discussed. Finally, the global and single addressed beam arrangements for the 729-nm system are described.

### 3.6.1 Frequency Stabilisation

All lasers exhibit frequency noise due to unavoidable environmental coupling (e.g. temperature fluctuations, and acoustics), and to instabilities in control signals (e.g. voltage and current controls). For the lasers corresponding to dipole transitions used in this thesis, we are relatively lenient and require slow frequency excursions of $< 1$ MHz with respect to the qubit transition, and laser linewidths less than the natural linewidth of the dipole transitions, which are often multiple MHz.

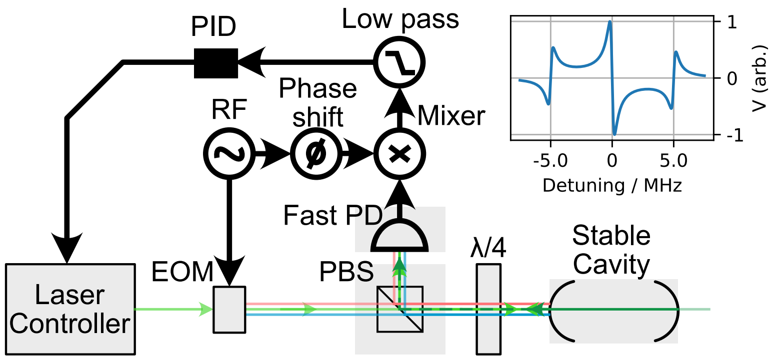



**Figure 3.8:** PDH optical and control schematic. Green optical path represents the carrier frequency, blue and orange lines represent the upper and lower sidebands respectively, generated by an EOM. The reflected light is directed onto a fast photodiode, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The signal is finally low pass filtered to recover the error signal, an example of which is seen in the upper right panel for a finesse $F \sim 4000$, a free spectral range of $FSR = 1.5$ GHz, and a modulation frequency of $f_{mod} = 5$ MHz.

However, for the quadrupole transition at 729-nm, we require significantly more stringent frequency stability, with laser linewidths of $< 100$ Hz, and negligible phase modulation of the laser light over a few MHz of the qubit transition to ensure spectator transitions are not excited during qubit operations. Although the order of magnitude of the required frequency stability is different for the dipole and quadrupole transitions, the concepts and techniques used to achieve this stability are similar. Generally, a closed loop consisting of measurement of the optical frequency of the laser and feedback to the laser control is required. In this work, stable external optical cavities are used for referencing the laser frequency. Specifically, we use the Pound-Drever-Hall (PDH) techniques to generate the error signal for the feedback loop. The PDH technique is well described in the literature [27–29], and so here we only give a brief overview of the technique to contextualise the required hardware introduced in the following sections.

The conceptual goal of the PDH technique is to measure the frequency difference between the laser and a resonant frequency of a stable optical cavity to generate an error signal, preferably linear over typical frequency excursions (while *locked*). This error signal can then be modified by some controller (typically PID), to directly infer how to tune the laser controls to minimise this error signal, often termed *locking* the laser to the cavity. Further, the error signal should be insensitive to amplitude noise on the laser light, to avoid amplitude to frequency noise conversion due to the feedback loop (in contrast, side-of-fringe stabilisation is sensitive to amplitude drifts).

Practically this scheme consists of phase modulation of the incident laser light, here by electro-optical modulators (EOMs), to generate sidebands, which are used as phase references to compare the phase of the incoming laser field to that of the phase of the field inside the cavity. Figure 3.8 shows a schematic of the PDH scheme, highlighting the beam paths and closed loop. The phase modulated light is directed onto the cavity, with the total return signal being some combination of the reflected sidebands, carrier, and the cavity leakage field, with the latter providing the phase of the electric field inside the cavity. The reflected light is directed onto a fast photodetector, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The resulting signal is low pass filtered to remove high frequency components, yielding the final error signal, an example of which is seen in the upper right panel of Figure 3.8. The central region is anti-symmetric and linear, centered on the desired lock point. Using a tuned controller for feedback to the laser controller, this provides a stable lock of the laser to the cavity.

As the cavity is used as a frequency reference, both a high finesse and stable optical lengths are desirable. The gradient of the PDH error signal is proportioanl to finesse, leading to increased signal to noise with higher finesse (see Ref. [28] for more details). Optical length deviations are minimised through mounting the cavity in a vacuum chamber, and taking care in isolating it from temperature fluctuations and vibrations.

Particular details of the PDH hardware and control loops are given in the following sections for each laser system.

### 3.6.2 Intensity Stabilisation

The light-matter interactions considered in this thesis are all proportional either to intensity or amplitude of the optical electric field at the ion's position. Intensity fluctuations over time therefore lead to fluctuations in the interaction strength, which are detrimental to the fidelity of quantum operations. Intensity noise originates either from total beam power fluctuations, or due to beam pointing drift with respect to the ion position. The former is mitigated through a closed feedback loop between a pick-off photodiode signal before the ion monitoring the optical power, and the drive RF amplitude of an upstream AOM. For all beams, this optical pick off is one of the last elements before the beam enters the vacuum system to reduce the impact of lossy elements between the stabilise point and the ion position. The low-noise photodiodes used were designed by David Nadlinger (*QLNPD*), and the feedback system controller, known as *SUServo*, was written by Peter Drmota. Beam pointing drifts are mitigated through careful mechanical design of the optical paths for passive stability, and through automated intermittant beam pointing calibrations using the ion as a sensor (see section 4.2.3).

### 3.6.3 Diode Lasers

For transitions 397-nm, 854-nm, and 866-nm, we require light that is frequency stable to < 1 MHz level, well below the natural line widths of the dipole transitions, and power stable to < 1 µW. All beam paths and relevant control loops for the diode laser systems are shown in figure 3.9. The laser diode heads[31], controllers[32], and splitterboards[33] (fig 3.9 (a)) are housed in a single 19" rack. The splitterboard distributes the light between two ion-trap experiments, the PDH lock cavity (fig 3.9 (c)), and a diagnostic wavemeter for checking wavelength and lock status[34]. The splitterboard also houses an Electro-optic modulator (EOM) module for applying

[31] *Toptica* diodes. 854/866: MDL DL pro; 397: MDL DL pro HP; 423: DL pro
[32] *Toptica* DLC Pro with PDH boards
[33] Designed by *OITG Cavities* group
[34] *HighFinesse* WS7

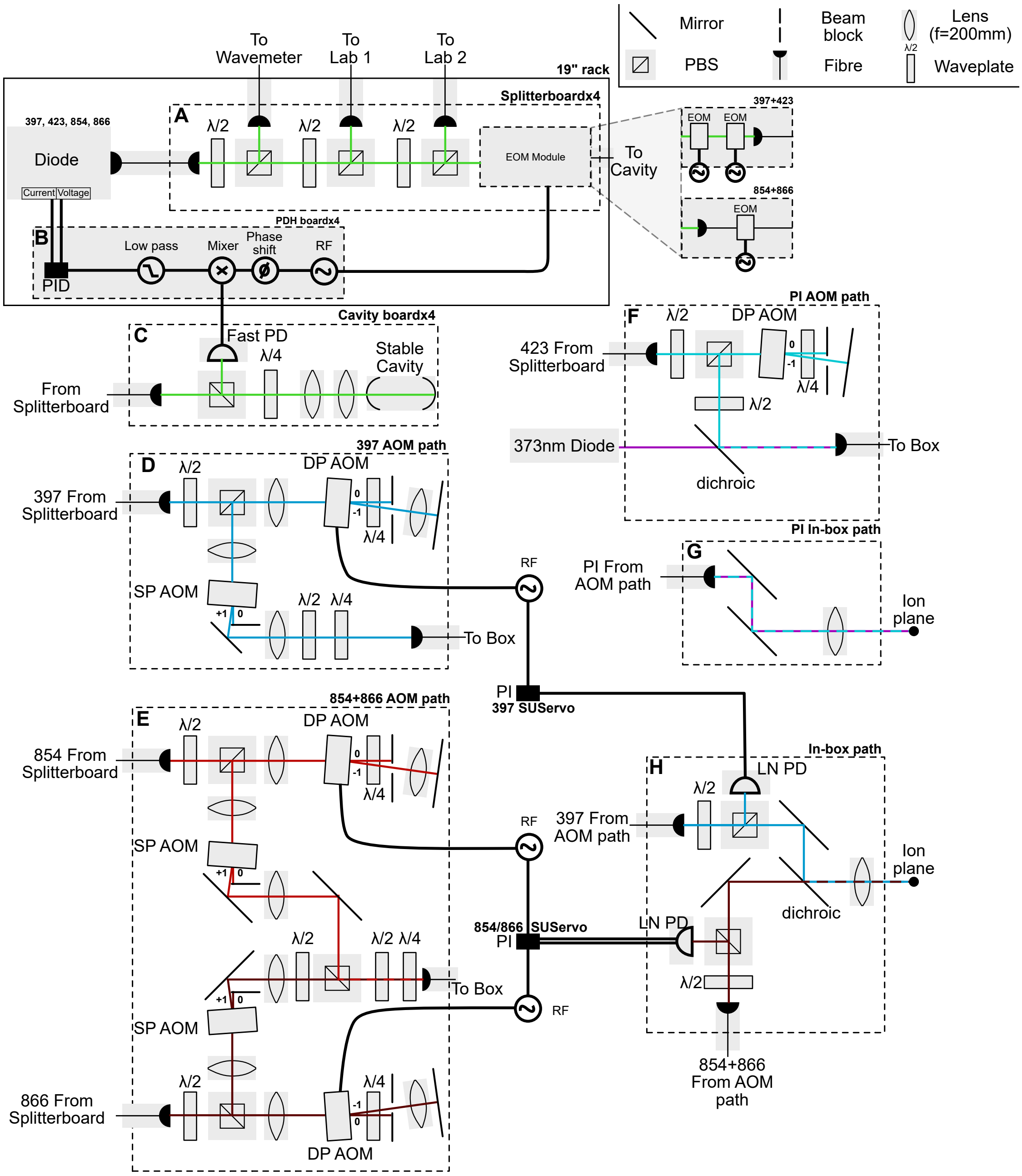



**Figure 3.9:** All beam paths for diode laser systems. Dotted boxes resemble modular optical paths with fibre and control lines in and out only. Frequency and optical power control loops are included.

both offset and PDH sidebands for an offset PDH lock. The offset sidebands may be arbitarily set (within the GHz range of the EOM) to provide a continuous choice of laser lock frequencies between cavity modes, rather than maintaining a fixed offset between the laser output frequency and the discrete cavity mode frequencies. For 397-nm and 423-nm, resonant free space EOMs[35] are used, for 854-nm and 866-nm broadband fibre EOMs[36] are used for convenience. The path on the reference cavity board (fig 3.9 (c)) consists of a four-bore reference cavity[37] under vacuum and temperature control, two lens for matching the collimated input beam to the cavity mode, and a beamsplitter and $\lambda/4$ waveplate for directing the reflected cavity light to a fast photodiode[38]. The beatnote due to the carrier and PDH sidebands are mixed down and low-passed by the *Toptica* PDH module to produce the signature antisymmetric PDH trace. Gains for PID controllers of both the diode current and voltage are set to stabilise the optical frequency.

The light from the “laser” laboratory is carried to the “trap" laboratory using 16 m polarisation maintaining fibres[39]. In the “trap” laboratory we use AOMs in a double pass configuration for both frequency shifting and power stabilising the light (fig 3.9 (d) and (e)).

As space inside the MuMetal enclosure is constrained, it is advantageous to couple multiple beams in a single fibre before the enclosure. The second stage photo-ionisation light, at 373-nm[40], is combined with the 423-nm beam by a dichroic (fig 3.9 (f)) before being coupled to a 16 m fibre that takes the light directly into the MuMetal enclosure. The respective refractive indices of fused silica for 373-nm and 423-nm differ to the extent that the two beams were separated by more than one waist at the ion plane after initial rough alignment. To ensure the two beams overlap at the ion plane, care was taken for the beam propagation to be perpendicular to both the lens and vacuum viewport window, and a parabolic mirror outcoupler[41]

[35] *Qubig* PM7-VIS-25 and PM9-VIS-2.25
[36] *EOSPACE* PM-0s5-05-PFA-PFA-850
[37] *Stable Laser Systems* SLS-6010-1-4bore
[38] *Thorlabs* PDA10A2
[39] ***Schafter-Kirchoff ...***
[40] *Toptica* iBEAM-SMART-375-S.
[41] *Thorlabs* RC02APC-F01

(rather than a lens) was used inside the MuMetal box.

The 854-nm and 866-nm light is combined after their respective double and single pass AOMs, and coupled into a 3 m PM[42] fibre carrying the light into the enclosure. Light from both the PI and the 397/854/866-nm paths are directed onto the ions via the in-vacuum prisms, discussed in section 3.4. The final mirrors prior to the vacuum system are motorised[43], so that beam alignment to the ions can be optimised when the MuMetal enclosure is closed. Practically we find that these beams rarely need to be realigned (few months), unless there is a significant ambient temperature change of a few degrees.

### 3.6.4 Narrow Linewidth 729 Laser

Lasers are a key tool for creating the highly localised, strong electric field amplitudes and gradients needed to drive both carrier and sideband transitions of the trapped ion.

As shown in Figure 4.1, two sublevels within the $4S_{1/2}$ to $3D_{5/2}$ manifolds define the qubit. This is an electric quadrupole transition as $\Delta l = 2$. For the Calcium ion this transition is at 729-nm, and so a near resonance 729-nm laser is used to implement single- and multi-qubit gates (sections 4.2.4 and 4.4.2). This transition is also used after Doppler cooling for resolved sideband cooling to prepare the motional mode close to its ground state (as discussed in section 4.3.1). This transition is narrow linewidth due the the long lived $3D_{5/2}$ level, and so, for power efficiency, a narrow linewidth laser must be used. For the "fast entangling gates" use-case, high intensity electric-fields are required at the ion to drive sideband transitions, this will be discussed further in the single-addressing and two-qubit gate sections, but here it is sufficient to say we require >100 mW of light at the ion plane. Here we describe the 729-nm system consisting of a Ti:Sapph laser system pumped by an Nd:YAG 532-nm laser.

An *M2 Solstis* Ti:Sapph is pumped via 18 W of 532-nm light from a *Coherent*

[42] ***SK ...***

[43] *Newport Picomotor* piezo mirror mount P/N 8821

*Verdi-V* system to produce around 5 W of 729-nm light. The Ti:Sapph is engineered to operate with a stable $< 50$ kHz linewidth. Ti:Sapph crystals have broadband gain profiles which make them amenable in research environments to create frequency tunable laser systems. They also often have lower intrinsic phase noise than diode lasers at the same wavelength, which is favourable for our application. For single mode operation, the *Solstis* employs multiple intracavity frequency selective elements which consist of (in order of coarse frequency selectivity), a birefringent filter, a tunable Fabry-Pérot etalon, and the surrounding bow-tie cavity. To lock the tunable etalon to one longitudinal cavity mode, an actively stabilised "dither" phase-lock loop is used. This dither consists of periodically varying the etalon spacing with a frequency of around 20 kHz. This dither frequency direclty phase modulates the light, which leads to the creation of sidebands which can interact with the ion causing errors in gates. This dither frequency (and harmonics of it), can be observed via composite pulse experiments on the ion, and are discussed in section 4.2.7.

Referencing the *Solstis* output with an external cavity and using PDH feedback can greatly suppress phase noise. Here an ulta high finesse cavity[44] is used, with a schematic of the 729-nm system shown in Figure 3.10. As opposed to the simpler controller implemented for the diode lasers, the 729-nm uses a cascaded PID lock with a fast feedback loop to the AOM situated after the Solstis, and an intermediate and slow feedback loop to piezos controlling the Solstis cavity length. With this system, a linewidth of $<< 1$ kHz for the 729-nm light is expected. The electronics for this control loop are provided by *Stable Laser Systems* in the form of their *FPGA Servo* lock box. For an effective PDH lock, both short and long term stability of the reference cavity is required. To ensure the cavity is insensitive to the environment, it is made of an ultra low expansion material, and mechanically isolated from the environment by an acoustic dampening enclosure and a vibration isolation platform. The cavity is housed in a vacuum system to prevent refractive index fluctuations due to air, and to further reduce thermal coupling to the environment. The cavity is temperature stabilised at the zero crossing temperature of 30.6°C. Long term

[44] *Stable Laser Systems* P/N 6020-4

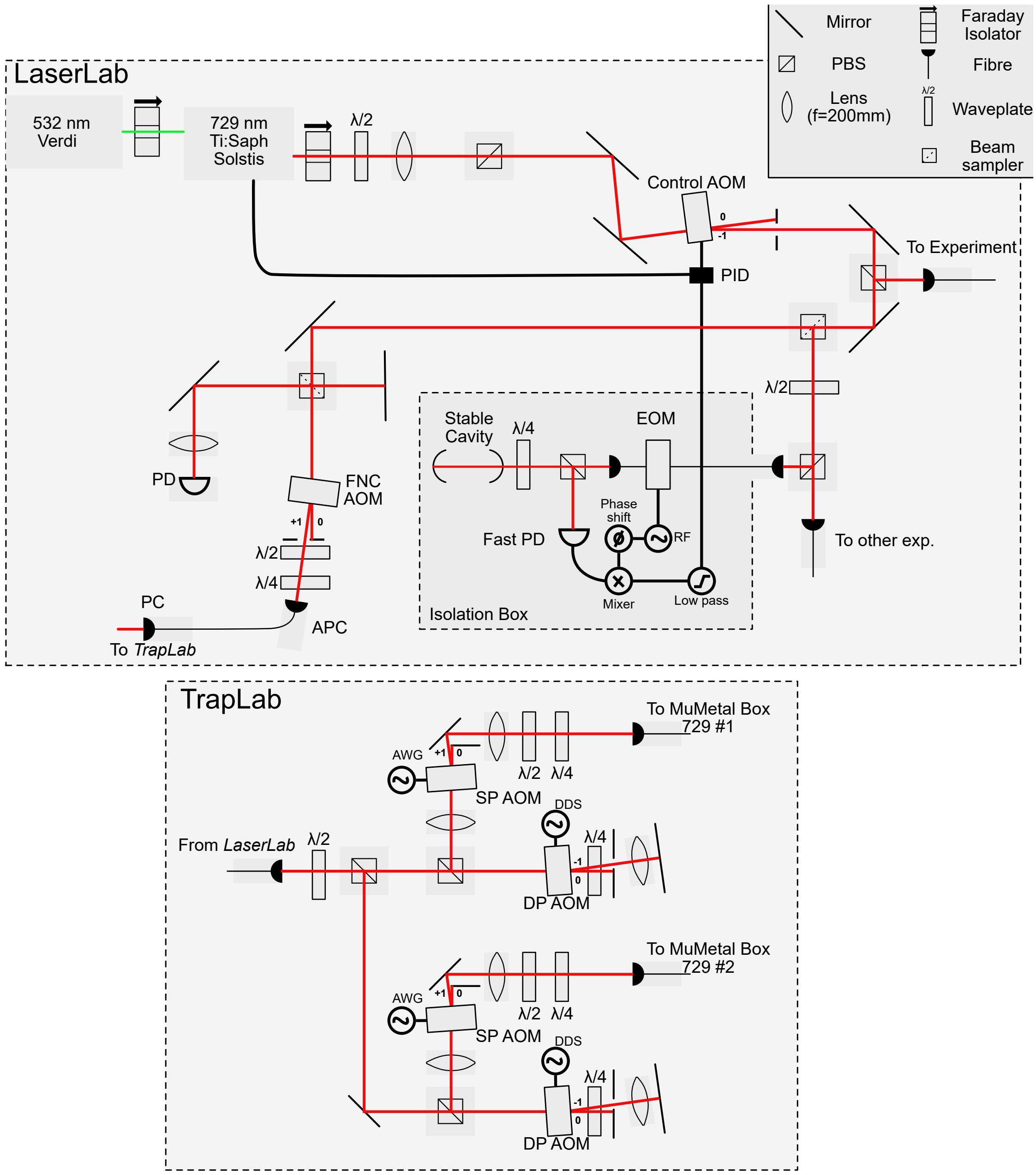



**Figure 3.10:** The 729-nm system. A Ti:Sapph laser tuned to 729-nm is pumped by a 532-nm source. Light is picked off at the first beam sampler to stabilise by PDH locking to a cavity. Light is distributed to two experiments, with the *FastGates* line including a fibre noise cancellation system.

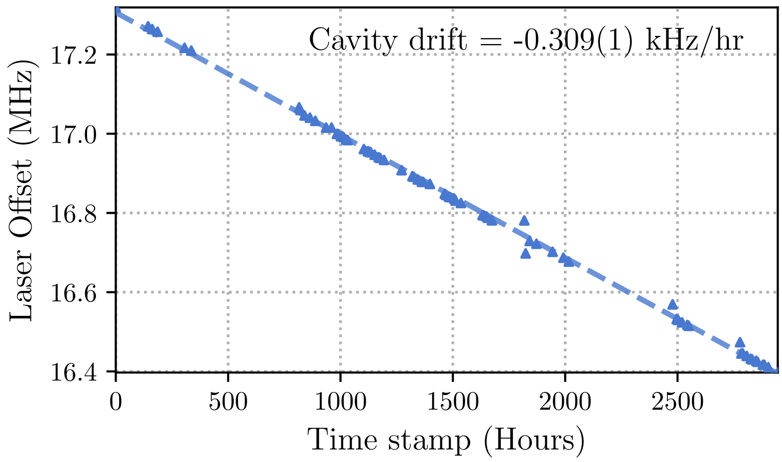



**Figure 3.11:** Cavity drift over 125 days measured by reference to the ion transition. The laser offset here is with respect to running all 729-nm AOMs at their central frequencies. Outliers are due to bad fits of the ion transition frequency, not due to changes in the cavity resonance.

cavity drifts of 309(1) Hz/hr are measured over 125 days, seen in figure 3.11. This measurement uses the ion as a frequency reference to probe the cavity frequency, with details given in section 4.2.3.

Figure 3.10 displays the other beam paths for our 729-nm system. Some light is picked off and sent to a wavemeter to continuously monitor the frequency. However, the majority is coupled to two output fibres for our experiment and another within the group. The 729-nm light is transported from a dedicated laser lab to a the trap apparatus lab by a 10 m single mode polarization maintaining fibre[45]. The fibre introduces phase noise due to mechanical and thermal pertubations along the 10 m length. To remove this introduced noise, passive stabilisation in the form of thick foam tubing along the fibre length, as well as active stabilisation by a fibre-noise-cancellation (FNC) technique [30], is used[46]. The efficacy of the FNC system is characterised in section 4.2.7 by measuring the spin coherence time of the ion.

[45] ***SK ...***

[46] FNC control consists of *Sinara Stabilizer* and *Pounder* mezzanine board with PID software developed by Ayush Agrawal.

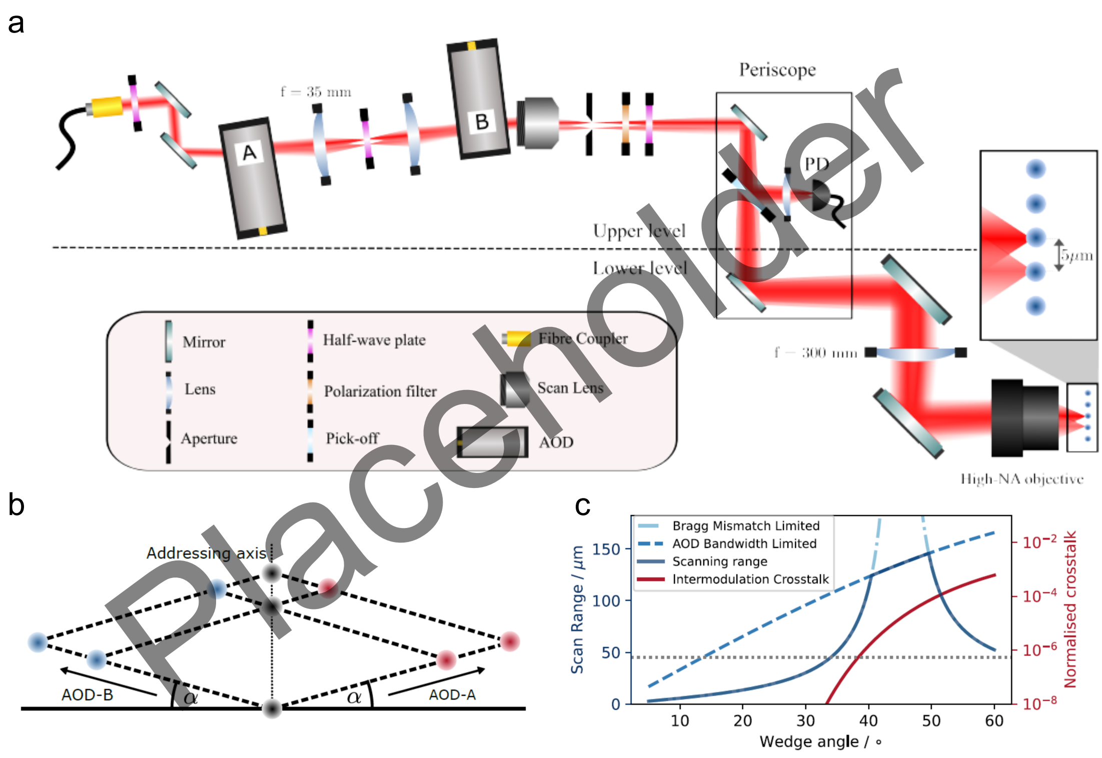



**Figure 3.12:** AOD setup

### 3.6.5 Single Addressing System

Single ion addressing is a requirement for executing arbitrary circuits on a register of ion-qubits. Furthermore, it is advantageous to focus the optical drive onto the ion, increasing the intensity of light, and so driving faster gates. A unique design feature of our system is the ability to produce an array of single ion addressing standing waves with individually contrallable spatial phase, thus enabling any-to-any connectivity for fast engangling gates based on the scheme demonstrated in Ref. [3]. The design and construction of the single addressing system was led by Iver Øvergaard and, as of writing, has been in constant use in the experiment for around 1 year. Here we shall quote the design specification and relevant details of the single addressing system. In section 4.2.6 we present the characterisation data upon installation. Further technical information can be found in IRO's Transfer report [31], availible upon request.

Multiple methods for single ion addressing have been developed in the ion trap community, including waveguide arrays [32, 33], micro-electronic-mechanical mirror arrays [34, 35], spatial-light-modulators [36], and acousto-optic deflectors [37, 38]. Each method has its own advantages and disadvantages in terms of speed, resolution, ease of integration, and scalability, and so choice of hardware is often a matter of experimental preference and constraints. We require:

- Addressing of up to two ions in a $\sim$ 10 ion chain with low crosstalk to neighbouring ions.
- Fast reconfigureability of the beam position within a single experimental shot (i.e. between different gates within a single circuit execution).
- Good optical power efficiency to ensure high intensity electric fields at the ion.
- Flexibility in beam positioning to allow for fine tuning of the overlap of two counterpropagating beams used to create the standing wave.
- Individual phase control of each addressed beam to stabilise the standing wave phase at the ion plane.
- Beam position stability to avoid power fluctuations.
- A compact design to fit within the magnetic shielded region of the experiment.
- Good telecentricity (fixed beam k-vector direction across the scanned area) to avoid unwanted travelling wave component in a standing wave gate scheme.
- Zero frequency shift of the beam at the ion plane.

Crossed Acousto-Optic Deflectors (cAODs) are a popular choice for single ion addressing, offering fast rise times ($\sim$ 10 µs), and good resolution over a large scanning area ($\sim$ 500 resolvable spots with 2° angular range) with zero frequency shift along the addressing axis. AODs operate through the same mechanism as AOMs, but are optimised for beam deflection rather than frequency modulation. As an addressing system cAODs have poor scaling of power efficiency with the number of addressed spots, and so are not ideal for large scale systems. As we are interested in addressing $\leq 2$ ions at a time, this is not a significant issue for our system.

Figure 3.12 (a) shows a schematic of the constructed addressing system. Due to space constraints within the magnetic shielded region, the optical elements are located on two height levels with a periscope steering the beam between them. The beam is delivered via single-mode polarisation maintaining fibre to the upper level where it is steered through the first AOD, with beam angle optimised for maximum diffraction efficiency into the $-1^{st}$ order. Two lenses[47] in a 4f configuration re-image the deflection origin of the first AOD[48] onto the second AOD with the second AOD angled for maximum diffraction efficiency into the $+1^{st}$ order. The cAOD arrangment is termed "crossed" as the second AOD is rotated by some angle $\alpha$ with respect to the first, so that the deflection from the second AOD has some orthogonal component to that of the first. When the two AODs are driven with RF tones of the same frequency, $f$, the high efficiency diffracted order combinations are: (0,0), ($-f$,0), (0,$+f$), ($-f$, $+f$). The last combination of ($-f, +f$) has zero overall frequency shift yet retains some frequency dependent deflection angle, which is suitable for addressing ions in a 1D line. To translate this deflection angle into some physical displacement at the ion plane, a telecentric scanning lens[49] is used. The telecentricity ensures that the beam k-vector direction is fixed across the scanned area, which is important for standing wave gate schemes to avoid unwanted travelling wave components. At the intermediate focus after the scanning lens, the unwanted orders are blocked by an adjustable slit aperture and the beam polarisation is cleaned and selected using a polarisation filter and half-wave plate. A plate beam splitter[50] is used within the persicope to pick off $\sim 5\%$ of the light for monitoring on a photodiode[51], which is used in a feedback loop to stabilise the intensity at the ion plane (as discussed in section 3.6.2). The scan lens, in combination with a relay lens, forms a beam expander to increase the beam NA for the final objective lens to

[47] **P/N**
[48] **P/N**
[49] **P/N**
[50] **P/N**
[51] **P/N**

focus to a beam waist of $\approx 1.5$ µm at the ion plane. Characterisation of the system (details given in section 4.2.6) showed that the addressing beam could be focused to a Gaussian waist of 1.50(1) µm, measured by scanning a single trapped ion with Rabi experiments. The same beam was then tested in a five-ion chain, where the nearest-neighbour intensity crosstalk was measured to be $4.16(2) \times 10^{-4}$**unit?**.

### 3.6.6 Global Addressing System

The single addressing provides the utility of spatially selecting a few ions to drive with the 729-nm light, however, the power efficiency scales as $O(n^2)$, with $n$ being the number of addressed spots. As such, we limit the single addressing to simultaneously illuminate only one or two ions at a time. It is therefore convenient to also have a global beam that has dimensions larger than the ion chain to address all the ions in parallel. This global branch is currently counterpropagating to the single addressing path, and shares the same optical path as the imaging system (see figure 3.7). To increase the beam intensity at the ions, the global branch uses an anamorphic prism pair[52] to elliptically shape the beam, with a 4:1 aspect ratio, with major axis along the ion chain direction. The beam is focussed onto the ion plane using an $f = xx$ lens[53], resulting in a major axis beam waist radius of 52(3) µm[54]. As with the other beams directed to the ion, we use a final piezo controlled mirror[55] for fine alignment and control when the MuMetal enclosure is closed.

## 3.7 Control System

**Could tie up loose end here of control FPGA blocks.**

[52] *ThorLabs* PS883-B

[53] **P/N**

[54] The beam profile was measured by extracting per-ion Rabi frequencies while driving on resonance quadrupole transitions (see section 4.2.4) and assuming a Gaussian cross section.

[55] *ThorLabs POLARIS*-K2S2P

## 3.8 Ion Loading

Atoms are provided to the trapping region by heating Calcium flakes housed in a resistive tube oven with the oven mount shown in figure 3.5. A thermocouple is tag welded onto the tube oven to provide a temperature reading for current feedback for fast stabilisation to the target temperature of $\simeq 200°$C. A steady state of $\simeq 2$ A across the 0.5 $\Omega$ load of the oven tube is required to maintain 200°C. The tube oven has a pinhole that is aligned to the ion flux aperture, visible in figure 3.5 with atom flux shown in green, and the atomic shield pinhole shown in figure 3.2. These apertures were included in the design to improve the operational lifetime of the trap by ensuring neutral Calcium metal cannot accumulate on the trap surface and cause shorts between RF, ground, and DC electrodes. The aperture design *luckily* failed. It failed as we observe atomic flux at all electrode locations suggesting the aperture is too large. It was *lucky* as the planned trapping region of *dc10* became non-viable after electrode *Fr dc11* developed a continuity fault. As such, ions are loaded at *dc5*.

The neutral atom flux is ionised using a two-step photoionisation scheme [26] that can selectively load the desired isotope of Calcium. Natural abundance Calcium, with typical isotope proportions of 96.9% : $^{40}$Ca, 2.09% : $^{44}$Ca, 0.647% : $^{43}$Ca [26], was appropriate as we aim to exclusively load $^{40}$Ca$^{+}$. The first stage involves a frequency stabilised 423-nm beam that couples the 4s$^2 \rightarrow$ 4s4p transition in neutral calcium. The natural linewidth of this transition ($\simeq$ 35 MHz) is much smaller than the isotope shifts between abundant isotopes of Calcium (typically $>$ 100 MHz shifts), as such, through frequency selectivity, only the desired isotope can be excited. The second ionisation stage couples 4s4p $\rightarrow$ continuum through a 373-nm UV laser source (although the exact wavelength does not matter as long as $<$389.8-nm). We practically see no loading of other Calcium isotopes, and only see rare non-fluorescent ions (once every few weeks) after collisions with residual gas (we suspect this is CaH$^{+}$).

The 373-nm laser causes charge build-up on the trap electrodes, which can affect the trapping potential, and so a physical shutter, controlled by a TTL signal, is used to block the beam when not actively loading ions.

# 4
# Experimental Characterisation

## Contents



Before we can proceed with our planned experiments involving the motion and spin of the atoms, the apparatus must be characterised. This characterisation serves two purposes: it allows us to benchmark the performance of our system against state-of-the-art results, and it reveals any current limitations that must be addressed before we can successfully carry out the experiments.

As discussed above, we are interested in two paradigms: discrete variable quantum computing (DVQC), that concerns the manipulation of qubits (or spins), and continuous variable quantum computing (CVQC), that concerns the manipulation of qumodes (or harmonic motion).

For DVQC, we must have access to the following elementary operations:

- Qubit state preparation, here via optically pumping the electronic state of the ion into one sublevel of the $4S_{1/2}$ manifold — the lower level of our qubit, section 4.2.1.
- Measurement of the qubit state, here via state-dependent fluorescence, section 4.2.2.
- Single qubit rotations, here via interaction with the 729-nm laser, section 4.2.4
- Two qubit entangling-gates, here via spin-dependent forces (SDFs), using the 729-nm laser and the motional modes of the ion, section 4.4.2.

For CVQC, we need:

- Motional state preparation, here via Doppler cooling and sideband cooling, section 4.3.1.
- Measurement of the motional state, which is not yet implemented on our system but will follow [**fluhmann2020direct**].
- Single and two qumode gates, which are also not yet implemented, but will utilise SDFs following [**sutherland2021universal**], section 4.4.1.

We split this chapter into interactions involving only the spin, then those focussed only on the motion, and finally we describe spin-motion interactions. First, however, we describe the ion, its level structure, and the relevant lasers to manipulate its state.

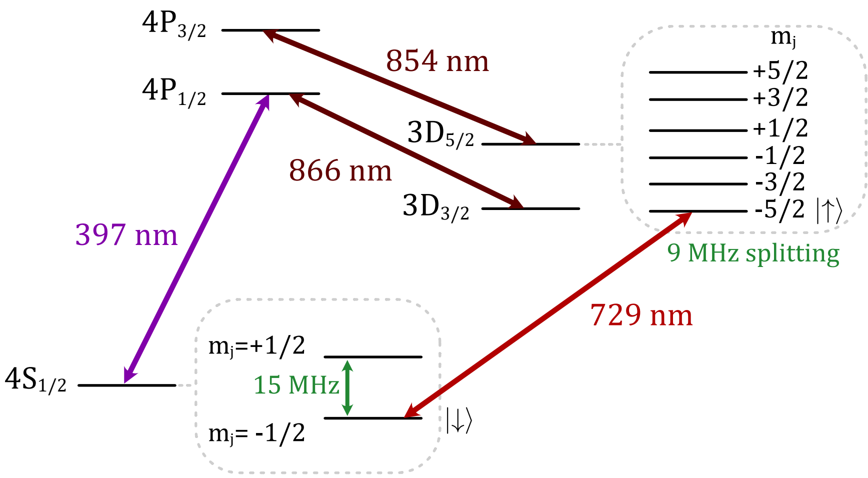



**Figure 4.1:** Electronic energy levels of $^{40}Ca^+$, which will be used in this thesis. The levels are split by the Zeeman effect due to a 5.4 G external magnetic field (they are shown explicitly only for the qubit manifolds). The transitions marked are required for cooling and control over the ion. The chosen qubit levels are labelled with $|\downarrow\rangle$, $|\uparrow\rangle$.

## 4.1 The Ion

The $^{40}Ca^+$ ion has a rich level structure, shown in figure 4.1, with many accessible quadrupole transitions $4S_{1/2} \leftrightarrow 3D_{5/2}$. The quadrupole transition is ideal for use as a qubit due to the long-lived metastable level ($\tau = 1.1$ s) [**barton2000measurement**], and the ability to optically couple the transition via a laser. As mentioned in section **??**, the 729-nm beam is used for this purpose. By selecting the appropriate frequency of the 729-nm beam, qubit operations such as state preparation (section 4.2.1) and single qubit gates (section 4.2.4) may be driven, or by setting the detuning such that a motional sideband is near resonance, to couple the spin and motional degrees of freedom (section 4.4.1). Here, we discuss which quadrupole transitions are used, and the rationale behind their choice.

Zeeman splitting of the $4S_{1/2}$ and $3D_{5/2}$ states leads to 2 and 6 non-degenerate sublevels, respectively. Due to the magnetic field strength of 5.4 G (section 3.3), the $S$-levels are split by $\sim$15 MHz, and the $D$-levels by $\sim$9 MHz. This splitting is considerably larger than both the natural linewidth of the 729-nm transition ($\sim$Hz), and the Ti:Sapph laser linewidth (<1 kHz), meaning the transitions can be selectively addressed by frequency tuning of the laser. There is some freedom

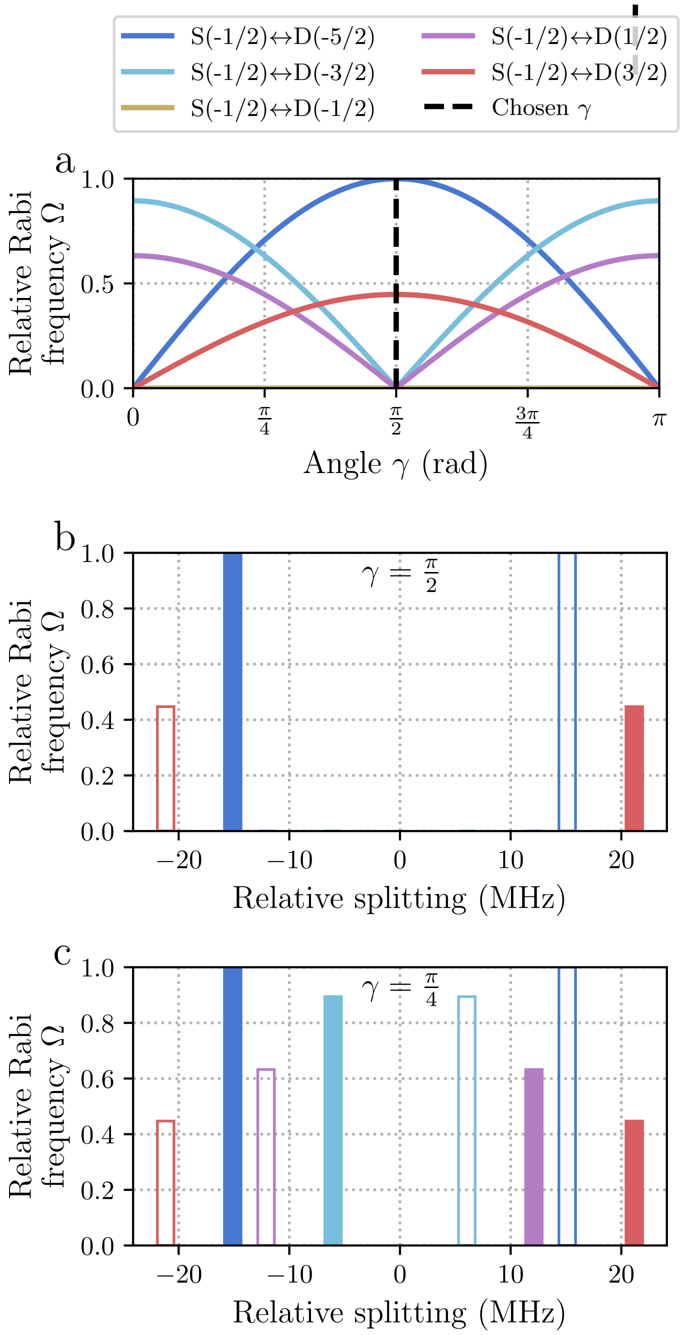



**Figure 4.2:** Normalised quadrupole Rabi frequencies, $\Omega$, and frequency splittings for the $4S_{1/2} \leftrightarrow 3D_{5/2}$ Zeeman sublevels in $^{40}Ca^{+}$. Solid bars correspond to transitions with lower state $|4S_{1/2},\ m_j = -1/2\rangle$, while open bars correspond to transitions with lower state $|4S_{1/2},\ m_j = +1/2\rangle$. **a)** Relative Rabi frequencies as angle $\gamma$ between the polarisation vector and the magnetic field is varied. The k-vector of the light is assumed to be perpendicular to the B-field. **b)** Relative frequency splittings and strengths when $\gamma = \pi/2$. This is the polarisation chosen for the current and future experiments to reduce the number of transitions which can cause off-resonant interactions when the qubit defined by $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -5/2\rangle$ is used. **c)** Relative frequency splittings and strengths when $\gamma = \pi/4$. At this angle, all transitions are accessible apart from those with $\Delta m_j = 0$. This is the polarisation chosen for our initial experiments to allow access to all transitions for characterisation.

in choosing which of these 10 possible (with $\Delta m_j \leq 2$) quadrupole transitions we define as our qubit. Each transition has a varying coupling strength, here denoted by the relative Rabi frequency, $\Omega$, depending on the polarisation and $k$-vector of the interacting light with respect to the B-field axis. The desired criteria for qubit and polarisation choice are as follows:

- Access to a closed-cycle transition for sideband cooling, (for details on this, see section 4.3.1). Closed-cycle here means $|4S_{1/2},\ m_j = -1/2\rangle \to |3D_{5/2}\rangle \to |4P_{3/2},\ m_j = -3/2\rangle \to |4S_{1/2},\ m_j = -1/2\rangle$.

- The qubit transition needs a large quadrupole matrix element, leading to a high Rabi frequency, $\Omega$, to enable laser-power-efficient, fast, gates. Polarisation, $\gamma$, may be optimised to increase this coupling.

- Ideally the qubit transition should have a low magnetic field sensitivity, to reduce the effect of magnetic field noise. Using a MuMetal enclosure around the experiment does relax this constraint, see section 3.3.

- To reduce the effects of off-resonant interactions due to nearby transitions, it is desirable to limit the number of accessible transitions near to the qubit transition. This is done by selecting a polarisation, $\gamma$, such that unwanted transitions are suppressed.

Given these criteria, we use the transition $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -5/2\rangle$ with magnetic field sensitivity $-0.446$ MHz G$^{-1}$. Closed-cycle sideband cooling is possible with this same transition, and state preparation via optical pumping is done via $|4S_{1/2},\ m_j = +1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -3/2\rangle$ with magnetic field sensitivity $-0.624$ MHz G$^{-1}$.

Figure 4.2 *a*, shows the normalised Rabi frequencies of each transition as the angle $\gamma$ between polarisation vector and B-field is varied. Here only linearly-polarised light with k-vector perpendicular to the B-field direction is considered. The polarisation of the 729-nm laser is set to $\gamma = \pi/2$ (black dashed line), which maximises the Rabi frequency of the qubit transition, while suppressing nearby

| Parameter | Value |
|---|---|
| 729-nm laser power | 1000 µW |
| 729-nm laser detuning | 0 MHz* |
| 729-nm pulse duration | 5 µs |
| 854-nm laser power | 60 µW, $250 \times I_{\mathrm{SAT}}$ |
| 854-nm laser detuning | 0 MHz |
| 854-nm pulse duration | 40 µs |
| 866-nm laser power | 15 µW, $\sim 70 \times I_{\mathrm{SAT}}$ |
| 866-nm laser detuning | 0 MHz |
| 866-nm pulse duration | 30 µs |

**Table 4.1:** * With respect to qubit transition frequency.
Experimental parameters used for state preparation of the ion into the $|\downarrow\rangle = |4S_{1/2},\ m_j = -1/2\rangle$ state.

transitions apart from that needed for state preparation. Figure 4.2 *b*, shows the relevant frequency splitting of the transitions at $\gamma = \pi/2$, which is the set polarisation for all experiments apart from initial quadrupole characterisation. For initial characterisation of the quadrupole transitions, the polarisation was set to $\gamma = \pi/4$, which allowed access to all transitions apart from those forbidden by the beam geometry, $\Delta m_j = 0$, see figure 4.2 *c*.
Other required lasers for manipulating the ion are: the 397-nm and 866-nm lasers used to form a closed-cycle for fluorescence, the 854-nm laser used to deshelve the ion from the $3D_{5/2}$ level for qubit state preparation. Details on these lasers can be found in section 3.6.

## 4.2 Spin

Discrete variable quantum computing consists of manipulating many two-level systems, which we refer to either as spins, or qubits. In this section the methods used to manipulate these spins are described, and their performances are benchmarked.

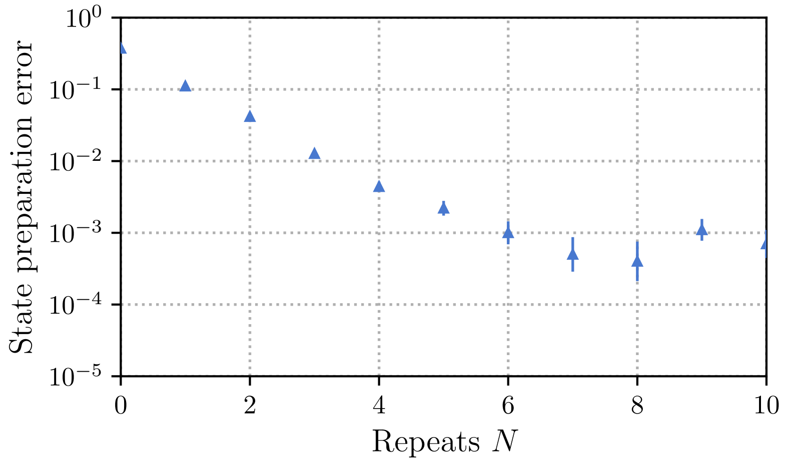



**Figure 4.3:** State preparation error as a function of the number of state preparation pulses, $N$. The error is defined as population not in the $|\downarrow\rangle$ state after the preparation sequence has been applied. The error is reduced to below $10^{-3}$ after $N = 7$ pulses, which is used in our experiments.

### 4.2.1 State Preparation

To utilise two levels of the ion as a qubit, the electronic state must first be prepared into the qubit manifold. Here, the ion is prepared into the $|\downarrow\rangle = |4S_{1/2},\ m_j = -1/2\rangle$ state. This is done via effective optical pumping by pulsing the quadrupole transition $|4S_{1/2},\ m_j = +1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -3/2\rangle$, followed by deshelving pulses using the 854-nm, 866-nm lasers on resonance. The 729-nm laser pulse duration and intensity is set to invert the populations, i.e. perform a $\pi$-pulse. Details for laser parameters are summarised in table 6.1. These three pulses are repeated $N$ times sequentially. Figure 4.3 shows the state preparation error with increasing $N$. The error is found by applying a $\pi$-pulse on the state preparation transition directly after the state preparation sequence and looking at the population in the $|\downarrow\rangle$ state. Any reduction from $|\downarrow\rangle = 1$ indicates residual population in the $|4S_{1/2},\ m_j = +1/2\rangle$ state, or the $D$ levels. After $N = 7$ pulses, which is used in our experiments, there is a state preparation error of $< 10^{-3}$. This is further verified by randomised benchmarking in section 4.2.4. The state preparation error is likely limited by accidental excitation of the $|4S_{1/2},\ m_j = -1/2\rangle$ state to the $D_{5/2}$ level. This may

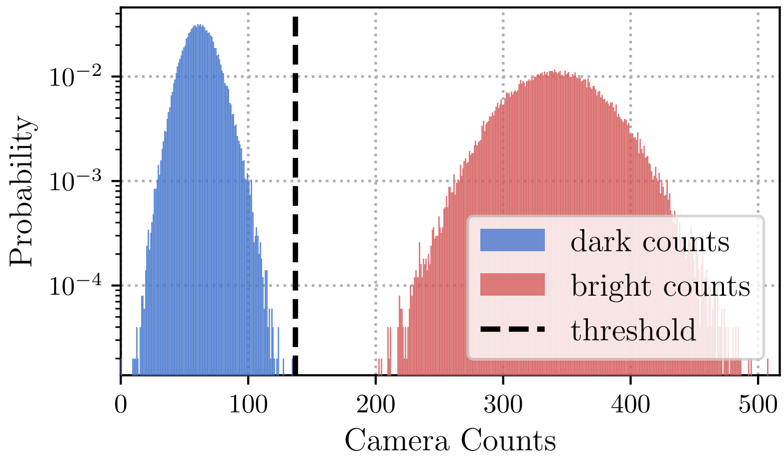



**Figure 4.4:** Readout histogram with 100,000 individual measurements for bright (red) and dark (blue) populations. The camera counts are presented as a histogram and fitted to Poissonian distributions to calibrate a threshold for discerning if the ion is bright or dark with the lowest probability for false measurement. Dark counts are mainly due to 397-nm light scattered from the nearest trap electrodes. With 100 µs readout duration, and laser parameters set to Doppler cooling parameters, a threshold count of 137 photons is found, giving an expected statistical error of $6 \times 10^{-9}$ given the fitted Poissonians.

be due to off-resonant excitation of another transition, or due to spectral purity of the 729-nm laser.

Another protocol for state preparation is optical pumping on the dipole transition (397-nm), which often is much faster. However, in this setup it is impractical as polarisation selectivity of the 397-nm transition is not possible due to the constraint of beam geometry from the in vacuum optics, see section 3.4, and frequency selectivity is not possible due to the low magnitude fixed B-field providing Zeeman shifts less than the natural linewidth of the 397-nm transition, see section 3.3.

### 4.2.2 Measurement

To measure the qubit state, the 397-nm and 866-nm lasers are applied, and the scattered 397-nm photons are imaged by the objective onto a camera (see section 3.5). 397-nm light will only be scattered if the qubit is in $|\downarrow\rangle$. Use of a high numerical aperture (NA = 0.6) objective greatly increases the collection efficiency. This efficiency allows the 397-nm beam to be red-detuned to Doppler cooling parameters

| Parameter | Value |
|---|---|
| 397-nm laser power | 3 µW, $1 \times I_{\mathrm{SAT}}$ |
| 397-nm laser detuning | -17.5 MHz |
| 866-nm laser power | 15 µW, $\sim 70 \times I_{\mathrm{SAT}}$ |
| 866-nm laser detuning | 0 MHz |
| Doppler cooling duration | 1000 µs |
| Readout duration | 100 µs |
| Camera threshold | 137 counts |

**Table 4.2:** Experimental parameters used for Doppler cooling and state-selective fluorescence of the ion.

(see section 4.3.1), while still maintaining low readout duration (100 µs) and low readout error. Cooling during readout increases the experiment's repetition rate by reducing the amount of dedicated cooling time required.

The parameters used for readout are summarised in table 4.2, and a typical histogram of readout counts for one ion can be seen in figure 4.4. The readout threshold is calibrated by taking bright counts with the 397-nm and 866-nm lasers on, and dark counts with the 397-nm laser on and the 866-nm laser off. This assumes dark counts are due to 397-nm scattering from the trap electrodes. Fitting Poissonian distributions to the bright and dark measurements, an optimal count threshold of 137 is found, shown by black dashed line. This threshold gives an expected statistical readout error of $\epsilon_{\downarrow} = \epsilon_{\uparrow} = 6 \times 10^{-9}$, where $\epsilon_{\downarrow,\uparrow}$ are the expected error associated with measuring bright or dark. This measurement ignores errors due to the finite lifetime of the metastable $3D_{5/2}$ level, which dominate $\epsilon_{\uparrow}$. The expected error due to the finite lifetime, $\tau$, is $\epsilon_{\mathrm{decay}} = 1 - e^{-t/\tau}$, where $t$ is the readout duration. For $^{40}Ca^{+}$, $\tau \sim 1.1$ s, $\epsilon_{\uparrow} \approx \epsilon_{\mathrm{decay}} = 10^{-4}$. Assuming the metastable lifetime is limited only by spontaneous decay overlooks leakage from the 854-nm deshelving beam, which could significantly shorten $\tau$ [**sotirova2024high-fidelity**].

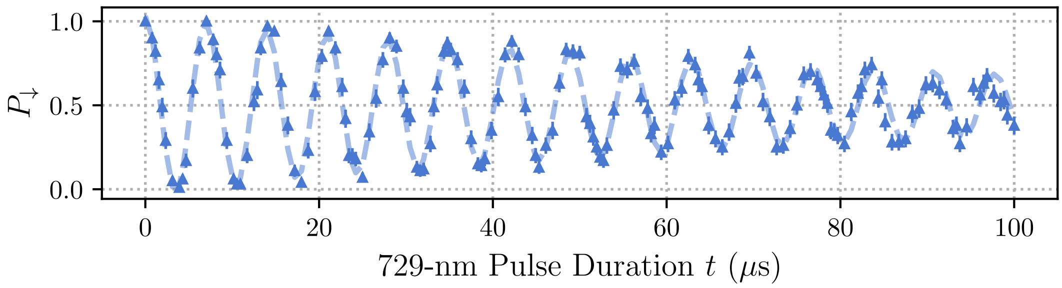



**Figure 4.5:** Long duration, $t$, Rabi flop on the qubit transition to extract Rabi frequency. Estimated probability of being in $|\downarrow\rangle$, $P_\downarrow$, at each time step using 200 individual experimental shots. Fitted dotted line using decaying oscillation model. Extracted Rabi frequency $\Omega/2\pi = 0.2864(2)$ MHz, and decay rate $\lambda = 0.0107(7)$ $\mu s^{-1}$.

### 4.2.3 Calibration Routines

#### Micromotion Compensation

#### Magnetic Field and Laser Offset

#### Beam Pointing Calibration

### 4.2.4 Single Qubit Rotations

A qubit is fully described by $|\psi\rangle = \alpha\,|\downarrow\rangle + \beta\,|\uparrow\rangle$, where $\alpha$ and $\beta$ are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$. The qubit state can be manipulated by applying single qubit rotations, $R_{x,y}(\theta)$, commonly referred to as gates. In the Bloch sphere, $x$- and $y$-rotations are performed by pertubations resonant with the qubit transition, here, using the 729-nm laser. The rotation angle, $\theta$, is given by $\theta = \Omega t/2$, where $\Omega$, is the Rabi frequency, and $t$ is the pulse duration. The rotation axis, is determined by the relative phase of the light field and the qubit. We define the initial relative phase to correspond to a $y$-rotation.

#### Gate Time Calibration

The pulse duration, for a $R(\pi/2)$ ($\pi/2$-pulse), is calibrated by Rabi oscillations. Figure 4.5 shows the population in $|\downarrow\rangle$, $P_\downarrow$, calculated from 200 experimental shots. To find the Rabi frequency, $\Omega$, the data is fitted to a decaying oscillation model,

$$P_\downarrow = \frac{1 + e^{-\lambda t}\cos(\Omega t/2)}{2}, \tag{4.1}$$

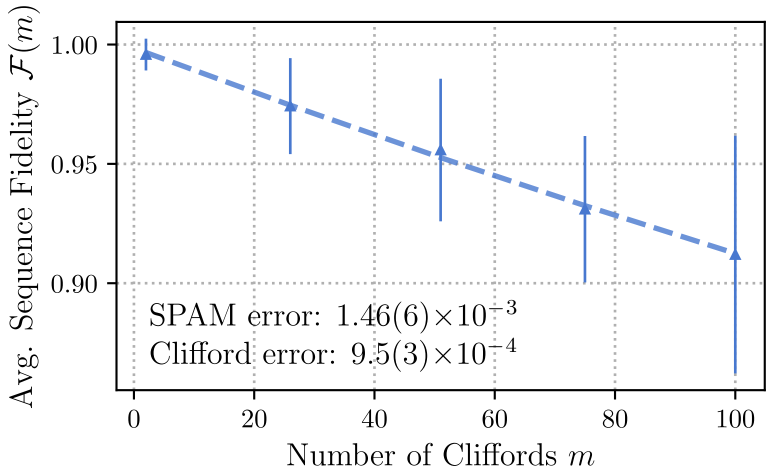



**Figure 4.6:** Estimating single qubit Clifford fidelities, $\mathcal{F}$, and errors due to state preparation and measurement, using randomised benchmarking. The data points are the average survival population after applying a sequence of $m$ random Clifford gates, and a final inverting Clifford gate. The dashed line is the fit to the decay model given by equation 4.2. The error bars are given by the standard deviation of the survival populations.

where $\lambda$ is the decay rate due to decoherence effects [**wineland1998experimental**]. The Rabi frequency is found to be $\Omega/2\pi = 0.2864(2)$ MHz, and the decay rate $\lambda = 0.0107(7)\ 1/\mu$s. From this $R(\pi/2) = 1.7$ µs can be estimated. These measurements were taken with 1 mW of 729-nm power, with beam radius ∼15µm at the ion.

### Randomised Benchmarking

To evaluate the quality of both our state preparation and single qubit rotations, randomised benchmarking (RB) [**knill2008randomized**] is employed. RB consists of applying random combinations of gates from the single-qubit Clifford group to estimate an average error per gate. The single-qubit Clifford group is the set of unitaries which map the Pauli matrices to one another through conjugation. The RB protocol used here is described in [**hughes2021benchmarking**]. The sequence of operations is as follows:

1. Prepare the qubit in $|{\downarrow}\rangle$ via optical pumping, section 4.2.1.

2. Apply a random sequence of $m$ single qubit Clifford gates, where $m$ is the length of the sequence.

3. Apply a final "inverting" Clifford gate, which is chosen such that the full sequence performs the identity operation.

4. Measure the population in $|{\downarrow}\rangle$, $P_{\downarrow}$, via state-selective fluorescence, section 4.2.2.

5. Repeat steps 1-4 multiple times to find the average survival population, $\langle P_{\downarrow}\rangle$.

6. Repeat steps 1-5 for many different random sequences with a range of sequence lengths, $m$.

7. Fit the decay of the average survival population, $\mathcal{F}(m) = \langle P_{\downarrow}(m)\rangle$, to a decay model to find the average error per Clifford gate.

The decay model used to find the fidelity versus number of Clifford gates is given by[**hughes2021benchmarking**],

$$\mathcal{F}(m) = \frac{1}{2}\left(1 + (1 - 2\epsilon_{\mathrm{SPAM}})(1 - 2\epsilon_c)^m\right), \tag{4.2}$$

where $\mathcal{F}(m)$ is the fidelity of the sequence of length $m$, $\epsilon_{\mathrm{SPAM}}$ is the state preparation and measurement error, and $\epsilon_c$ is the average error per Clifford gate. The Clifford gates are decomposed into sequences of $\pi/2$- and $\pi$-pulses about either the $x$- or $y$-axes. Up to $m = 100$ Clifford gates are probed, and the decay of the fidelity, $\mathcal{F}(m)$, is fit using the above model. The error per Clifford is found to be $\epsilon_c = 9.5(3) \times 10^{-4}$, while the SPAM error is $\epsilon_{\mathrm{SPAM}} = 1.46(6) \times 10^{-3}$. The decay plot for this RB sequence can be seen in figure 4.6. The independently characterised state preparation (section 4.2.1) and measurement (section 4.2.2) errors of $\sim 10^{-3}$, and $\sim 10^{-4}$ respectively, are consistent with the measured $\epsilon_{\mathrm{SPAM}}$.

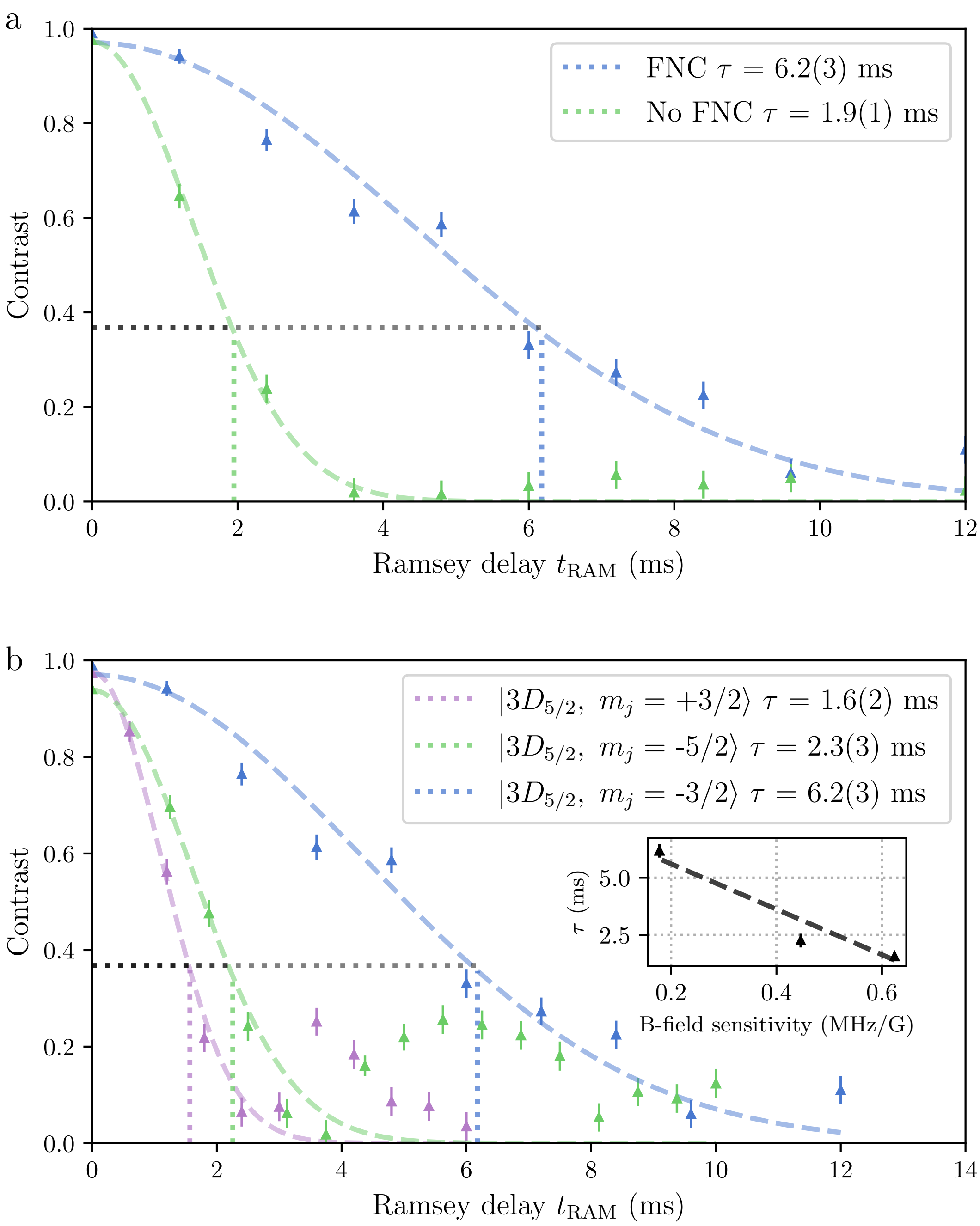



**Figure 4.7:** Coherence time measurements. **a)** Coherence time of the $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -3/2\rangle$ transition with and without fibre noise cancellation (FNC). **b)** Coherence time of three quadrupole transitions with varying magnetic field sensitivities. The inset plot shows a linear fit between the coherence time and the magnetic field sensitivity of the transition.

### 4.2.5 Robust Phase Estimation

### 4.2.6 sec:Single Addressing char

### 4.2.7 Spin Coherence Times

**Add section of spin-lock scans of laser.** Decoherence, as can be seen in figure 4.5, can be caused by laser frequency and power fluctuations, magnetic field noise, and spontaneous emission from the $3D_{5/2}$ level. The coherence time of the qubit is a useful diagnostic for finding unwanted noise sources, and for benchmarking the performance of the qubit as a store of quantum information.

To measure the coherence times, we perform Ramsey sequences with varying delay durations. The Ramsey sequence is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 4.2.1.
2. Apply a $R_y(\pi/2)$ pulse to prepare the qubit in $|+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle)$ superposition state.
3. Wait for a time $t_{\mathrm{RAM}}$.
4. Apply a second $R_\phi(\pi/2)$-pulse, with a variable axis of rotation, $\phi$.
5. Measure the population in $|\downarrow\rangle$, $P_\downarrow$, via state-selective fluorescence, section 4.2.2.
6. Repeat steps 1-5 for different values of $\phi$ and fit the resulting fringe to find the contrast.
7. Repeat steps 1-6 for different values of $t_{\mathrm{RAM}}$ to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

With this tool, the efficacy of the fibre noise cancellation (FNC, see section **??**) can be measured. Figure 4.7 *a* shows the coherence time of the $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -3/2\rangle$ transition with and without FNC. The coherence time is increased from 1.9(1) ms to 6.2(3) ms with FNC, showing that without FNC, the coherence times are limited by laser phase noise.

With FNC enabled, to see if the coherence time is now limited by magnetic field noise, transitions with varying magnetic field sensitivities are compared. Figure 4.7 *b* shows the coherence time of three quadrupole transitions with varying $D_{5/2}$, $m_j$ numbers: $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = \{-5/2, -3/2, +3/2\rangle$. These transitions have magnetic field sensitivities $-0.446$, $-0.178$, $+0.624$ $\mathrm{MHz\,G^{-1}}$ respectively. The contrast decays for all three qubits are shown in the figure. The inset plot shows a well correlated linear fit between the coherence time and the magnetic field sensitivity, suggesting that the coherence time is limited by magnetic field noise. For both $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -5/2\rangle$ and $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = +3/2\rangle$, revivals in contrast can be seen which suggest that the dominant noise source is oscillatory and coherent over the Ramsey scan duration.

These measurements were taken without the MuMetal enclosure (section 3.3) fully assembled around the vacuum system, which we expect will greatly reduce external magnetic field noise, and therefore increase the coherence times.

## 4.3 Motion

Motion of ions trapped in the same potential can be described by a set of normal modes [**james1998quantum**]. These modes are labelled with respect to the ion chain geometry: axial modes oscillate parallel with the chain, and radial modes oscillate perpendicular to the chain. Due to the geometry of the 729-nm beam access with respect to the ion chain (see section 3.4), only the radial modes can be addressed in our system. For the *NPL* trap (see section 3.2), the radial modes frequencies are non-degenerate. In our system, for a single ion, the radial mode frequencies are $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and $\omega_{\text{lower}}/2\pi = 2.5$ MHz, and the axial mode frequency is $\omega_{\text{axial}}/2\pi = 1.6$ MHz. All work here is using the upper radial mode.

The motional mode, $i$, leads to sidebands in the 729-nm spectrum at $\omega_0 \pm \omega_i$ [**ozeri2011tutorial**], where $\omega_0$, the carrier, is the frequency of the 729-nm laser when on resonance with the qubit transition. The $+\omega_i$ sideband is referred to as the blue sideband (BSB), and the $-\omega_i$ is the red sideband (RSB). The RSB, BSB, and RF fields near to resonance with the motional modes, are the tools used in this section to manipulate the motion of the ion.

The motional state preparation, heating rate, frequency stabilities, and coherence times are characterised in this section.

### 4.3.1 Cooling

For any interaction that involves the motion of the ion, it is essential to both prepare and measure the motional state with high fidelity. For all motional interactions considered in this thesis, the ion is initially prepared as close as possible to the motional ground state, with a mean phonon number $\bar{n} \approx 0$. This is done by an initial stage of Doppler cooling, followed by pulsed sideband cooling on the 729-nm transition. The final motional state is characterised by measuring the population in Fock states using RSB and effective BSB pulses.

#### Doppler Cooling

Doppler cooling relies on the fact that light incident on a moving ion appears frequency-shifted in the ion's rest frame. For Doppler cooling of $^{40}Ca^{+}$, both the 397-nm and 866-nm lasers are applied. The 397-nm laser is red-detuned from the dipole $|4S_{1/2}\rangle \leftrightarrow |4P_{1/2}\rangle$ transition, leading to preferential scattering of 397-nm photons that have a net cooling effect [**wineland1979laser**]. The excited $4P_{1/2}$ level decays via two channels: emission of a 397-nm photon back to $|4S_{1/2}\rangle$, or an 866-nm photon to $|3D_{3/2}\rangle$, with a branching ratio of 14.5 [**ramm2013precision**]. Since many photon scattering events are required, the cooling cycle must be closed. To prevent population trapping in the metastable $|3D_{3/2}\rangle$ level, the 866-nm laser is applied on resonance during cooling.

Equilibrium is reached when the cooling rate balances photon recoil heating. Assuming the 397-nm transition is broadened only by its natural linewidth $\gamma$, the minimum Doppler temperature is,

$$T_{\text{Doppler}} \approx \frac{\hbar\gamma}{2k_B}, \tag{4.3}$$

| Parameter | Value |
|---|---|
| 729-nm laser power | 3.0 mW |
| 729-nm laser detuning | -4.0 MHz* |
| 729-nm pulse duration $N = 1$ | 2.5 µs |
| 729-nm pulse duration $N = N_{\text{max}}$ | 33 µs |
| Repumping 866-nm duration | 14 µs |
| Deshelving 854-nm duration | 5 µs |
| Number of cooling cycles $N_{\text{max}}$ | 50 |

**Table 4.3:** *729-nm detuning is relative to the $|4S_{1/2},\ m_j = -1/2\rangle \leftrightarrow |3D_{5/2},\ m_j = -5/2\rangle$ transition frequency.
Experimental parameters used for sideband cooling. Other parameters for the repump and deshelve pulses are the same as for state preparation, see table 6.1. The 729-nm laser is detuned to the RSB resonance for the upper radial mode, $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and the pulse duration for $N = N_{\text{max}}$ is chosen to be the expected RSB $R(\pi)$-pulse duration for $n = 1$.

where $\hbar$ is the reduced Planck constant, and $k_B$ is Boltzmann's constant [**wineland1979laser**]. For $^{40}$Ca$^+$, the natural linewidth of the 397-nm transition is $\frac{\gamma}{2\pi} = 21$ MHz[**morton2003atomic**], giving $T_{\text{Doppler}} = 0.5$ mK. For a radial mode frequency of $\frac{\omega}{2\pi} = 4$ MHz, the expected mean phonon number is,

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_BT} - 1}, \tag{4.4}$$

yielding $\bar{n} = 2.3$ at the Doppler limit. The parameters used here for Doppler cooling are summarised in table 4.2.

### Sideband Cooling

To further cool the ions toward their motional ground state, resolved sideband cooling is used. As mentioned above, the motion of the ion modulates the absorption spectrum of the ion, leading to red and blue sidebands (RSB and BSB). For the $|4S_{1/2}\rangle \leftrightarrow |3D_{5/2}\rangle$ transition, at appropriate motional mode frequencies, interaction strengths, and using a narrow linewidth laser (section **??**), these sidebands can be resolved spectroscopically. The pulsed sideband technique employed consists of RSB pulses on the 729-nm transition, followed by deshelving, and repumping pulses on the 854-nm and 866-nm transitions respectively. For efficient cooling, initial short

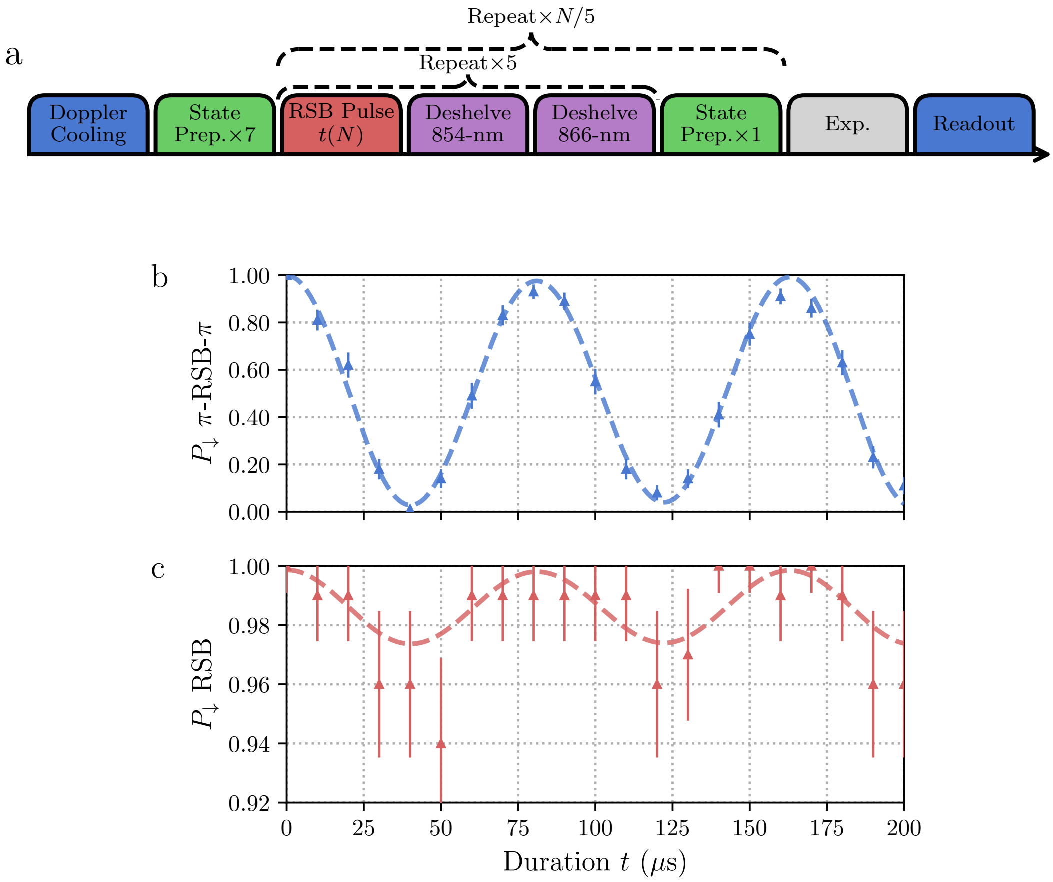



**Figure 4.8:** **a)** Example pulse sequence for one experimental shot including pulsed sideband cooling. The *Exp.* block represents an arbitrary sequence of operations, for the experiment under study, such as a thermometry sequence. The blue blocks represent Doppler cooling and readout (section 4.2.2), the green blocks represent qubit state preparation (section 4.2.1), the red blocks are RSB pulses with varying durations $t(N)$, and the purple blocks are deshelving pulses. After every 5 sideband cooling pulses, there is an interleaved state preparation sequence to ensure there is no population trapped outside of the closed cooling cycle. The RSB length is linearly increased with pulse number $N$, upto the expected RSB $R(\pi)$-pulse duration for $n = 1$. **b, c)** Thermometry scans after Doppler and sideband cooling. The red points are the populations measured via on-resonance RSB pulses, and the blue points are the populations after a sequence of $R(\pi)$-RSB-$R(\pi)$ with varying RSB probe durations. The dashed lines are fits to a thermal Fock state distribution, with truncation at $n = 100$. The extracted mean occupation number is $\bar{n} = 0.03(1)$, and $\eta\Omega/(2\pi) = 11.6(1)$ kHz.

729-nm pulses are applied to preferentially excite the thermal population with high phonon number, as the pulse duration for RSB $R(\pi)$ is proportional to $\sqrt{n}$, where $n$ is the phonon number. The RSB pulse length is linearly increased to the expected RSB $R(\pi)$-pulse duration for $n = 1$.

An example pulse sequence for a single experimental shot using sideband cooling, can be seen in figure 4.8 *a*. Experimental parameters used are summarised in table 4.3.

To verify the efficacy of our sideband cooling, thermometry experiments are performed by probing Fock state populations. This is done using on resonance RSB pulses and effective BSB pulses composed of a $\pi$-RSB-$\pi$ pulse sequences. The time dynamics of population flopping is measured as RSB pulse length, $t$, is varied. In the case of Fock state $|0\rangle$, there should be full contrast “BSB” oscillations, and no visible oscillations when probed with RSB pulses. A thermal Fock state distribution [**meekhof1996generation**] (with truncation at Fock state = 100) is fitted to these signals to extract the mean occupation number, and $\eta\Omega$, the carrier Rabi frequency multiplied by the Lamb-Dicke parameter. A typical thermometry scan after Doppler and sideband cooling can be seen in figure 4.8 $b, c$, where $b$ shows the population after effective “BSB” pulses, and $c$ shows the population after RSB$(t)$ pulses with duration $t$. These oscillations are fit to thermal Fock state distributions, shown by the dashed lines. The mean occupation number after sideband cooling is found to be $\bar{n} = 0.03(1)$, and $\eta\Omega = 11.6(1)$ kHz.

### 4.3.2 Heating Rates

After cooling the motion, the ion will return to a temperature in equilibrium with the environment. However, this process can be slowed by adequate isolation of the ion. In UHV ion trap systems (see section 3.4), motional heating is predominantly caused by electric field noise from the ion trap [**wineland1998experimental**]. Noise due to thermal charge fluctuations at the trap surface can be mitigated by increasing ion-electrode distances. The *NPL* trap (section 3.2), used here, has an

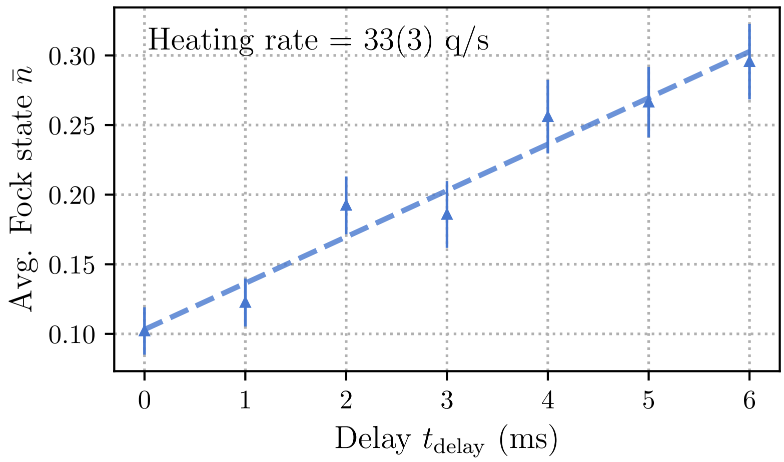



**Figure 4.9:** Heating rates of upper radial mode, measured by varying the delay between cooling and thermometry pulses. The dashed line is a linear fit to the data, with slope 33(3) quanta/s. The error bars are given by the standard deviation of the fitted $\bar{n}$.

ion-electrode distance of 250 µm, larger than most surface traps, but less than that of a macroscopic blade or rod style trap [**milne2021construction**]. Heating rates were probed by a series of thermometry experiments, explained above, with varying delay times, $t_{\mathrm{delay}}$, between cooling and thermometry pulses. A typical plot can be seen in figure 4.9. The heating rate of the system was found by a linear fit, shown by the dashed line, to be 33(3) quanta per second on the upper radial 4 MHz mode on one ion. As will be discussed in section 4.3.4, the heating rate sets a limit for the motional coherence time.

### 4.3.3 Motional Mode Stability

Due to thermal drifts and microphonics introducing amplitude and frequency noise on the RF chain, drifts of the radial motional mode frequencies are seen over time. As will be discussed in the following section 4.3.4, this can lead to dephasing of the motional state. Here, the current motional mode stability is characterised to estimate its contribution to dephasing, and to set a benchmark for future stability improvements.

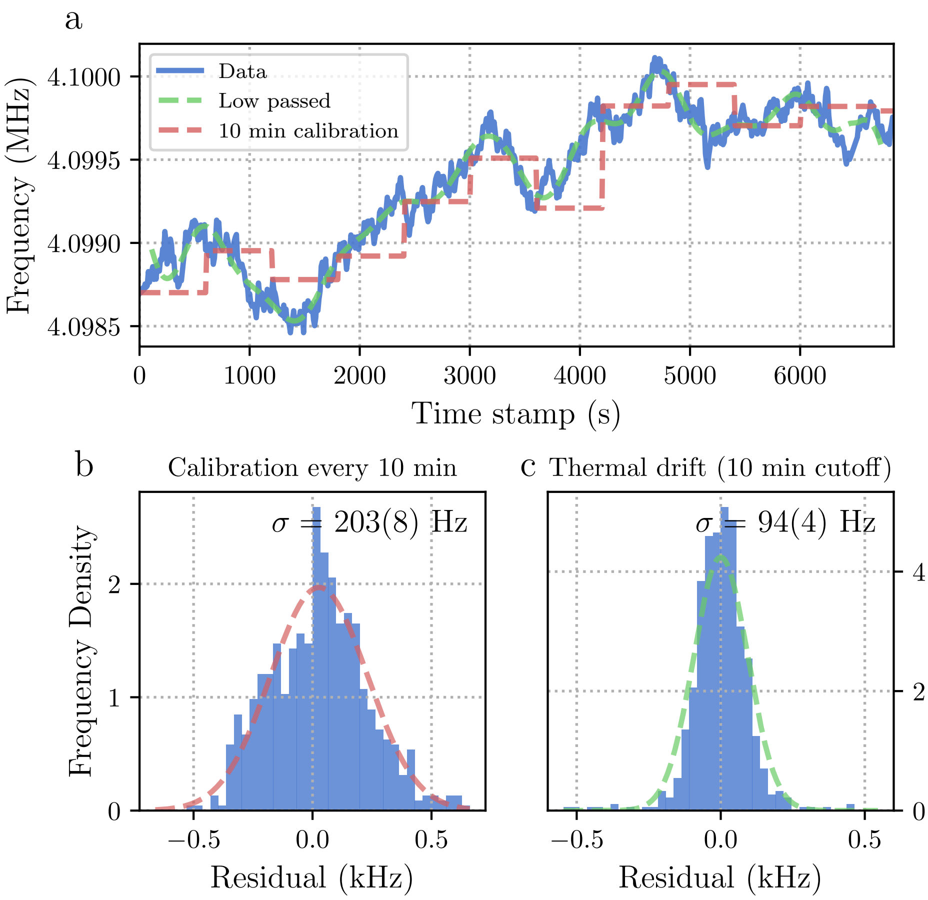



**Figure 4.10: a)** Motional mode frequency drift over time. The radial motional mode frequency is measured by exciting motion using an RF field in resonance with the motional frequency, and looking at increased fluorescence. The resulting frequency trace is shown in blue. The green line shows a low pass filter of the data, with cut-off frequency of 1.67 mHz, corresponding to an assumed 10 minutes slow thermal drift. The motional mode frequency is typically calibrated every 10 minutes, which is indicated by the red line. **b)** Residuals from the measured data and expected calibrated mode frequencies. The standard deviation of these residuals suggests the typical mis-set detuning of the mode frequency in current experiments with automated mode frequency calibrations every 10 minutes. **c)** Residuals from the measured data and the low passed trace. The standard deviation of these residuals are due to noise with period less than 10 minutes.

The motional mode frequency is measured via an RF "tickle" experiment. This consists of short ($\sim 5$ µs) RF field pulses applied to one of the trap DC electrodes which has a non-zero electric field projection to the mode we wish to probe. The frequency of the RF field is scanned around the expected motional mode frequency, and when resonant, drives the motion of the ion. This resonance is measured by observing an increase in fluorescence counts using a 397-nm laser red-detuned by 100 MHz. To measure mode frequency drift, this sequence is repeated every 10 seconds over multiple hours. A typical plot can be seen in figure 4.10 *a*. Over this two hour period, the frequency had an absolute drift of 1.5 kHz. Typically, we calibrate the mode frequency every 10 minutes, which is indicated with the red dashed line. Residuals between this calibrated line and the data, shown in *b*, have a standard deviation of $\sigma/(2\pi) = 203(8)$ Hz, giving the expected uncertainty of the mode frequency detuning given our current periodic calibration routine. A slow drift with frequency cut-off of 1.67 mHz is also fitted to the data, shown in green. The residuals from this fit, shown in *c*, have a standard deviation of $\sigma/(2\pi) = 94(4)$ Hz. This suggests that the motional mode frequency is relatively stable over time scales shorter than 10 minutes, but has a slow drift on longer time scales, which is likely due to thermal fluctuations.

### 4.3.4 Motional Coherence Times

We will utilise the motional modes of the ion to store quantum information. As with the section on spin coherence times 4.2.7, the fidelity of operations, and success probability of algorithms will be limited by the coherence time of the motional state. To measure motional coherence, a Ramsey sequence is performed between states $|\downarrow, 0\rangle$ and $|\downarrow, 1\rangle$, where the first element is the qubit state, and the second element is the motional Fock state. To prepare these states the following pulse sequence is applied:

1. State prepare system to $|\downarrow, 0\rangle$ via optical pumping and sideband cooling.

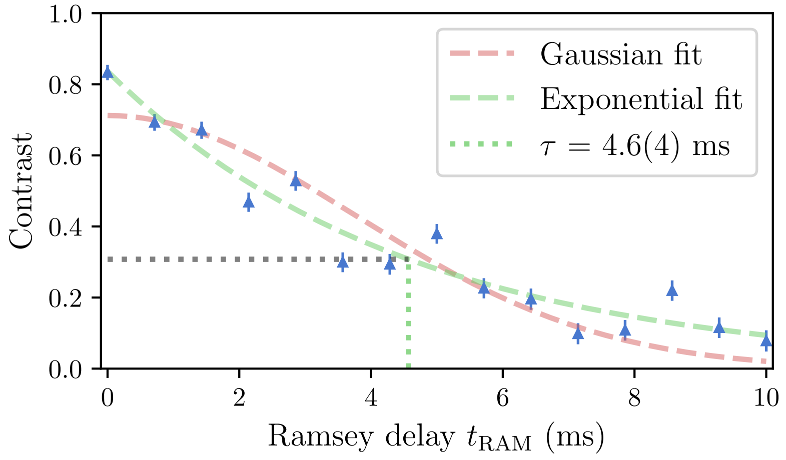



**Figure 4.11:** Motional coherence time measured by Ramsey sequence on the motional state. The blue points are the population in $|{\downarrow}\rangle$ after a Ramsey sequence with varying delay time, $t_{\mathrm{RAM}}$. The dashed line is an exponential decay fit, with a coherence time $\tau = 4.6(4)$ ms.

2. Apply $R_y(\pi/2)$-pulse on carrier transition to prepare the qubit in $(|{\downarrow}, 0\rangle + |{\uparrow}, 0\rangle)/\sqrt{2}$.

3. Apply $R(\pi)$-pulse on the RSB to prepare the motional state in $(|{\downarrow}, 0\rangle + |{\downarrow}, 1\rangle)/\sqrt{2}$.

4. Delay for some time $t_{\mathrm{RAM}}$.

5. Apply $R(\pi)$-pulse on the RSB to map accumulated phase onto the qubit state.

6. Apply $R_\phi(\pi/2)$-pulse on the carrier, with variable axis of rotation, $\phi$.

7. Measure the population in $|{\downarrow}\rangle$ via state-selective fluorescence, section 4.2.2.

8. Repeat steps 1-7 for different values of $\phi$ to find the contrast of the Ramsey fringe.

9. Repeat steps 1-8 for different values of $t_{\mathrm{RAM}}$ to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

There are two main mechanisms for motional decoherence[**turchette2000decoherence**]: motional heating, as characterised in section 4.3.2, and motional dephasing due to mode frequency instability, as discussed in section 4.3.3.
A heating rate dominated coherence time is given by, $\tau_{\text{HEAT}} = (\sqrt{e}-1)/\dot{\bar{n}}$ [**turchette2000decoherenc** where $\dot{\bar{n}}$ is the heating rate. A dephasing dominated coherence time, if noise is correlated on timescales long compared to the Ramsey delay [**omalley2015qubit**], is characterised by a Gaussian decay profile with a coherence time of $\tau_{DEPH} = \sqrt{2}/\sigma_\omega$. Evaluating both of these models with $\dot{\bar{n}} = 33(3)$ q/s, and $\sigma_\omega/(2\pi) = 94(4)$ Hz, the expected motional coherence times of $\tau_{\text{HEAT}} \approx 20$ ms and $\tau_{\text{DEPH}} \approx 2.3$ ms respectively, are found. From these auxiliary characterisations of the motional modes, it is expected that the coherence time will be limited by motional dephasing. Figure 4.11 shows the decaying coherence when probed by the above sequence. It can be seen that the contrast has a good fit assuming an exponential decay (green dashed line), defined by $C(t_{\text{RAM}}) = C(0)e^{-t_{\text{RAM}}/\tau}$, where $C(t_{\text{RAM}})$ is the contrast at time $t_{\text{RAM}}$, and $\tau$ is the coherence time. The fit yields a motional coherence time of $\tau = 4.6(4)$ ms, defined as the time at which the contrast decays to $1/e$ of its initial value. An exponential decay suggests that the noise source causing the dephasing is correlated only for durations much shorter than the Ramsey delay times [**omalley2015qubit**]. The Gaussian decay, (red dashed line), which was appropriate for the previous spin coherence time measurements in section 4.2.7, is a poor fit here.

## 4.4 Spin-Motion

For full control of the spin-motion hybrid system, we require interactions that couple the two. Perhaps the simplest of this class of interactions are the red- and blue-sidebands (RSB and BSB respectively). The RSB interaction was used previously in the thermometry and sideband cooling sections 4.3.1. The RSB (BSB) consists of a single frequency laser tuned at the carrier frequency minus (plus) the motional mode frequency, $\omega_m$. The interaction is well described by the (Anti-)

Jaynes-Cummings Hamiltonian, and effectively couples spin flips with the addition or subtraction of a motional quanta depending on the initial spin state. Here, another such interaction coupling spin and motion, known as the spin-dependent force, SDF, is introduced. While the RSB and BSB interactions couple the spin and motion in the motional Fock basis, the SDF couples them through spin-dependent displacements, thus in a coherent state basis. The direction of the displacement is determined by the spin state and an effective detuning parameter.

### 4.4.1 Spin-Dependent Forces

The optical Mølmer-Sørensen (MS) scheme [**sorensen2000entanglement**] is used to generate the SDF via a bichromatic laser field. Bichromatic refers to the simultaneous application of two tones symmetrically detuned around the qubit carrier frequency, with absolute detuning approximately equal to the motional mode frequency, $\delta \approx \omega_m$. The resulting interaction, when ignoring off-resonant and higher order couplings, is given by,

$$\hat{H}_{MS} = i\hbar\eta\Omega\hat{\sigma}_\phi \cos(\delta t)\left(\hat{a}e^{-i\omega_m t} + \hat{a}^\dagger e^{i\omega_m t}\right), \tag{4.5}$$

where $\eta$ is the Lamb-Dicke parameter, $\Omega$ is the carrier Rabi frequency, $\hat{a}(\hat{a}^\dagger)$ is the lowering (raising) operator, and $\hat{\sigma}_\phi$ is the Pauli operator with $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$. Applying the rotating wave approximation, and defining $\delta_g = \delta - \omega_m$, the interaction Hamiltonian can be approximated to,

$$\hat{H}_{MS} \approx \frac{i\hbar\eta\Omega}{2}\,\hat{\sigma}_\phi\left(\hat{a}e^{-i\delta_g t} + \hat{a}^\dagger e^{i\delta_g t}\right). \tag{4.6}$$

The trajectory of this displacement can be controlled by varying $\delta_g$: on resonance, $\delta_g = 0$, corresponds to linear trajectories, whilst off resonance, $\delta_g \neq 0$, corresponds to cyclic trajectories where after some duration $t = 2\pi/\delta_g$, the motion returns to the initial state (with perhaps some phase shift [**haljan2005spin-dependent**]). The dynamics of the SDF may be measured by projective spin measurements, the population in $|\downarrow\rangle$, $P_\downarrow$, after applying the SDF for duration $t$ is described by [**burd2020squeezing**],

$$P_{\downarrow,\mathrm{th}} = \frac{1}{2}\left[1 + e^{-4\left(\bar{n}+\frac{1}{2}\right)|\alpha(t,\delta_g)|^2}\right], \tag{4.7}$$

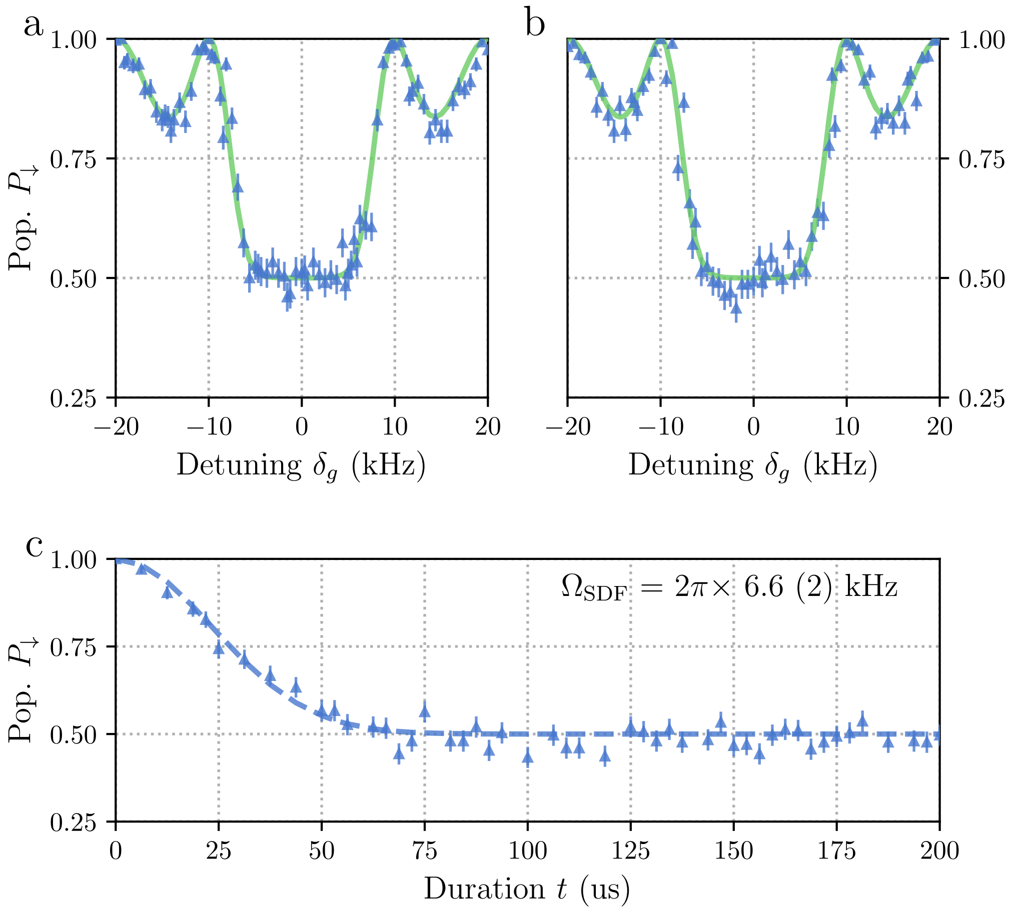



**Figure 4.12:** Detuning and duration scans of the spin-dependent force applied to an initial thermal state with $\bar{n} = 0.03(1)$. **a)** Detuning scan, where the detuning, $\delta_g$, is varied, and the duration is fixed at $t = 100$ µs. The green line is a fit to the expected behaviour given by equation 4.7. **b)** Detuning scan with a mis-set qubit frequency of 0.5 kHz. The green line is the same fit as in $a$ to guide the eye. **c)** Duration scan, where the duration, $t$, is varied, and the detuning is fixed at $\delta_g = 0$. The dashed line is a fit to the expected behaviour given by equation 4.7 with only $\Omega_{\mathrm{SDF}}$ as a free parameter.

where $\bar{n}$ is the initial thermal phonon occupation, and $|\alpha(t, \delta_g)| = \Omega_{\mathrm{SDF}} \sin(\delta_g t/2)/\delta_g$, is the absolute displacement of the initial state. This control is utilised in both two-qubit entangling gate experiments, as well as in the creation of squeezed states.

### Calibrating the SDF

Geometric phase gates, such as the MS interaction, are widely employed in ion trap experiments due to their robustness against variations in initial motional state. However, the SDF is sensitive to various frequency and power miscalibrations and drifts of these between calibrations. Here, we describe the workflow for calibrating and optimising the SDF behaviour.

In practice, the SDF interaction involves relatively few degrees of freedom: the central qubit frequency, the motional mode frequency, the power in each tone, and the durations of the pulses must all be calibrated. The complexity comes from the fact that the ion is a multi-level system, possesses multiple motional modes, and that each experimental parameter can only be controlled with finite precision.

The effect of nearby motional modes is negated by either selecting the interaction mode to be well separated from the others either in frequency, by beam geometry, by operating the SDF near to resonance of the desired mode, or by using amplitude shaping of the pulse. Smoothing a square pulse through amplitude shaping narrows the pulse frequency bandwidth, and thus reduces unwanted off-resonant excitation of other modes.

The power balance of the SDF tones is complicated due to our use of AOMs. AOMs generate frequency shifts in the laser beam at the expense of small frequency-dependent angular shifts. After the AOM, the beam is coupled into a single mode fibre (see figure 3.10), and so the AOMs effectively introduce a frequency-dependent loss. To calibrate the power balance, a pick off of the bichromatic beam after the ion is monitored on a high bandwidth photodiode. The beatnote contrast is measured for the two tones and the power balance is optimised by maximising this contrast.

The far off-resonant levels of our ion lead to light-shifts of our qubit frequency. To account for the light-shift the central qubit frequency must be calibrated at the optical power used for the interaction. This is done with the SDF interaction itself. When the qubit frequency is set incorrectly, the two tones will no longer have the same absolute detuning from their respective RSB or BSB. This manifests as a "skewness" of the SDF detuning trace. By varying the qubit frequency and inspecting these detuning scans, the desired SDF behaviour can be found.

To verify the behaviour of the calibrated SDF, the experimental data is compared with theory. Both "detuning" and "duration" scans are used. The "detuning" scan is performed by varying the detuning, $\delta_g$, of the interaction, whilst keeping the SDF duration, $t$, constant, and vice-versa for the "duration" scan.

Figure 4.12 *c* shows the measured duration scan described by equation 4.7, with $\delta_g = 0$ and the the motional state assumed to be thermal with average Fock state $\bar{n} = 0.03(1)$ which was found previously from thermometry measurements. Here, the displacement, $|\alpha(t)| = \Omega_{\text{SDF}} t/2$, where $\Omega_{\text{SDF}}$ is the SDF amplitude. The dashed line fit is given by floating $\Omega_{\text{SDF}}$ only, and finds $\Omega_{\text{SDF}}/(2\pi) = 6.6(2)$ kHz. These measurements were performed with total 729-nm power of 2 mW and $\delta_{\text{LS}}/(2\pi) =$ 3.5 kHz, the light shift due to the 729-nm laser relative to the unperturbed qubit frequency.

The detuning scan, shown in figure 4.12 *a*, has the same experimental parameters, but with fixed duration $t = 100$ us. The fit in green is taking the displacement to be $|\alpha(t)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, given the value for $\Omega_{\text{SDF}}$ measured from the duration scan. Here, a qualitatively good agreement between the measured data and the expected behaviour is seen, with the main features being the "closure" at $\delta_g = 2\pi/t$ where the motional state returns to the initial state, and the central flat region around $P_{\downarrow,\text{th}} = 0.5$ where the two motional wave packets are non-overlapping.

Figure *b* shows an example detuning scan when the qubit frequency is mis-set by 0.5 kHz. The central region now displays a clear asymmetry around $\delta_g = 0$. The green line here is the fit from *a* to guide the eye.

### 4.4.2 Two-Qubit Entangling Gates

We perform two-qubit entangling gates using the Mølmer-Sørensen (MS) interaction [**sorensen2000entanglement**]. This interaction is the same described SDF from the previous section, but applied globally to two ions. The MS interaction relies on the spin dependent geometric phase accumulated during the motional displacement [**ozeri2011tutorial**]. To create a two-qubit entangled state, a

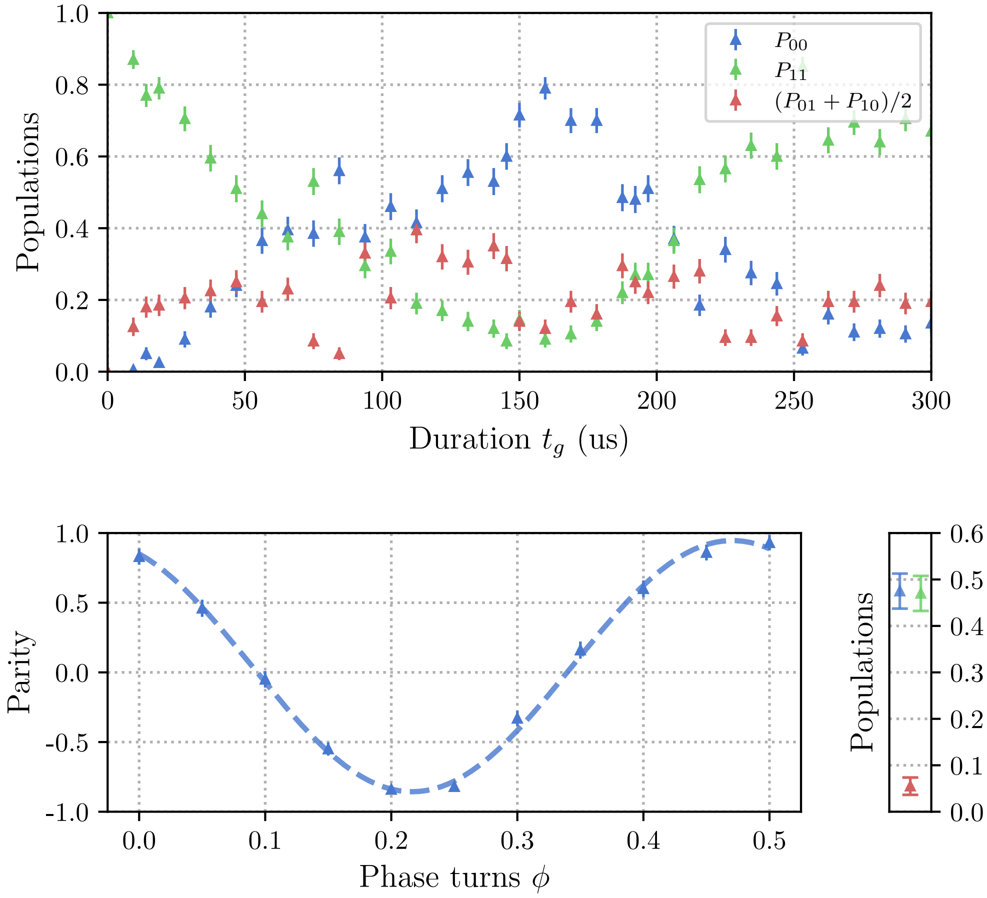



**Figure 4.13:** Experimental data for $\mathcal{F} = 92(2)\%$ Mølmer-Sørensen (MS) two-qubit entangling gate. **a)** Populations versus duration scan of the MS interaction. Each point corresponds to the average of 200 shots, and the error bars are given by the standard deviation of these averages. The blue points are the populations of both ions being in the $|0\rangle$ state, $P_{00}$, the green points are the populations of both ions being in the $|1\rangle$ state, $P_{11}$, and the red points are the ions being the the opposite states, $(P_{01} + P_{10})/2$. The desired entangling gate corresponds here to $t_g = 75\mu s$. **b)** Parity oscillations after the MS interaction. The parity is defined as $P_{00} + P_{11} - P_{01} - P_{10}$. The fitted contrast here is 0.90(2). **c)** Populations at the desired entangling gate duration, $t_g = 75$ us.

differential geometric phase of $\pi/2$ must be accumulated between the two-qubit basis states [**ozeri2011tutorial**]. To ensure there is no residual motional entanglement, the final motional state must return to the initial state. In practise, using an SDF detuned by $\delta_g$, this is achieved by applying the MS interaction for a duration $t_g = 2n\pi/\delta_g$, where $n$ is the number of "loops". The MS gate is a universal two-qubit gate, and along with only single qubit gates, constitutes a universal gate set for discrete quantum algorithms.

Here we quote the current fidelity of experimentally demonstrated two-qubit gates on our system. The fidelity serves to quantify the similarity of two density matrices. Here for a two-qubit entangling gate, we target the creation of the Bell state $|\Phi^+_{\phi_0}\rangle = 1/\sqrt{2}\left(|00\rangle + e^{i\phi_0}|11\rangle\right)$, from an initial state of $|00\rangle$, where $phi_0$, is some fixed phase. The fidelity between our mixed state $\rho$, and the pure Bell state may be given by,

$$\mathcal{F} = \langle\Phi^+_{\phi_0}|\,\rho\,|\Phi^+_{\phi_0}\rangle = \frac{1}{2}\left(\rho_{00,00} + \rho_{11,11}\right) + \frac{1}{2}\left(e^{i\phi_0}\rho_{11,00} + e^{-i\phi_0}\rho_{00,11}\right), \tag{4.8}$$

where $\rho_{ij,kl} = \langle ij|\rho|kl\rangle$. To extract the fidelity experimentally, a partial state tomography protocol is used [**sackett2000experimental**], where the first bracketed term of equation 4.8 is measured by performing projective measurements of the two ions after the gate sequence to extract populations, and the second bracketed term, known as the coherence terms, are measured by applying a global analysis $R_\phi(\pi/2)$ pulse to the two ions and applying parity measurements. Parity measurements are defined as $P = P_{00} + P_{11} - P_{01} - P_{10}$, where $P_{ij}$ is the population of the two ions in state $|i\rangle\,|j\rangle$ after the gate sequence and variable $\pi/2$-pulse. Creating a Bell state is essential for two-qubit entanglement, but its basis can vary as long as it is consistent across experiments. This allows the Bell state phase, $\phi_0$, to float, which in practice means using only the amplitude of the fitted parity oscillations to compute fidelity, ignoring any phase offset.

The best two-qubit entangling gate fidelity currently achieved on our system is $\mathcal{F} = 92(2)\%$. As shown in figure 4.13, the magnitude of the parity scan was measured to be 0.90(2), while the populations $(\rho_{00,00} + \rho_{11,11}) = 0.95(2)$. Each population point in these figures are found by taking the average of 200 shots of the gate sequence.

These results serve as both a proof of principle for full spin control on our system, but also as a benchmark to which we can compare future system improvements. For future work, we will require the use of this entangling gate either as Bell state preparation for input to analogue simulation experiments [**bazavan2023synthetic**], or as a primitive gate for the spin component of hybrid algorithms [**varona2024towards**, **brenner2024factoring**]. In both these cases, especially any use that requires multiple concatenated entangling gates, we will likely require improved gate fidelities. Current gate fidelities are limited by the motional dephasing. The gate error due to motional dephasing follows (see section 4.4.6 in [**ballance2017high-fidelity**])

$$\epsilon_{\mathrm{DEPH}} = 0.686 \frac{t_g}{\tau_{\mathrm{DEPH}}}, \tag{4.9}$$

where for these experiments we estimate a motional coherence time of approximately $\tau_{\mathrm{DEPH}} = 1$ ms. Given the gate duration of $t_g = 75$ µs, this gives a dephasing error of $\epsilon_{\mathrm{DEPH}} = 0.05$, or approximately 5%. This is the dominant error source in our current gate fidelity, and can be improved by decreasing the gate duration, or by improving the mode frequency stability. We have not yet attempted to optimise these gates, nor have we attempted to fully account for the observed errors. However, this work serves as a benchmark for future improvements, and a demonstration of the full spin control capabilities of our system.

# 5

# *k-copy* Randomised Benchmarking

## Contents



**Check framing carefully.**

To construct useful scale quantum processors we must house many qubits physically close together, and must ensure the full register operates with “low error rates”.

But how should we define our errors in a large system? Naively, one might hope that the error rate of a multi-qubit system could be inferred from the error rates of its constituent qubits, but unfortunately, not all errors are commensurable — there are a zoo of possible error channels. They may be coherent, incoherent, continuous, stochastic, local, correlated, time dependent or independent (and I am sure a host of further adjectives exist). In response, a zoo of error correction and mitigation techniques have been developed each with their own scaling overheads and assumptions on these noise models they treat [39–41]. It is a large and intimidating search space, and so, to better equip ourselves for navigation, we require tools to characterise what errors are relevant in our physical systems. Crucially, errors which induce correlations across multiple qubits, termed spatial correlations or crosstalk errors [42, 43], require mitigation and correction strategies built on noise models that faithfully represent their correlated structure [44]; assuming globally depolarising noise models (white noise), can lead to systematic underestimation of logical error rates [45, 46], or systematic bias of the corrected observable outcome [47]. In this chapter we introduce a scalable method for characterising correlated errors in multi-qubit systems: *k-copy randomised benchmarking*. This technique extends single-qubit Clifford RB [48–50] (discussed in chapter **??**) to parallel application across multiple qubits, where now symmetry resolved dynamics provides evidence of, and quantifies various multi-qubit error channels.

This chapter is structured as follows: we start by giving an overview on the problem of *characterisation*, and what techniques already exist to address it. We then outline the *k-copy* protocol and discuss the relevant symmetric subspace structure. We derive the population dynamics when subject to simple noise models and discuss variations of the protocol that isolate specific subspaces. We provide experimental demonstrations of the protocol on a trapped-ion processor through injecting known correlated and uncorrelated errors. We end this chapter by demonstrating the scalability of the protocol on an 8-qubit register, and discuss the limitations and potential of this method for further scaling to larger systems.

## 5.1 Errors and Characterisation

Errors, which we shall give a broad definition as any deviations between the realised and ideal state of a quantum information processor (QIP), are due to unintentional environmental coupling and control imperfections. They are unavoidable, and if not carefully considered or mitigated, lead to unreliable or unverified computation. The process of modelling and quantifying these errors is known as *characterisation*. Characterising errors has two practical advantages: 1) for the engineer, quantifying an error allows for a clear signal to improve hardware and calibration routines, and 2) for the theorist, understanding physically motivated error channels allows for the development of error correction and mitigation techniques that are robust to real world noise. Both users are concerned ultimately with the same question: How faithfully can we execute arbitrary circuits on our physical devices? Extracting metrics to this end is a wide research area in quantum information, with techniques of varying scalability and implicit assumptions on underlying noise models [51]. On one end of the spectrum sits full tomography [52–55] which reconstructs quantum channels, here represented as completely positive, trace preserving (CPTP) maps on the space of density matrices,

$$\mathcal{E}(\rho) : \mathbb{C}^{d\times d} \to \mathbb{C}^{d\times d}, \tag{5.1}$$

where the dimension of the Hilbert space is $d = 2^n$ for $n$ qubits. The CPTP map is rich in information, capturing all Markovian (those which do not depend on history) error channels, but has poor scaling with system size, generally requiring $O(d^4)$ parameters to describe, and $O(d^4)$ measurements to reconstruct. This scaling fundamentally limits tomography to small system sizes. The goal then shifts to how to extract useful information in predicting the behaviour of a system executing a circuit, without full channel reconstruction. Now, two strategies emerge: 1) we make assumptions on the underlying noise model, or 2) we actively simplify the structure of the error channel through averaging. The first strategy leads to techniques such as noise learning [56, 57] and the latter leads to random circuit methods, such as randomised benchmarking (RB) [48, 58, 59]. Both approached dramatically reduce the number of parameters required to describe the system, and the number of measurements required to extract them.

The assumptions on the noise model are cleanly motivated by considering the modularity of quantum circuits, and how errors may violate this modularity. Following the description of imperfect computation and crosstalk errors given in references [60], and [43], respectively, we define the following models of QIPs: *Ideal QIPs*, *Markovian QIPs* (containing the subset of *Crosstalk-Free Markovian QIPs*), and *Non-Markovian QIPs*.

The most desirable, yet technically unfeasible, is the *Ideal QIP* — described by a closed quantum system where the qubits comprise a quantum state represented by a single vector in Hilbert space. The state is isolated from the environment and may be decomposed into a collection of subsystems that remain uncoupled unless explicit multi-qubit operations are applied. Operations are all unitary evolutions, conveniently described by single and two-qubit unitaries, which we term *gates*. Errors may exist, but are limited to reversible unitary interactions.

*Markovian QIPs* include external and control field noise due to an environment with no history, i.e. a source of Markovian noise. We must depart from the quantum state vector and unitary evolution description, and instead use density matrices and CPTP maps to describe the state and dynamics of the QIP. We highlight a subset of these models, termed *Crosstalk-Free Markovian QIP*, where the dynamics of distinct subsystems are governed by locally acting CPTP maps. To satisfy this condition, operations are said to be *local* and *independent* [43].

- *Local* is the assumption that only explicitly applied multi-body operations lead to correlations between qubits that participate in the ideal multi-body operation. For a bipartite system, a local Markovian QIP executing a circuit with no explicit coupling between subsystems, $A$, and $B$, can be described by the product channel CPTP map,

$$\mathcal{E}_{AB}(\rho_A \otimes \rho_B) = \mathcal{E}_A(\rho_A) \otimes \mathcal{E}_B(\rho_B), \tag{5.2}$$

where $\mathcal{E}_A$ and $\mathcal{E}_B$ are CPTP maps acting locally on the $A$ and $B$ subsystems respectively.

- *Independent* is the assumption that the action of an applied gate on a subset of qubits is independent of any other gates applied in parallel on the other qubits. This implies that the CPTP map of gate $G$, acting on subsystem $A$ in layer $i$, given by $\mathcal{E}_G^{(i)}(\rho_A)$, is identical to the action of the same gate acting on subsystem $A$ in layer $j$,

$$\mathcal{E}_G^{(i)}(\rho_A) = \mathcal{E}_G^{(j)}(\rho_A), \tag{5.3}$$

  where parallel applied gates on the other subsystems may be different between layers.

With these two criteria satisfied, we may retain the composibility of gates, conveniently allowing us good prediction of the behaviour of a many-qubit deep circuit, using only characterisations of each gate in a universal gate set applied to small subsystems. If these criteria are not met, to accurately predict the behaviour of a deep circuit, each unique layer must be characterised, which is intractable.

The last class of models, included for completeness, are *Non-Markovian QIP*, where we relax the assumption that the effect of the environment has no history dependence. In this case, any instantaneous state may still be able to be described by a density matrix, however the dynamics can no longer be described by the application of a sequence of CPTP maps. To accurately characterise the system, full process tomography would need to be implemented for each possible circuit.

Practically, to ensure we can reasonably characterise our physical systems, and predict their performance on novel circuits, we must design and appropriately isolate hardware to be well approximated by the *Crosstalk-Free Markovian* models. The next section provides some background on existing techniques to verify crosstalk-free performance, and provide some quantification when the assumption is violated.

## 5.2 Existing approaches to correlated-noise characterisation

Existing approaches to correlated-noise characterisation span a trade-off between expressiveness and scalability. The most expressive, and arguably least scalable, are tomographic techniques, such as gate set tomography [53] or the crosstalk sensitive variation [61]. Here, the implicit assumption that the noise environment is Markovian is sufficient, and no other assumptions of underlying noise models are required. These techniques are valuable in producing detailed CPTP maps that inform not only the presence of crosstalk-free violating noise, but also offer evidence on the underlying noise process. Demonstrations are however limited to small system sizes of two-qubits in [61].

Noise learning protocols are substantially more scalable by imposing a restricted set of noise models, and fitting experimental data to these models. Correlated-noise specific variations of this approach [57, 62] apply repeated random circuits that act to simplify the action of noise channels to Pauli type errors, and then through measurements and stastitical inference, likely Pauli error models are found, and crosstalk-free violating errors may be quantified. Noise learning can also be adaptable by first fitting simple models, for example when twirling over random circuits yields structured error channels, and progressively increasing the complexity of the inferred model when arbitrary circuits are run on the same device [63]. This nested-model approach provides a framework for robust characterisation of the system, however, demonstrations are still currently limited to small systems.

Similar to enforcing Pauli-type errors, randomised benchmarking techniques average the action of noise channels to impose structure to general error channels. This is achieved by twirling the action of the channel over deep circuits comprised of a chosen gate group. Clifford twirling techniques, notably the standard RB protocol [48–50], have been adapted for detecting crosstalk induced correlations by simultaneous RB [64], and correlated RB [65] protocols. Correlated RB uses independent local twirling of multiple system copies to extract multi-qubit survival decay rates with Clifford sequence length. The weight and locality of correlated errors may then be inferred through a distance of the measured channel from a product (uncorrelated) channel. Other, non-Clifford twirled RB protocols, such as cycle-benchmarking [59] and EPLG [66] apply parallel gate sequences on multiple subsystems to extract some error applied per layer, that can then be used to infer error contributions due to correlated-noise channels. These methods have been shown to scale to 10s of qubits, and provide system level diagnostics of device performance.

We note two further streams of current research, first the application of RB in determining the unitarity of noise [67], and the extension of this work to quantifying bipartite unitary noise [68], which can induce correlations across the qubit register. Second, the use of custom circuits designed to probe crosstalk-free violations [43, 69, 70]. **Either expand section, to include why these citations are relevant, or compress section to only the highly relevant papers. Dear reader, what do you think?**

The present protocol we have developed, and demonstrated experimentally, sits firmly in the RB methodology. It however differs from simultaneous and correlated RB by providing not only a quantification of crosstalk-free violating noise, but also providing evidence for the action of the error channel on a multi-qubit system. This extra information is inferred through direct observation of population decay and coupling between the subspaces enforced by twirling over the diagonally applied single-qubit Clifford group. The inspiration of this work comes from correlation spectroscopy [71, 72] and subspace-leakage techniques [73, 74]. Correlation spectroscopy demonstrates that common-mode noise can be revealed through its action on collective and differential degrees of freedom, while subspace-leakage approaches provide a framework for defining error models not by full CPTP maps, but rather by their action on a discrete set of physically relevant subspaces. In such settings, symmetry is not merely a simplifying assumption providing structure to arbitrary channels, but rather it is the very mechanism that separates crosstalk-free violating errors from local noise. The next section details the subspaces enforced by diagonally

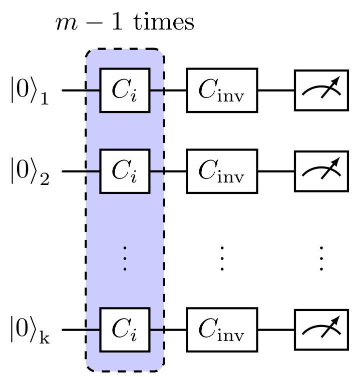



**Figure 5.1:** Circuit diagram for *k-copy* randomised benchmarking. A sequence of $m-1$ random Clifford gates are applied in parallel to $k$ qubits, with a final inverting Clifford applied to ideally recover the initial $|0\rangle$ state. All qubits are measured, and correlations between qubits that successfully return to the initial state are recorded.

applied single-qubit Clifford twirling, and provides the protocol for *k-copy* RB.

## 5.3 The *k-copy* RB scheme

This section starts with a circuit description of the *k-copy* RB protocol. We then develop the representation-theoretic framework required to analyze the decay curves arising from the correlated measurements. We finish this section by deriving the correlated populations observed experimentally when the system is subject to common random-unitary single-qubit, local depolarising, and entangling type noise.

### 5.3.1 Protocol

The *k-copy* RB protocol is implemented through standard single-qubit Clifford RB sequence (see section **??**) applied simultaneously to multiple qubits, with the same randomly sampled Clifford applied to each qubit at every step, as shown in figure 5.1. A single sequence is composed of $m-1$ parallel gates sampled randomly from the Clifford group. As in standard RB, the final applied layer is an inverting Clifford, which, due to the Clifford group being closed, may be selected to take the final state back to the initial computational eigenstate. The full sequence of random

Clifford gates is then equivalent to the identity operation with some unknown noise overlaid[1]. Practically, each gate layer can be executed using a global field, or through parallel local control lines addressing each qubit. A final $\langle Z \rangle$ measurement is made in parallel on all qubits to probe the successful execution of the Clifford sequence. Qubits measured in the target state are labelled $\mathcal{S}$, and qubits measured in the opposite state are labelled $\mathcal{F}$, for success and failure respectively. To build statistics of success and failure rates, $r$ random circuits are trialled for each length $m$, and each circuit is repeated $s$ times to reduce shot noise. To retain information on multi-qubit correlations, for every experimental shot, data from each qubit pair $(i, j)$ is assigned to one of four outcomes, $\mathcal{P}_{SS}^{(i,j)}$, $\mathcal{P}_{SF}^{(i,j)}$, $\mathcal{P}_{FS}^{(i,j)}$, or $\mathcal{P}_{FF}^{(i,j)}$, where $S$ and $F$ denote success and failure. As with other RB protocols, the success probabilities versus sequence length are then fit to explicitly defined decay curves inferred by the action of the twirled error models.

By limiting the analysis to pairwise comparisons, the *k-copy* protocol is both efficient in the number of trials required on the device, and in the postprocessing of the recovered data. For a register of $k$ qubits, the same shot record contains outcomes for all $k(k-1)/2$ pairs. The two-body error models are derived in the following sections for prevalent noise mechanisms in both ion trapping and superconducting information processors.

### 5.3.2 Representations of $Cl_1^{\otimes 2}$

Here, we discuss the representations of the single-qubit Clifford group acting diagonally on two copies, $Cl_1^{\otimes 2}$, first presented in reference [75]. The irreducible representations give a natural (Schur) basis to consider how possible error channels are simplified by *k-copy* Clifford twirling.

We consider a group $\mathcal{G}$ and a representation $r : \mathcal{G} \to GL(\mathcal{L})$, where $GL$ is the general linear group (consisting of the set of invertible matrices and the matrix multiplication operation), and $\mathcal{L}$ denotes the Liouville (vectorised operator) space

[1]Note: To further symmetrise state preparation and measurement errors, we may allow the final target state to be randomised between $|0\rangle$ and $|1\rangle$ for each sequence [49].

of two qubits. Twirling over the elements $g$ in $\mathcal{G}$ enforces symmetry on a general error channel, $\Lambda$, given in Liouville form,

$$\tilde{\Lambda} = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} r(g)\Lambda r(g)^{\dagger}, \tag{5.4}$$

where $|\mathcal{G}|$ is the number of elements comprising the finite group $\mathcal{G}$. The twirling enforces commutation with the group action,

$$r(g)\tilde{\Lambda}r(g)^{\dagger} = \tilde{\Lambda} \quad \forall g \in \mathcal{G}. \tag{5.5}$$

The representation of the space, $\mathcal{L}$ decomposes to irreps $\lambda$ as,

$$\mathcal{L} = \bigoplus_{\lambda}(V_{\lambda} \otimes \mathbb{C}^{m_{\lambda}}), \tag{5.6}$$

where $V_{\lambda}$ is the irreducible representation space, and $\mathbb{C}^{m_{\lambda}}$ is a complex valued multiplicity space with dimension of the multiplicity $m_{\lambda}$. From Schur's lemma, equation 5.5 implies the twirled error channel has the form,

$$\tilde{\Lambda} = \bigoplus_{\lambda}(I_{\lambda} \otimes \mathcal{M}_{\lambda}), \tag{5.7}$$

where $\mathcal{M}_{\lambda}$ is the multiplicity matrix of dimensions $m_{\lambda} \times m_{\lambda}$.

The 2-copy single-qubit Clifford group, analysed in detail in references [75, 76], decomposes the general 16-dimensional two qubit space into 4 irreducible representations, irreps, with dimensions $\{1, 2, 3, 3'\}$, (the notation $3'$ is used to contrast the other $3-$dimensional irrep), and multiplicities $\{2, 1, 3, 1\}$. The sub-representations are summarised in table 5.1. **Could move character table and conjugacy classes to appendix?** The irreps are uniquely determined by their character on the conjugacy classes of the Clifford group. Two group elements $g, h \in G$ are said to be in the same conjugacy class if they are related by conjugation,

$$g = khk^{-1}, \tag{5.8}$$

where $k \in G$. With this definition, the 24 single qubit Clifford elements are divided into five permutation classes, shown in table 5.2, with a representative name and

| Sub-representation | irrep | Basis operators |
|---|---|---|
| Identity | 1 | $II$ |
| Left qubit | 3 | $XI, YI, ZI$ |
| Right qubit | 3 | $IX, IY, IZ$ |
| Trace | 1 | $(XX + YY + ZZ)/\sqrt{3}$ |
| Diagonal traceless | 2 | $(XX + YY - 2ZZ)/\sqrt{6}$, $(XX - YY)/\sqrt{2}$ |
| Symmetric off-diagonal | $3'$ | $(XY + YX)/\sqrt{2}$, $(XZ + ZX)/\sqrt{2}$, $(YZ + ZY)/\sqrt{2}$ |
| Antisymmetric | 3 | $(XY - YX)/\sqrt{2}$, $(XZ - ZX)/\sqrt{2}$, $(YZ - ZY)/\sqrt{2}$ |

**Table 5.1:** Sub-representations of the $Cl_1^{\otimes 2}$ group. The irrep, dimension of each sub-representation, and a set of operators in the Pauli basis which span the subspace dimension is given. The irrep basis vectors are from equation 56 in reference [76].

| Class | Representation | Member |
|---|---|---|
| Identity | e | I |
| Swap | (12) | H |
| Two-swap | (12)(34) | X |
| Three-cycle | (123) | HS |
| Four-cycle | (1234) | S |

**Table 5.2:** Conjugacy classes for the single qubit Clifford group.

| irrep | e | (12) | (12)(34) | (123) | (1234) |
|---|---|---|---|---|---|
| Identity/Trace | 1 | 1 | 1 | 1 | 1 |
| Left/Right/Antisymmetric | 3 | -1 | -1 | 0 | 1 |
| Diagonal traceless | 2 | 0 | 2 | -1 | 0 |
| Symmetric off-diagonal | 3 | 1 | -1 | 0 | -1 |

**Table 5.3:** Character table for the Clifford group given the irreps in table 5.1, and with conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle. The "sign" irrep of the single-qubit Clifford group has zero multiplicity in the two-copy case, and so is excluded here.

example member following each class. These classes make obvious the known isomorphism between $Cl_1$ and the symmetric group $S_4$ (modulo phase). Character tables [77], are useful tools in representation theory, to uniquely describe irreps by the traces of the representative matrices for group elements of each conjugacy class. The table is presented as a 2D matrix with rows corresponding to group representations and columns to group conjugacy classes. The character, $\xi$, is defined by,

$$\xi_\lambda(g) = \mathrm{Tr}[r_\lambda(g)], \tag{5.9}$$

where $r_\lambda$ is the irreducible component of the representation of $G$ acting on some vector space,

$$r_\lambda(g)_{i,j} = \mathrm{Tr}[B_i^\dagger g B_j g^\dagger], \tag{5.10}$$

where $B_i$ are the orthonormal basis vectors for irrep $\lambda$. Taking the two-qubit sub-representations reproduced in table 5.1, the character table calculated is shown in table 5.3. We can see confirmed that the left, right, and antisymmetric sub-representations form a multiplicity of the $3D$-irrep labelled 3, while the trace sub-representation is a multiplicity of the $1D$ trivial irrep.

Given the structure of a general twirled map is enforced to be block diagonal in the irrep basis, given by equation 5.7, we can do a simple count to find generally how many parameters are required to fully describe the CPTP map. Accounting for the multiplicities, and the constraint that the map is trace preserving, we find that at most $2^2 + 3^2 + 1 + 1 - 2 = 13$ independent scalars are needed to describe $\tilde{\Lambda}$. The diagonal terms, here labelled $\vec{\alpha} = (\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as})$, describe a decay rate contribution from each sub-representation, whilst off-diagonal terms, $\vec{m} = (m_{id\to tr}, m_{L\to R}, m_{R\to L}, m_{L\to as}, m_{R\to as}, m_{as\to L}, m_{as\to R})$ describe coupling between the multiplicity spaces.

In practice, different physical error mechanisms populate only a subset of the coefficients $\alpha$ and $m$, leading to simplified dynamics with characteristic signatures of the underlying process. Table 5.4 provides possible physical mechanisms capable of coupling isotypic representation subspaces. We devote some time in the next

| Coupling | irrep | Mechanism |
|---|---|---|
| $m_{id\to tr}$ | 1 | Decays (non-unital) |
| $m_{L\to R}$, $m_{R\to L}$ | 3 | Qubit Swap, Correlated decays (non-unital) |
| $m_{L\to as}$, $m_{R\to as}$, $m_{as\to L}$, $m_{as\to R}$ | 3 | Entanglement |

**Table 5.4:** Possible mechanisms that result in the coupling of sub-representation sectors of the $Cl_1^{\otimes 2}$ twirled error channel.**Is this too strong a claim?**

two sections to describing noise due to fluctuations of a common field, and to local depolarising type errors. In appendix **??** we discuss a $ZZ$-type noise model, relevant to the superconducting qubit platform.

### 5.3.3 Common random unitary noise

It can be seen that the error channels that lead to couplings neatly discern the physical origin of the noise channel. For example, in transmon superconducting qubit platforms, always on $ZZ$-couplings are a dominant crosstalk error source [78–80], and so the couplings between single qubit subspaces and the antisymmetric subspace can be quantified; in the neutral atom platform, correlated non-unital errors may be observed if the physical qubit separation is of similar order to the qubit transition wavelength [81]; however, in ion trap platforms, environmental noise is often dominated by magnetic and electric field fluctuations, which may act globally on all qubits if they are stored in the same trapping potential. These field fluctuations induce both common and independent single-qubit rotations, but do not couple irrep subspaces together (i.e $m = 0$ for $m \in \vec{m}$). Thus, the general error map of this type is now parametrised by at most the 6 scalars, $\vec{\alpha}$, and results in Liouville form diagonal in the Schur basis, $\mathcal{S}$, (ordered by table 5.1),

$$\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}) = diag\left(I_1, \alpha_L I_3, \alpha_R I_3, \alpha_{tr} I_1, \alpha_{dt} I_2, \alpha_{so} I_3, \alpha_{as} I_3\right), \tag{5.11}$$

where $I_d$ is the $d$-dimensional identity.

We term noise due to common small-angle single-qubit rotations acting on $k$-qubits *common noise*. For two qubits the general form is given by the separable channel,

$$\mathcal{E}_{\mathcal{A}}(\rho) = \sum_{a\in\mathcal{A}} p_a(U_a \otimes U_a)\rho(U_a \otimes U_a)^\dagger, \tag{5.12}$$

where $\mathcal{A}$ specifies the ensemble of unitaries applied with probabilities $p_a$, with $\sum_{a\in\mathcal{A}} p_a = 1$, and $U_a(\theta_a, \hat{n}_a) = e^{-i(\theta_a/2)\hat{n}_a\cdot\hat{\sigma}}$, with rotation angle, $\theta_a$, axis, $\hat{n}_a$, and Pauli vector, $\hat{\sigma}$. A subset of error channels following this description cannot be described by a product channel map on a bipartite system, and so violate the *local* condition given in equation 5.2. These errors therefore cannot be modelled by a crosstalk-free Markovian QIP although originating from separable channels. This is an important distinction: non-product channel are not limited to direct entangling noise.

To find the twirled error channel of the common unitary channel, $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_{\mathcal{A}})$, we first consider the effect of twirling on a single common unitary channel, $\mathcal{E}_a(\rho) = (U_a \otimes U_a)\rho(U_a \otimes U_a)$. We outline two approaches to find $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_a)$:

First, as the basis sets for the relevant sub-representations are known (see table 5.1), and as the single qubit Clifford group only contains 24 elements, we may construct the common unitary channel in the Pauli transfer matrix (PTM) representation[60, 82], and, using equation 5.4, we may explicitly calculate the twirled channel $\tilde{\Lambda}^{\mathcal{P}}(\mathcal{E}_a)$ utilising a symbolic computation program[2]. Here the superscript $\mathcal{P}$ denotes we are working in the Pauli basis. The $\alpha$ parameters are then simply read from the diagonal elements after transforming the PTM representation to the known basis operators for the sub-representations, i.e. transforming to $\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_a)$.

The second approach utilises the symmetries of the irrep spaces, and the separability of the error channel to construct the PTM projectors onto the irrep spaces. This method is outlined in appendix **??**, and provides further insight into the symmetries each irrep possesses, and the naming convention for the sub-representations.

[2]Such as *Mathematica* by *Wolfram Research, Inc.*, or the *Python* package *SymPy*.

Both approaches ultimately yield the alpha parameters,

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1+2c), \tag{5.13}$$

$$\alpha_{tr} = 1, \tag{5.14}$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1-c)^2\kappa), \tag{5.15}$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1-c)^2\kappa), \tag{5.16}$$

where $c = \cos(\theta_a)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Taking these scalar products to second order in error angle $\theta_a$, we find that the anisotropic term may be ignored for small errors,

$$\alpha_L = \alpha_R = \alpha_{as} = 1 - \theta_a^2/3, \tag{5.17}$$

$$\alpha_{tr} = 1, \tag{5.18}$$

$$\alpha_{dt} = \alpha_{so} = 1 - \theta_a^2. \tag{5.19}$$

Reintroducing the stochastic application of common unitaries, given by equation 5.12, we find,

$$\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_{\mathcal{A}}) = \sum_{a\in\mathcal{A}} p_a \tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_a). \tag{5.20}$$

As such, the dynamics for the twirled channel of an ensemble of common unitaries is entirely described by the error angle magnitudes, with alpha parameters,

$$\alpha_i = \sum_{a\in\mathcal{A}} p_a \alpha_a. \tag{5.21}$$

This definition leads to a convenient (and conforming to previous RB work) definition for the correlated error per Clifford given by,

$$\varepsilon_{\text{corr}} = \frac{1}{6}\sum_{a\in\mathcal{A}} p_a \theta_a^2, \tag{5.22}$$

where the correlated error per Clifford, $\varepsilon_{\text{corr}}$, quantifies the single-qubit error contribution due to *common noise*. Comparing to equation 5.11, the general ensemble of common single-qubit rotations channel, after twirling over $Cl_1^{\otimes 2}$, is given by the diagonal matrix in the Schur basis,

$$\begin{aligned}\tilde{\Lambda}^{\mathcal{S}}(\mathcal{E}_{\mathcal{A}}) = \mathrm{diag}\Big(I_1,\ (1-2\varepsilon_{corr})I_6,\ I_1,\\ (1-6\varepsilon_{corr})I_2,\ (1-2\varepsilon_{corr})I_6\Big).\end{aligned} \tag{5.23}$$

### 5.3.4 Local depolarising noise

Ultimately, environmental noise will not perfectly couple symmetrically to both qubits, and so independent local single-qubit error channels must also be considered. Assuming an independent noise channel acting on each qubit, Clifford twirling leads to a fully depolarising noise channel of the form,

$$\mathcal{E}_{dp,i}(\rho) = (1-2\varepsilon_i)\rho + \varepsilon_i I_2 \mathrm{Tr}(\rho), \tag{5.24}$$

where $\rho$ is for a single qubit, and the subscript $i$ signifies which qubit we are considering. For the "left" and "right" qubits we describe the decay rates by $\varepsilon_L$ and $\varepsilon_R$, respectively. Parallel local depolarisation is given by the channel, $\mathcal{E}_{dp} = \mathcal{E}_{dp,L}\otimes\mathcal{I}+\mathcal{I}\otimes\mathcal{E}_{dp,R}$, with $\mathcal{I}$ being the identity channel on a single qubit. Comparing to equation 5.11, this channel may be expressed in PTM form as the diagonal matrix,

$$\Lambda^{\mathcal{P}}(\mathcal{E}_{dp}) = \mathrm{diag}\Big(1, (1-2\varepsilon_L)I_3, (1-2\varepsilon_R)I_3, (1-2\varepsilon_L)(1-2\varepsilon_R)I_9\Big). \tag{5.25}$$

We find that $\Lambda(\mathcal{E}_{dp})$ already commutes with the representations $r(g)$ for all elements $g$ in the single qubit Clifford group, as such the channel already obeys the symmetries enforced by twirling over $Cl_1^{\otimes 2}$ (see equation 5.5). Transforming to the Schur basis, we extract $\alpha$ parameters,

$$\alpha_L = 1-2\varepsilon_L, \tag{5.26}$$

$$\alpha_R = 1-2\varepsilon_R, \tag{5.27}$$

$$\alpha_{tr} = \alpha_{dt} = \alpha_{so} = \alpha_{as} = 1-2\left(\varepsilon_L+\varepsilon_R\right). \tag{5.28}$$

### 5.3.5 Combined model

Combining the depolarisation channels with the twirled common single-qubit rotation channel,

$$\Lambda_T(\varepsilon_L, \varepsilon_R, \varepsilon_{corr}) = \Lambda_{dp}(\varepsilon_L, \varepsilon_R)\tilde{\Lambda}_{corr}(\varepsilon_{corr}), \tag{5.29}$$

we find scalar parameters,

$$\alpha_L = 1 - 2\left(\varepsilon_L + \varepsilon_{corr}\right), \tag{5.30}$$

$$\alpha_R = 1 - 2\left(\varepsilon_R + \varepsilon_{corr}\right), \tag{5.31}$$

$$\alpha_{tr} = 1 - 2\left(\varepsilon_L + \varepsilon_R\right), \tag{5.32}$$

$$\alpha_{dt} = \alpha_{so} = 1 - 2\left(\varepsilon_L + \varepsilon_R + 3\varepsilon_{corr}\right), \tag{5.33}$$

$$\alpha_{as} = 1 - 2\left(\varepsilon_L + \varepsilon_R + \varepsilon_{corr}\right). \tag{5.34}$$

To reproduce the decay dynamics observed in the RB sequence, we must consider an initial prepared state, and the final measurement basis. For the RB sequence given in figure 5.1, the initial state is prepared as $|00\rangle$, and the final correlated populations are measured in the computational eigenstates, $\{|00\rangle \leftrightarrow SS, |01\rangle \leftrightarrow SF, |10\rangle \leftrightarrow FS, |11\rangle \leftrightarrow FF\}$, where $S$ and $F$ denote "success" and "failure" of recovering the target state outcome for each qubit. The correlated populations may be given analytically using the derived Liouville operators, and by the vectorized initial and final states,

$$\mathcal{P}_{LR} = \langle\langle L, R|\Lambda_T^m|00\rangle\rangle, \tag{5.35}$$

where $L, R$ denote the successful recovery of the target final state for the "left" and "right" qubit respectively, and $m$ is the number of Clifford gates in the applied sequence. Using this equation, and the explicit transfer matrices, the final populations decays are found,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \tag{5.36}$$
$$\mathcal{P}_{SF} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \tag{5.37}$$
$$\mathcal{P}_{SF} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \tag{5.38}$$
$$\mathcal{P}_{FF} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m. \tag{5.39}$$

Considering the final state of the single qubit subspace, as probed by standard RB, we find the expected single exponential decay,

$$\mathcal{P}_S^{(L)} = \mathcal{P}_{SS} + \mathcal{P}_{SF} = \frac{1}{2}\left(1 + \left(1 - 2\left(\varepsilon_L + \varepsilon_{corr}\right)\right)^m\right), \tag{5.40}$$

where the superscript $(L)$ refers to considering the left qubit only. Comparing to standard single qubit RB parameterisation,

$$\mathcal{P}_S = \frac{1}{2}\left(1 + (1 - 2\varepsilon_c)^m\right), \tag{5.41}$$

where $\varepsilon_c$ is the "error-per-Clifford", we find the observed decay of each qubit is now separable into a local, and correlated error component,

$$\varepsilon_c^{(L)} = \varepsilon_L + \varepsilon_{corr}. \tag{5.42}$$

**Asymptotes**

Here we consider the asymptotic structure of the correlated decay curves given in 5.36. Under dominant local depolarisation, the two-qubit state will approach the maximally mixed state, indicated by the asymptote $\mathcal{P}_{SS}|_{m\to\infty} = 1/4$, as seen in figure 5.2 (a). Under dominant *common noise*, with decay profile seen in (b), an asymptote of $\mathcal{P}_{SS}|_{m\to\infty} = 1/3$ is approached due to the invariance of the symmetric trace sector, $\alpha_{tr}$. Panel (c) shows the case where both local depolarising and *common noise* is present. When $\varepsilon_{corr} > \varepsilon_L = \varepsilon_R$ a clear intermediate asymptote is seen before a final decay to the fully depolarised state. This intermediate asymptote is the central experimental signature of correlated noise exploited throughout this chapter, and is explored experimentally in section 5.6.2 below.

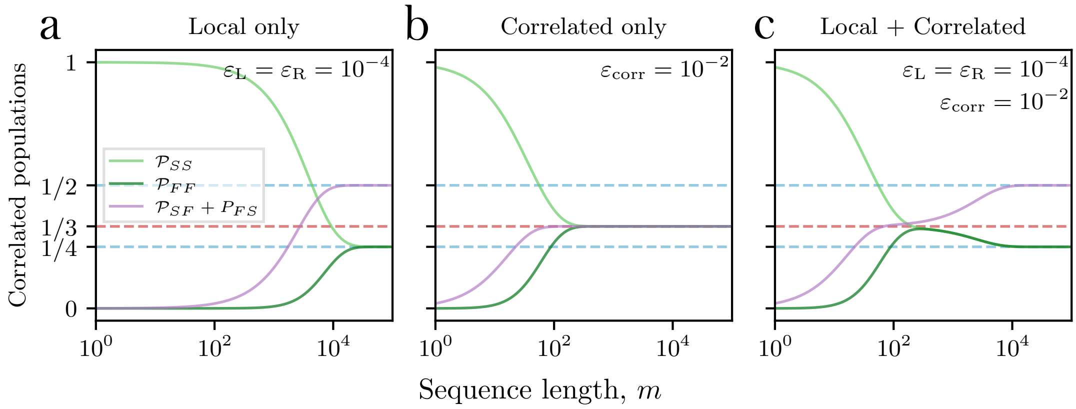



**Figure 5.2:** The expected multi-exponential decay profiles for two-qubit correlated success and failures recovered during *k-copy* RB. Panels (a), (b), and (c) display the structured decays for local depolarising only, *common noise* only, and the combined case, respectively. Dashed lines indicate the expected asymptotes for depolarising (blue) and for *common noise* (red).

The mixed state reached after the application of *common noise* to the initial state $|00\rangle$ can be understood by directly inspecting the space of possible pure two qubit states reached through common only rotations. The general pure state after common rotations is the separable identical two-qubit states,

$$|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle \,, \tag{5.43}$$

with $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$ for a single qubit general state. Depolarising into the symmetric separable two-qubit states is equivalent to averaging over the full Bloch sphere,

$$\begin{aligned} \rho_{\mathrm{sym}} &= \frac{1}{4\pi}\int_0^{\pi}\int_0^{2\pi} \left(\sin(\theta)\,|\Psi\rangle\langle\Psi|\right) \mathrm{d}\phi \mathrm{d}\theta \\ &= \frac{1}{3}|00\rangle\langle 00| + \frac{1}{3}|11\rangle\langle 11| + \frac{1}{3}|\Psi^+\rangle\langle\Psi^+| \,, \end{aligned} \tag{5.44}$$

where the third term is the Bell state $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. The presence of this Bell-state component reflects residual coherence induced between the two qubits by the shared rotation error, and motivates the alternative strategy of a coherence-sensitive probe developed in section 5.4.

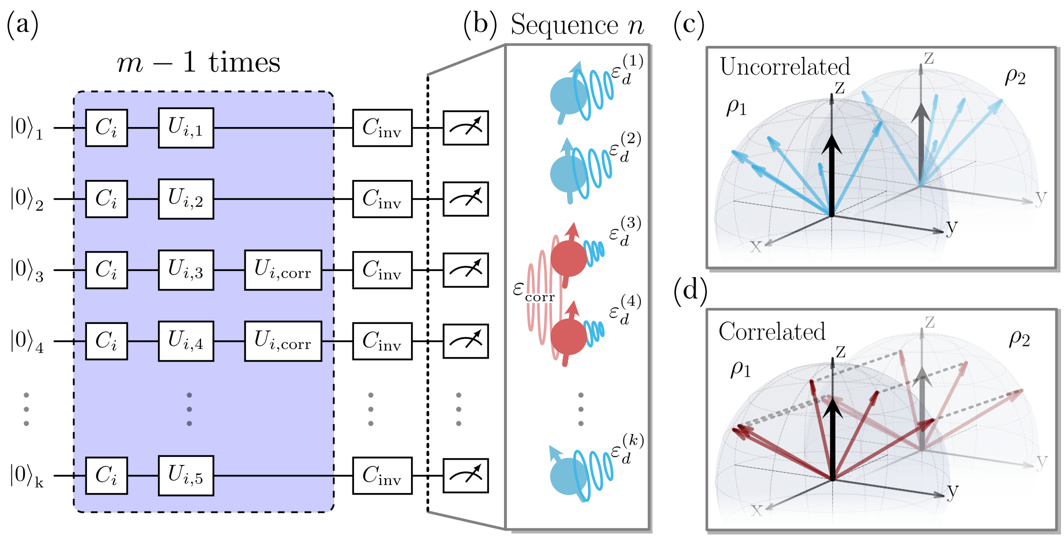



**Figure 5.3:** k-copy RB applies identical single-qubit Clifford sequences in parallel to all qubits. (a) shows the circuit diagram for a single RB random sequence with random unitary errors included explicitly. (b-d) illustrate the distinction between uncorrelated and *common noise* (a type of correlated noise) in the final-state trajectories.

### Schematic explanation

The distinction between correlated and uncorrelated noise is illustrated schematically in figure 5.3. We consider a particularly simple unitary noise mechanism to highlight the qualitative difference of information gained between standard and *k-copy* RB. After each Clifford gate we consider an independent unitary rotation errors on each qubit, and a common unitary rotation error on qubits $i = 3$ and $i = 4$. Each unitary rotation axis is randomly drawn (isotropically from $SO3$), and the angle of the rotation is small. The error channels are explicitly included in the circuit diagram in (a). The state of the qubits after the twirling procedure, and before measurement, are shown in (b). We see that all qubits are no longer in the eigenstate $|0\rangle$ (up arrow) due to the unitary noise. Panels (c) and (d) display the final vectors resulting from 5 random sequences. All qubits here exhibit identical single-qubit RB error rates, which are probed by standard RB, and shown here through the identical depolarised Bloch vectors (black arrows). However, looking at pairwise correlations, we see: independent noise produces uncorrelated sequence-to-sequence deviations, whereas common noise produces shared deviations between qubits acted on by the same random Clifford sequence. The parallel twirling during *k-copy* RB preserves exactly this symmetry information that would be averaged away under independently sampled local Clifford sequences.

### 5.3.6 State preparation errors

Similar to previous work on subspace leakage benchmarking [73], state preparation errors can lead to non-trivial modifications of the correlated population decay curves. First, we consider independent state preparation on two qubits. We model independent errors as probabilistic bit flips. To first order in error, on a single qubit we have,

$$\rho_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)}) \left|0\right\rangle \left\langle 0\right| + \varepsilon_{sp}^{(i)} \left|1\right\rangle \left\langle 1\right|, \tag{5.45}$$

where the superscript refers to which qubit, $L$, or $R$, the preparation error is relevant to. This error may be modelled by $\mathcal{E}_{sp}^{(i)}$, a probabilistic $X$ channel applied to an arbitrary state,

$$\mathcal{E}_{sp}^{(i)}(\rho) = \left(1 - \varepsilon_{sp}^{(i)}\right)\rho + \varepsilon_{sp}^{(i)} X\rho X. \tag{5.46}$$

Transforming to the Liouville form, denoted by $\Lambda_{sp}^{(i)}$, this error is simply incorporated into the population dynamics given by equation 5.35, by acting on the initial state,

$$\mathcal{P}_{LR} = \langle\langle L, R|\Lambda_T^m \Lambda_{sp}^{(L)} \Lambda_{sp}^{(R)}|00\rangle\rangle. \tag{5.47}$$

In this picture, the state preparation errors scale only the overlap of the initial state with the sub-representations. The overlaps are multiplied by,

$$A_{sp}^{(i)} = (1 - 2\varepsilon_{sp}^{(i)}), \tag{5.48}$$

if the qubit is involved non-trivially in the subspace projector, i.e. for the left and right qubit sectors $A_{sp} = A_{sp}^{(i)}$. Relevant to this work, the correlated survival populations with state preparation errors included are,

$$\begin{aligned} \mathcal{P}_{SS} = \frac{1}{4} + \frac{A_{sp}^{(L)}}{4}\alpha_L^m + \frac{A_{sp}^{(R)}}{4}\alpha_R^m \\ + \frac{A_{sp}^{(L)}A_{sp}^{(R)}}{6}\alpha_{dt}^m + \frac{A_{sp}^{(L)}A_{sp}^{(R)}}{12}\alpha_{tr}^m. \end{aligned} \tag{5.49}$$

Similar forms exist for populations, $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$. Significantly, this state preparation error does not add projection on the antisymmetric or symmetric off-diagonal sectors, which would introduce another exponential component into the observed decay curves.

Correlated state preparation errors may be considered by adapting equation 5.46 to a two-qubit stochastic error channel, such as correlated spin-flips given by

$$\mathcal{E}_{sp}^{corr}(\rho) = (1 - \varepsilon_{sp}^{corr})\rho + \varepsilon_{sp}^{corr}(X \otimes X)\rho(X \otimes X). \tag{5.50}$$

The subspaces involving both qubits non-trivially are now unaffected by the state preparation error, shown explicitly by,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{A_{sp}^{corr}}{4}\alpha_L^m + \frac{A_{sp}^{corr}}{4}\alpha_R^m + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \tag{5.51}$$

where $A_{sp}^{corr} = 1 - 2\varepsilon_{sp}^{corr}$. These correlations thus appear similar to *common noise* errors in the invariance of $\alpha_{tr}$ to this channel. Therefore correlated state prep errors can lead to a systematic bias in quantification of correlated error per Clifford if not accounted for in the fit by an initial state preparation error offset.

## 5.4 Parity contrast: an independent probe of coherence

Population decays described by equation 5.36 indicate an intermediate asymptote corresponding to the suggested state, $\rho_{sym}$, given by equation 5.44. Here we propose a parity-based measurement to independently verify the coherent structure seen in $\rho_{sym}$. Specifically, a reduced tomography method based on [83] is used to quantify overlap of the final measured state with the Bell state, $|\Psi^+\rangle$, component in $\rho_{sym}$.

The circuit for probing this coherence is shown in figure 5.4 (a): differential rotations of the form $\hat{R}_{+\phi}(\pi/2) \otimes \hat{R}_{-\phi}(\pi/2)$, with rotation axis $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$, are applied prior to measurement, and the contrast of the resulting fringe (as $\phi$ is varied) is extracted.

Figure 5.4(b) show the expected contrast of parity measurements as a function of Clifford sequence length. The analytic model of the parity contrast can be simply

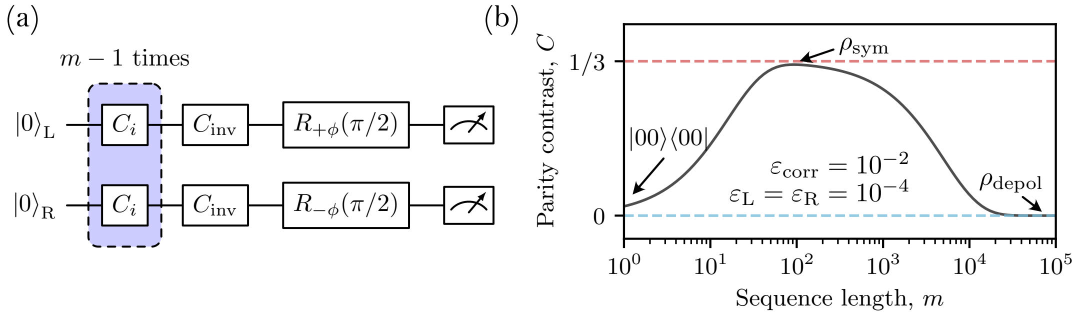



**Figure 5.4:** Probing coherence of the mixed state after *k-copy* RB. Panel (a) shows a circuit that first twirls the state over $Cl_1^{\otimes 2}$, and then applies differential $\pi/2$-pulses with variable phase, $\phi$, to recover parity oscillations. Panel (b) shows the expected contrast of the oscillations as a function of sequence length when both local depolarising and *common noise* is present. The dashed lines indicate expected asymptotes for depolarising (blue) and for *common noise* (red).

calculated by extending equation 5.35 to include the final differential rotation, in transfer matrix form, $\Lambda_\Pi$,

$$\begin{aligned}\Pi(\phi) &= \langle\langle \pi | \Lambda_\Pi(\phi) \Lambda_T^m | 00 \rangle\rangle, \\ &= C\cos(2\phi),\end{aligned} \tag{5.52}$$

where $\Pi$ is the parity signal, $|\pi\rangle\rangle = |00\rangle\rangle + |11\rangle\rangle - |01\rangle\rangle - |10\rangle\rangle$, and $C$ is the parity contrast. Evaluating $\Pi(\phi)$ for *common* and depolarising noise, we find the parity contrast follows,

$$C = \frac{1}{3}(\alpha_{tr}^m - \alpha_{dt}^m), \tag{5.53}$$

with $\alpha_{tr}$ and $\alpha_{dt}$ given by equation 5.30. This method is demonstrated experimentally in section 5.6.3 below.

Furthermore, we may now isolate some $\alpha$ decay parameters by combining the contrast measurements with the computational state measurements made in *k-copy* RB. Namely, if we define the computational parity, $\Pi_Z$, as

$$\begin{aligned}\Pi_Z &= \mathcal{P}_{SS} + \mathcal{P}_{FF} - \mathcal{P}_{SF} - \mathcal{P}_{FS} \\ &= \frac{1}{3}(2\alpha_{dt}^m + \alpha_{tr}^m),\end{aligned} \tag{5.54}$$

then in combination with $C$, we may isolate single exponential decays originating from the independent sectors named “symmetric trace”, and “diagonal traceless”.

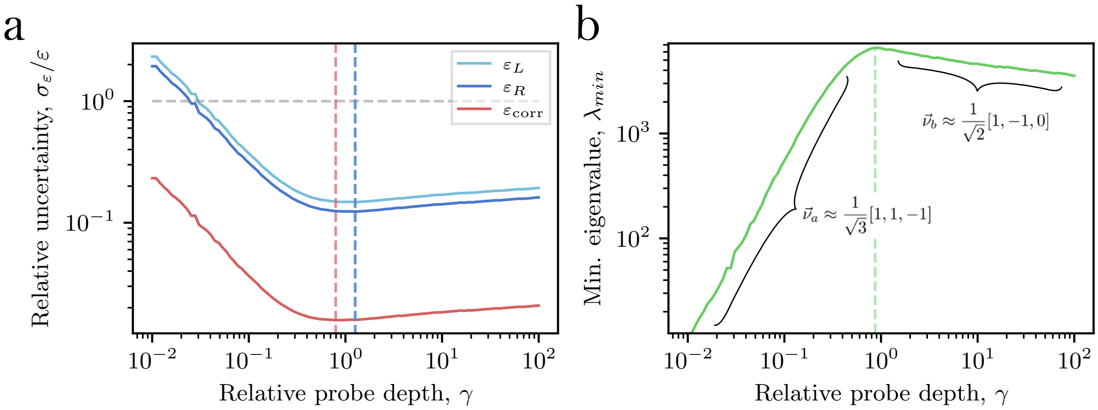



**Figure 5.5:** The performance of the *k-copy* fitting as a function of maximum Clifford sequence length (probe depth). Probe depth, $\gamma$, is relative to the characteristic decay length for the slowest mode, $\alpha_{tr}^m$. Panel (a) uses the Cramér-Rao derived relative uncertainty for each extracted $\varepsilon$, while (b) shows the minimum eigenvalue of the Fisher information matrix, with associated eigenvectors highlighted.

Specifically the combinations

$$\begin{aligned} \Pi_Z - C &= \alpha_{dt}^m, \\ \Pi_Z + 2C &= \alpha_{tr}^m. \end{aligned} \tag{5.55}$$

## 5.5 Sampling strategy

In this section we find the Fisher information gained from a single shot of the *k-copy* RB protocol, and use this to find a strategy for setting the sequence lengths, $m$. We then explore the shot efficiency of extracting error per Clifford parameters for a range of underlying noise magnitudes, using both the *k-copy* protocol described in section 5.3, and using the parity measurement protocol described in section 5.4.

The Fisher information [84], $\mathcal{I}$, quantifies what we can learn from random observable, $X$, about a set of underlying unknown parameters, $\theta$, that models the probability of observing $X$. It is particularly informative to consider the Cramér-Rao bound, which states that the inverse of $\mathcal{I}$ is a lower bound for the variance of any unbiased estimator of $\theta$,

$$\mathrm{Cov}(\theta) \gtrsim \mathcal{I}(\theta)^{-1}. \tag{5.56}$$

The Fisher matrix is defined as,

$$\mathcal{I}_{ij}(\theta) = \mathrm{E}_X\left[\frac{\partial}{\partial\theta_i}\log f(X;\theta)\frac{\partial}{\partial\theta_j}\log f(X;\theta)\right], \tag{5.57}$$

where $f(X;\theta)$ is the likelihood function, and $\theta$ is a set of parameters. Here, we are interested in a multinomial probability function, with each shot resulting in a categorical outcome $X \in \{SS,\ SF,\ FS,\ FF\}$. The population functions given by equations 5.36, are then the probability of observing $X = x$ for a given set of parameters, $\theta = \{\varepsilon_L,\ \varepsilon_R,\ \varepsilon_{\mathrm{corr}}\}$, yielding,

$$f(X = x;\theta) = P_x(\theta). \tag{5.58}$$

We now find the Fisher matrix to be,

$$\mathcal{I}_{ij}(\theta) = \mathrm{E}_X\left[\frac{1}{P_X^2}\frac{\partial P_X}{\partial\theta_i}\frac{\partial P_X}{\partial\theta_j}\right]. \tag{5.59}$$

For a discrete random variable, the expectation is,

$$\mathrm{E}[g(X)] = \sum_x P_x g(x), \tag{5.60}$$

giving our final expression for the single-shot multinomial Fisher matrix,

$$\mathcal{I}_{ij} = \sum_{x\in\{SS,SF,FS,FF\}}\frac{1}{P_x}\frac{\partial P_x}{\partial\theta_i}\frac{\partial P_x}{\partial\theta_j}. \tag{5.61}$$

The information gained in any one shot depends on the underlying error magnitudes, as well as the chosen sequence length, $m$. As the correlated profiles, $P_x$, appear as multi-exponentials decays dependent on common parameters, $\theta$, we must choose a range of probe sequence lengths that yield large information over all the characteristic decays. As the characteristic decays are unknown, a good heuristic strategy for fitting an unknown exponential is logarithmic sampling. However, the total extent of sequence lengths to test is non-obvious. It would be experimentally simple if the initial gradient of the correlated profiles successfully estimated the parameter $\theta$, however, expanding the populations around $m = 0$, for small errors, we find that both purely uncorrelated and purely *common* noise decays are indistinguishable,

$$\mathcal{P}_{SS,LR}|_{m=0} \approx \mathcal{P}_{SS,\mathrm{corr}}|_{m=0} \approx 1 - 2\varepsilon, \tag{5.62}$$

where we have assumed identical error magnitudes, $\varepsilon_L = \varepsilon_R = \varepsilon_{\text{corr}} = \varepsilon$. This suggests that we must collect data near the asymptotic structure of the decays to distinguish uncorrelated and *common* noise sources. To verify this, we set experimentally reasonable parameters for an underlying *common noise* dominated decay, and vary the maximum sequence length, $m_{\max}$, while observing the expected uncertainty given by the Cramér-Rao bound of the Fisher information. We normalise by the magnitude of the parameter estimated to give the relative uncertainty, $\frac{\sigma_\varepsilon}{\varepsilon}$. We also find the minimum eigenvalue of the Fisher matrix, $\lambda_{\min}$, which corresponds to how accurately the least constrained vector in parameter space can be measured. Figure 5.5 uses $N = 5000$ samples per sequence length (e.g. $s = 50$ shots, and $r = 100$ random sequences), and 10 total sequence lengths sampled logarithmically from $m_{\min} > 2$ to $m_{\max} = \gamma\tau$. The scalar $\gamma$ is the relative probe depth and $\tau$ is the characteristic decay for the slow, $\alpha_{tr}$ mode,

$$\tau = \frac{-1}{\log(\alpha_{tr})} = \frac{-1}{\log(1 - 2(\varepsilon_L + \varepsilon_R))}. \tag{5.63}$$

We set error magnitudes of $\varepsilon_L = 1.0 \times 10^{-3}$, $\varepsilon_R = 1.2 \times 10^{-3}$, $\varepsilon_{\text{corr}} = 1.0 \times 10^{-2}$, which are similar magnitudes we observe experimentally (see section 5.6). It can be seen that the relative uncertainty for all $\theta$ are minimised when $\gamma \approx 1$. This length also corresponds to maximising $\lambda_{\min}$ — lowering the cost of estimating the least constrained vector of parameters, where the vector is written in the basis $[\hat{\varepsilon}_L, \hat{\varepsilon}_R, \hat{\varepsilon}_{\text{corr}}]$. It is interesting to note that the least constrained parameter vector, when $\gamma \ll 1$, roughly corresponds to $\vec{\nu} \approx \frac{1}{\sqrt{3}}[1, 1, -1]$, confirming our suspicion that *common* and depolarising noise is indistinguishable at small $m$. In the case that $\gamma \gg 1$, we find $\vec{\nu} \approx \frac{1}{\sqrt{2}}[1, -1, 0]$, corresponding to the depolarisation rates on the left and right qubit are tending to be indistinguishable, however we note that even over a couple of orders of magnitude, the distinguishability of $\varepsilon_L$ and $\varepsilon_R$ does not dramatically drop. Here, we draw the conclusion that a good strategy for efficient information per shot, is to set $m_{\max}$ to the characteristic decay of the slowest mode.

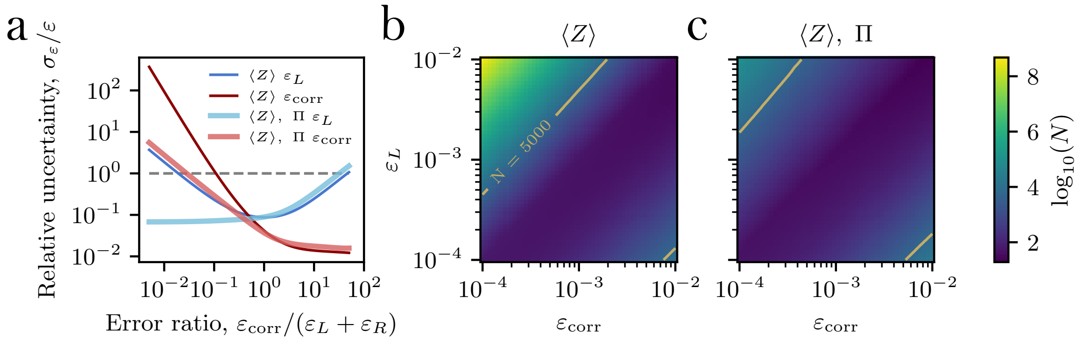



**Figure 5.6:** (a) The Cramér-Rao derived relative uncertainty for a range of correlated and uncorrelated error ratios. The thin lines correspond to expected uncertainties using $\langle Z\rangle$ measurements from *k-copy* RB, while thick lines use both $\langle Z\rangle$ and parity measurements, as described in section 5.4. (b-c) The required number of shots, $N$, per sequence length, $m$, for unity fractional uncertainty, for a range of correlated and uncorrelated errors. In (b), only $\langle Z\rangle$ measurements are used, while in (c) both $\langle Z\rangle$ and parity measurements are used. A contour is shown for 5000 samples, which is typically used in our experiments.

We now explore the ability for *k-copy* RB to fit $\theta$ for a range of underlying error magnitudes. Figure 5.6(a) shows the relative uncertainty as a function of the error ratio, defined here as $\varepsilon_{\rm corr}/(\varepsilon_L+\varepsilon_R)$. We fix $\varepsilon_L=\varepsilon_R=1\times10^{-3}$, and again set $N=5000$ samples per sequence length $m$, with $m_{\max}=\tau$, and vary $\varepsilon_{\rm corr}$ to vary the error ratio. The thin blue and red line show the expected relative uncertainty for $\varepsilon_L$ and $\varepsilon_{\rm corr}$ respectively. We can see that for the case that local noise dominates, $\varepsilon_L\gg\varepsilon_{\rm corr}$, the sufficient information for accurately estimating the correlated noise is not available. This limitation was found to be experimentally important, and motivates the injection of fixed *common noise* offsets in sections 5.6.4, and 5.6.5. Another strategy to alleviate this problem is by including data from the coherence probes described in section 5.4. The thick blue and red lines correspond to measurements where $N=2500$ samples are collected by the typical $\langle Z\rangle$ *k-copy* protocol, and $N=2500$ samples are using the parity protocol, denoted as $\Pi$. We find that inclusion of the parity samples dramatically reduce the relative uncertainty of the fitted parameters when $\varepsilon_L\gg\varepsilon_{\rm corr}$. Panels (b) and (c) show the required number of samples, $N$, for a relative uncertainty $\frac{\sigma_\varepsilon}{\varepsilon}=1$, for a range of local and *common* noise magnitudes. We highlight the $N=5000$ sample contour as this may be collected in reasonable time on our apparatus. Again, we see that including measurements using the parity protocol dramatically reduces the required number of samples in the regime of $\varepsilon_L \gg \varepsilon_{\mathrm{corr}}$ (top left corner of plots).

Here we have shown that the Fisher information gathered in an experimentally realistic *k-copy* protocol is sufficient to accurately determine the underlying error per Cliffords, $\theta = \{\varepsilon_L, \varepsilon_R, \varepsilon_{\mathrm{corr}}\}$, over many orders of magnitude of local and *common* errors. We shall show experimentally in section 5.6.4 that in the presence of experimental imperfections, and the fitting protocol (Levenberg-Marquadt), whose effects are not considered in this analysis, the proposed protocol still sufficiently recovers the magnitudes of the expected error per Cliffords with known noise injection.

## 5.6 Experimental Results

The results presented here constitute the first project-data output from the newly commissioned *FastGates* apparatus. This work was both necessary for confirming the utility of the *k-copy* RB protocol, as well as pushing the automation of the apparatus for robust operation, useful for all future experimental projects.

### 5.6.1 Physical implementation

We demonstrate *k-copy* RB using up to eight trapped $^{40}\mathrm{Ca}^+$ ions separated by an average spacing of $\sim 5$ µm. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ in a quantising field of $\sim 5$ G. Single-qubit rotations are driven by 729-nm light, with details given in section 4.2.4. The two-qubit coherence and Stark shift measurements use the individual addressing system, detailed in section 3.6.5, while all other measurements use the global elliptical beam, detailed in section 3.6.6.

The single-qubit Clifford gate set is decomposed into native $\pi/2$ single-qubit rotations of the form $\hat{R}_x(\pi/2)$ and $\hat{R}_z(\pi/2)$: $\hat{\sigma}_x$ rotations are performed natively on resonance with the qubit transition, with relative phase set by modulating the 729-nm beam

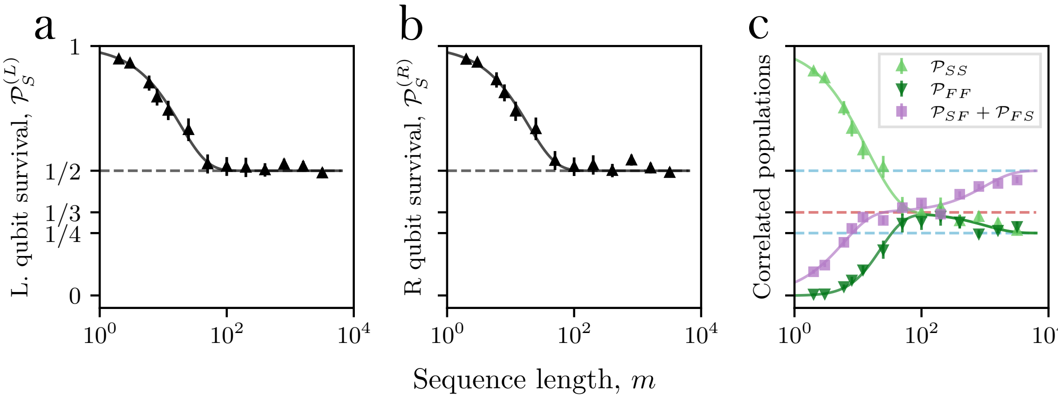



**Figure 5.7:** Correlated two-qubit probabilities after *k-copy* RB. Panels (a,b) show the single-qubit success probabilities, which decay with a single exponential (solid line) toward $\mathcal{P}_S|_{m\to\infty} = 1/2$. Panel (c) shows the correlated two-qubit populations, which exhibit an intermediate $\mathcal{P}_{SS} = 1/3$ asymptote before relaxing toward $\mathcal{P}_{SS}|_{m\to\infty} = 1/4$.

via an acousto-optic modulator (AOM), while $\hat{\sigma}_z$ rotations are performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations. Two decomposition strategies are used, crucially with varying average numbers of physical $\hat{\sigma}_x$ $\pi/2$-gates per Clifford. The first uses an average of $\kappa = 2.17$ $\pi/2$-gates, while the second uses $\kappa = 1.167$ gates.

We begin by reproducing the multi-exponential decays seen in simulation, figure 5.2, followed by parity measurements to observe the coherence of the intermediate symmetric state described in section 5.4. We then present data with known injected errors through phase and detuning offsets to show the ability for *k-copy* RB to discern correlated and uncorrelated errors for a range of magnitudes. We finish this section with evidence for scaling the protocol to more than two qubits. We show both data for three-qubit decay profiles, and pairwise analysis of an eight-ion chain with dominant correlated errors due to global drive field inhomogeneity.

### 5.6.2 $Z$-basis measurements

The *k-copy* RB protocol, with circuit diagram shown in figure 5.1, is executed on two qubits using a global field. To ensure a visually clear intermediate asymptote is seen, we set a fixed 10 kHz frequency detuning of the 729-nm laser compared to the bare qubit. As single qubits gates are calibrated for on-resonance pulses, this detuning will be a large *common noise* source. The Clifford sequence length is log-sampled up to $m = 3200$ total gates to capture both the intermediate asymptote due to this common unitary error, and the final decay to the fully depolarised state. The recovered multi-exponential decay from the correlated success and failure rates of the two qubits is shown in figure 5.7(c). Each solid point is inferred through $r = 50$ randomisations and $s = 50$ shots per randomisation. For each random sequence $j$, the population $\mathcal{P}_{SS,j}$ is estimated from the $s$ binary measurement outcomes, $P_{SS,j} = \frac{1}{s}\sum_{i=1}^{s} X_{ij}$, where $X_{ij} = 1$ if the $i$th shot is measured in state $SS$, and $X_{ij} = 0$ otherwise. The recovered population is then the mean over the $r$ random sequences, $\mathcal{P}_{SS} = \frac{1}{r}\sum_{j=1}^{r}\mathcal{P}_{SS,j}$. Error bars show the standard error of the mean across random sequences,

$$\mathrm{SE}(\mathcal{P}_{SS}) = \sqrt{\frac{1}{r^2}\sum_{j=1}^{r}\left(\mathcal{P}_{SS,j} - \mathcal{P}_{SS}\right)^2}. \tag{5.64}$$

The correlated populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|00\rangle$ approaching $\rho_{sym}$, equation **??**, with $\mathcal{P}_{SS} \to \frac{1}{3}$ asymptote highlighted by the red dashed line. The dashed blue line shows the fully depolarised $\mathcal{P}_{SS} \to \frac{1}{4}$ asymptote, seen clearly by $m \approx 1000$ gates. The solid lines are found through modelling errors by a combination of local depolarising errors and *common noise*, given by equation 5.36 generalised to include independent state-preparation and measurement error (SPAM), $\varepsilon_{sp}$ (see equation. 5.49 for example). The model is fit using the non-linear Levenberg–Marquardt method, yielding extracted parameters $\varepsilon_{sp} = 0(5) \times 10^{-3}$, $\varepsilon_{\mathrm{corr}} = 2.6(2) \times 10^{-2}$, $\varepsilon_L = 5(6) \times 10^{-4}$, and $\varepsilon_R = 0(6) \times 10^{-4}$, confirming that correlated errors dominate with this global field detuning. [3]

From the same shot data record we recover the single-qubit populations required for standard RB, as described in section **??**. To match the naming of the relevant participating irreps, we label the two qubits $L$ and $R$ for left and right respectively.

[3] The large relative uncertainty of both local depolarising rates is due to the large covariance between $\varepsilon_L$ and $\varepsilon_R$. The fitting may be constrained to the sum of local errors, to find $\varepsilon_L + \varepsilon_R = 5(1) \times 10^{-4}$. However, if precision measurements of this local error is of interest, either $r$, and $s$ could be increased, or the local error contributions could be isolated as discussed in section **??**.

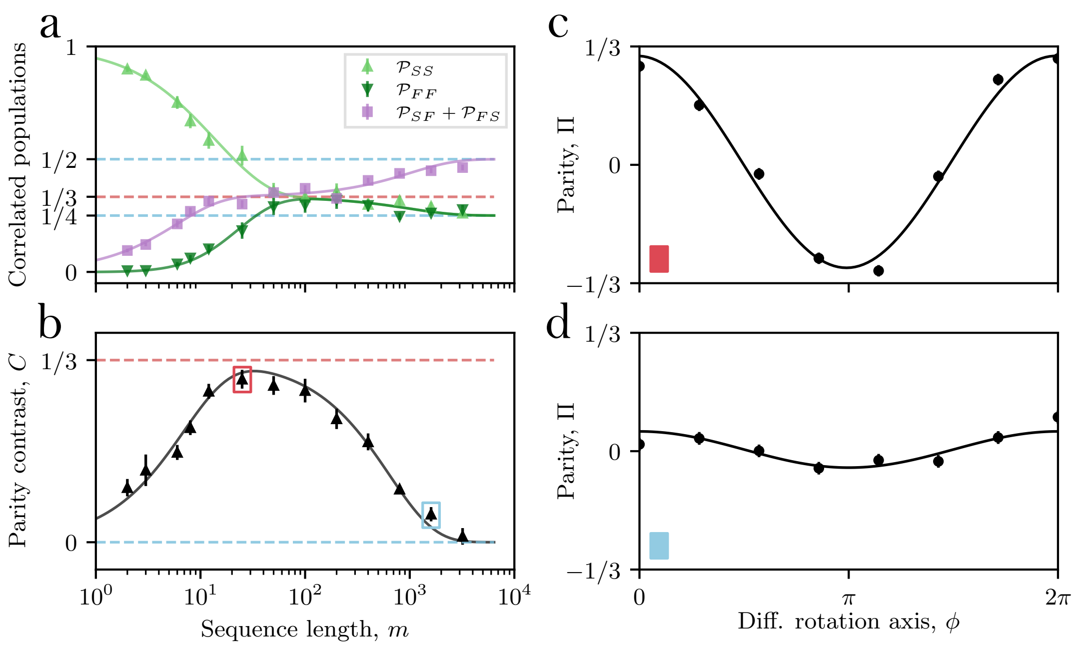



**Figure 5.8:** Parity contrast versus sequence length; (a) shows the standard $Z$ measured correlated populations. (b) shows parity contrast — a signal proportional to occupation of the symmetric-trace sector. Solid lines are multi-exponential decays given by equation 5.36 and equation 5.53 for (a) and (b) respectively. Dashed lines indicate the expected asymptotes. (c) and (d) show example parity fringes for sequences of $m = 25$ and $m = 1600$ Clifford gates respectively with solid lines given by equation 5.52.

The standard RB single exponential decay curves are shown in (a) and (b), with extracted single-qubit error per Cliffords of $\varepsilon_c^{(L)} = 2.6(2) \times 10^{-2}$ and $\varepsilon_c^{(R)} = 2.6(2) \times 10^{-2}$. As expected, *k-copy* RB provides evidence for this standard error per Clifford to be decomposed to local and correlated components, given by equation 5.42.

### 5.6.3 Coherence measurements

A partial-tomography parity measurement, given by the circuit seen in figure 5.4(a), is used to independently confirm occupation of the symmetric sector, quantifying overlap of the final state with Bell state, $|\Psi^+\rangle$. As the final differential rotation pulse requires local addressing of the ions, the single addressing system is used. Each random sequence is repeated for varying final rotation angle, $\phi$, to produce parity fringes, examples shown in (c) and (d) in figure 5.8 for $m = 25$ and $m = 1600$ Clifford gates respectively. The contrast is extracted by equation 5.53, and a plot against

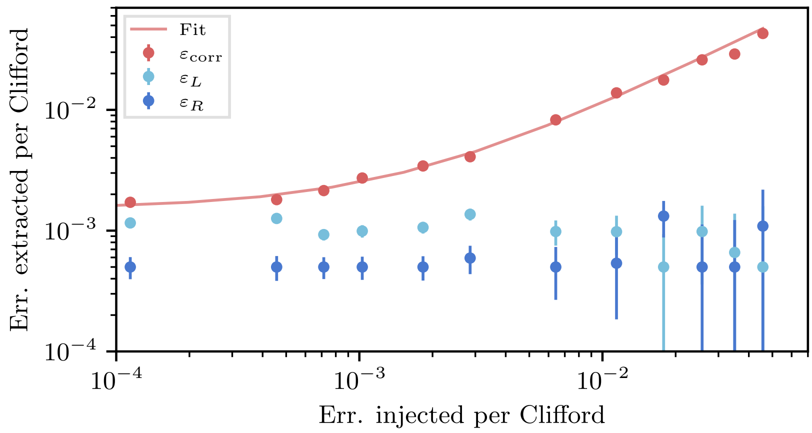



**Figure 5.9:** Injecting correlated noise between two qubits via common drive phase noise. Points correspond to fitted error per Clifford from full *k-copy* RB experiments. The solid red line is a linear fit between measured and injected error.

sequence length is shown in (b). The contrast asymptote $C \to 1/3$ is observed for $m \in [10, 100]$, corresponding to the residual coherence of $\rho_{sym}$, equation 5.44. The parity contrast data is fit (solid line) according to equation 5.53 to extract error per Cliffords of, $\varepsilon_{\text{corr}} = 2.1(2) \times 10^{-2}$ and $\varepsilon_L = \varepsilon_R = 3.9(3) \times 10^{-4}$.

### 5.6.4 Controlled noise injection

To verify the ability for *k-copy* RB to discern correlated and uncorrelated errors, we compare the magnitude of extracted errors to known injected errors. We limit the injected error method to those consisting of noise on common drive fields, and local depolarising type noise. Both of these noise models can be induced entirely through single qubit rotations, which are trivial to implement on our device.

#### Laser phase jitter

We first inject correlated errors via phase noise on a common drive field. This noise mechanism is prevalent in many platforms that use a common oscillator, such as laser, microwave, or radio-frequency (RF) fields to resonantly drive qubit transitions.

The injected error magnitude is inferred through considering the ideal and noisy unitary channel. A unitary qubit rotation is given by,

$$\hat{U} = e^{-i\frac{\theta}{2}\vec{n}\cdot\vec{\sigma}}, \tag{5.65}$$

where $\theta$ is the rotation angle, and $\vec{n}\cdot\vec{\sigma}$ is the rotation axis. We decompose the single-qubit Clifford gate set into a series of resonant $\theta = \pi/2$ qubit rotations around axes $\hat{\sigma}_x$ and $\hat{\sigma}_z$. Practically, phase noise on either the qubit frequency or drive field will introduce some deviation of this rotation axis. We shall consider the effect of zero-mean phase noise on $\hat{\sigma}_x$ rotations, leading to an erroneous rotations axis,

$$\hat{\sigma}_{\delta\phi} = \cos(\delta\phi)\hat{\sigma}_x + \sin(\delta\phi)\hat{\sigma}_y, \tag{5.66}$$

where $\delta\phi$ is the standard deviation of the noise in radians. We may find the expected error of a single Clifford gate using,

$$\varepsilon_{\delta\phi} = \kappa\left(1 - \mathcal{F}(V,U)\right), \tag{5.67}$$

where $\kappa$ is the average number of $\pi/2$ rotations in a Clifford gate, and $\mathcal{F}(V,U)$ is the average gate fidelity. The fidelity of two unitaries averaged over the Haar measure is given in appendix equation **??**, reproduced here,

$$\mathcal{F}(V,U) = \frac{1}{d(d+1)}\left(|\mathrm{Tr}(U^\dagger V)|^2 + d\right). \tag{??}$$

Considering $\theta = \pi/2$ rotations, we find,

$$\hat{U} = e^{-i\frac{\pi}{4}\hat{\sigma}_x} = \frac{1}{\sqrt{2}}\left(\hat{I} - i\hat{\sigma}_x\right), \tag{5.68}$$

$$\hat{V} = e^{-i\frac{\pi}{4}\hat{\sigma}_{\delta\phi}} = \frac{1}{\sqrt{2}}\left(\hat{I} - i\hat{\sigma}_{\delta\phi}\right). \tag{5.69}$$

The fidelity, given that dimension $d = 2$ for a single qubit, is found to be,

$$\mathcal{F}(V,U) = \frac{1}{6}\left((1+\cos(\delta\phi))^2 + 2\right). \tag{5.70}$$

For small phase noise standard deviations, $\delta\phi \ll 1$, we find an expected error per Clifford of,

$$\varepsilon_{\delta\phi} = \frac{\kappa}{3}(\delta\phi)^2. \tag{5.71}$$

A known phase error may be conveniently injected through phase offsets sampled from a normal distribution with zero mean and standard deviation $= \delta\phi$ applied to the RF drive of a control AOM on the 729-nm global addressing path. The phase of the AOM RF is imprinted directly onto the 729-nm light leading to the expected drop in fidelity of the Clifford gates. In this experiment $\kappa = 2.17\ \pi/2$ rotations. A full *k-copy* RB experiment is executed with logarithmic sampling up to $m = 512$ Clifford gates with $r = 50$ randomisations and $s = 50$ repeats. The resulting multi-exponential success and failure probability decay curves are fit to equation 5.36, again generalised to include state-preparation errors. The correlated and local error components are found for a range of injected phase noise standard deviations. The resulting measured errors are shown in figure 5.9. The $x$-axis is the expected error injected $\varepsilon_{\delta\phi}$, while the $y$-axis is the extracted error magnitudes. The fit, shown by the solid red line, is a direct proportionality of injected error plus a baseline correlated error rate not due to phase injection,

$$\varepsilon_{\text{corr}} = \varepsilon_{\text{corr},0} + \varepsilon_{\delta\phi}. \tag{5.72}$$

The baseline correlated error is therefore the only parameter to fit. With $\varepsilon_{\text{corr},0} = 1.71(4) \times 10^{-3}$, we find good agreement between the model and data. The local depolarising error on both qubits remains relatively constant, with the left qubit having a larger local error compared to the right qubit. We believe this asymmetry in local error is due to the global 729-nm beam not being centrally aligned to both qubits. The uncertainty in extracted error is found through Levenberg-Marquadt non-linear fit, it can be seen that the uncertainty in the depolarising errors increases substantially with the total correlated error. This may be explained by the available Fisher information gained by the chosen logarithmic sampling strategy, as explored in section 5.5. Figure 5.6 (a) supports that we expect an increase in depolarising parameter uncertainty for $\varepsilon_{\text{corr}} \gg (\varepsilon_L + \varepsilon_R)$.

**Laser detuning scan**

*k-copy* RB can isolate errors arising from shared control fields, making it useful both for device characterisation and for optimising global control parameters. A fixed detuning of the global drive, under the action of *k-copy* Clifford twirling, is well modelled by the common random unitary error channel described in section 5.3.3. We demonstrate this by scanning the 729-nm laser frequency used for single-qubit rotations away from the bare qubit resonance. As with the previous section, we consider the fidelity of the error channel compared to the ideal $\pi/2$ $\hat{\sigma}_x$ single qubit rotation.

A coherent detuning error is written as,

$$V = \exp\left(-i\frac{\pi}{4}\left(\hat{\sigma}_x + \frac{\delta}{\Omega}\hat{\sigma}_z\right)\right), \tag{5.73}$$

where $\delta$ is the detuning from resonance, and $\Omega$ is the Rabi frequency. Assuming $\delta/\Omega \ll 1$, the fidelity is found to be,

$$\mathcal{F}(V, U) = 1 - \frac{1}{3}\left(\frac{\delta}{\Omega}\right)^2 + \mathcal{O}\left(\frac{\delta}{\Omega}\right)^4. \tag{5.74}$$

To convert this to an error contribution to a single qubit Clifford gate, we introduce the scalar $\kappa_\delta$, to account for both the number of $\pi/2$ rotation in each Clifford gate, but also the duration and delays between pulses where phase tracking between qubit and laser also drift. The error contribution is then given by

$$\varepsilon_\delta = \kappa_\delta(1 - \mathcal{F}(V, U)) = \frac{\kappa_\delta}{3}\left(\frac{\delta}{\Omega}\right)^2 + \mathcal{O}\left(\frac{\delta}{\Omega}\right)^4. \tag{5.75}$$

Each data point is obtained from a complete *k-copy* RB experiment with 11 sequence lengths up to $m = 1024$, using $r = 50$ random sequences and $s = 50$ shots per sequence, with correlated populations fit to equation 5.36 to extract $\{\varepsilon_{\text{corr}}, \varepsilon_L, \varepsilon_R\}$ at each detuning. The upper panels of figure 5.10 show example datasets for the *k-copy* RB sequences, while the lower panel shows the extracted error per Cliffords as a function of detuning. The correlated error is observed to increase quadratically with detuning, while the local depolarising errors remain approximately constant, consistent with detuning of a shared drive producing *common noise*. To

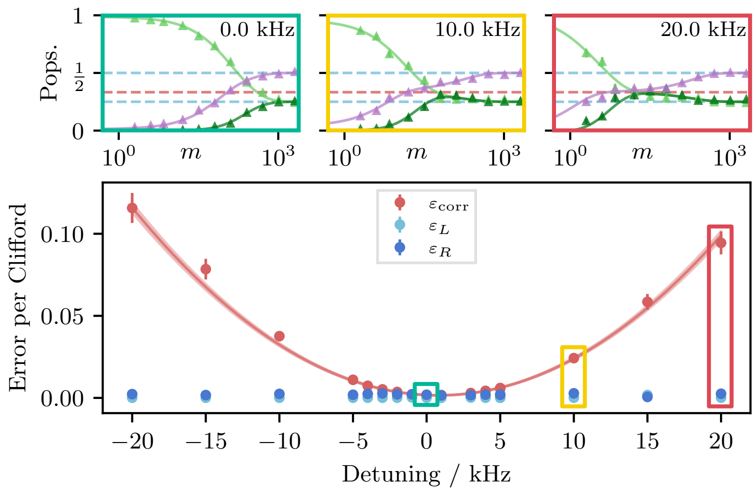



**Figure 5.10:** Correlated and single-qubit depolarising errors extracted as a function of shared-drive detuning. The correlated error (red points) rises quadratically with laser detuning, while the local depolarising errors (blue points) remain approximately constant. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. The red shaded region signifies the 68% confidence, generated through covariance-based parameter sampling.

fully parameterise the injected error, we include a background correlated error not originating from detuning, $\varepsilon_0$, and a shift of the light-shifted qubit versus the bare qubit frequency, $\delta_0$. The solid red line is therefore fit to the model,

$$\varepsilon_{corr} = \varepsilon_0 + \frac{\kappa_\delta}{3}\left(\frac{\delta_L - \delta_0}{\Omega}\right)^2. \tag{5.76}$$

Here, $\Omega = 2\pi \times 100(1)\,\text{kHz}$ is the independently measured Rabi frequency. Fitting gives $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $\kappa_\delta = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)\text{kHz}$.

### Single addressed Stark shifts

Uncorrelated noise injection requires the introduction of noise with spatial inhomogeneity. This is implemented experimentally through an off-resonant single addressing beam inducing Stark-shifts between the on-resonant parallel single qubit rotations executing the *k-copy* twirling. The single addressing beam is detuned from the qubit resonance by −15 MHz, and the induced Stark shift with beam power, $\gamma$, is calibrated using an RPE sequence (see section 4.2.5), with results shown in figure 5.11 (c) for both the left and right qubit. From the frequency shift versus

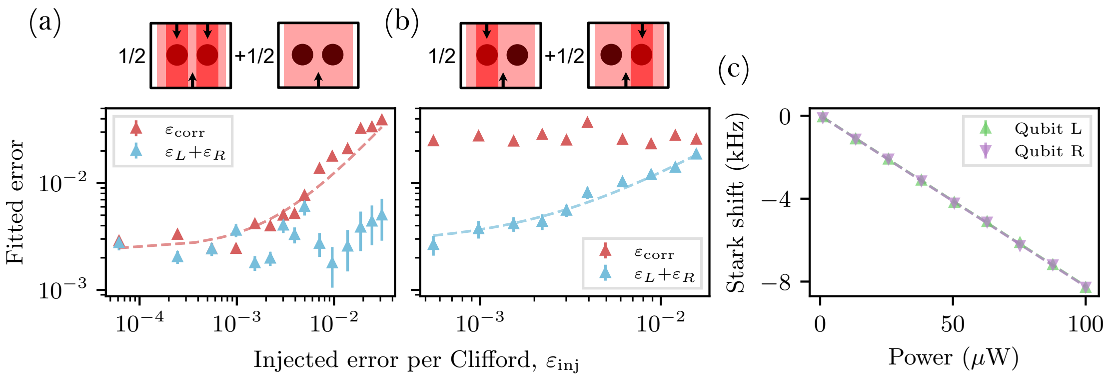



**Figure 5.11:** Correlated and uncorrelated errors are injected via a single-addressed Stark-shift beam, and quantified through fitting the correlated success populations. The four addressing configurations are depicted visually above the plot. The wide beam signifies the on-resonance global drive which executes the parallel single-qubit rotations, while the narrow beams represent the off-resonant Stark shift beams addressing single-qubits (black circles). In (a) the errors are correlated between qubits, while panel (b) shows the extracted errors after independent errors are injected. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. Panel (c) shows the calibration of Stark shift against total beam power used to infer the injected error.

power, we extract $\gamma_L = 8.24(5) \times 10^7 \mathrm{Hz\,W^{-1}}$, $\gamma_R = 8.25(4) \times 10^7 \mathrm{Hz\,W^{-1}}$ for the left and right qubit respectively. The injected error, $\varepsilon_{\mathrm{inj}}$, is then inferred by,

$$\varepsilon_{\mathrm{inj}} = \frac{\kappa}{3}\phi^2, \tag{5.77}$$

where the phase error, given in radians is, $\phi = P\tau\gamma$, with $P$ being the power in the single addressing beam, and $\tau$ the duration of the Stark-shift pulse. In these experiments, the duration of the Stark-shift pulse is fixed to $\tau = 20$ µs, while the total power of the beam is varied to supply a known error.

Two experimental sequences were run, the first, shown in figure 5.11(a), implements stochastic *common noise* through applying either two spots shifting both ions, or no Stark shift, with a 50 : 50 chance after each Clifford gate. The second experiment, shown in (b), applies local errors through a single spot either addressing the left or right qubit, again with 50 : 50 chance after each Clifford gate. These two arrangements are shown schematically in the upper panels. To ensure correlated errors dominate even with injected local errors, an additional constant correlated error offset of $\varepsilon_{corr,0} \approx 2 \times 10^{-2}$ is applied through random phase fluctuations,

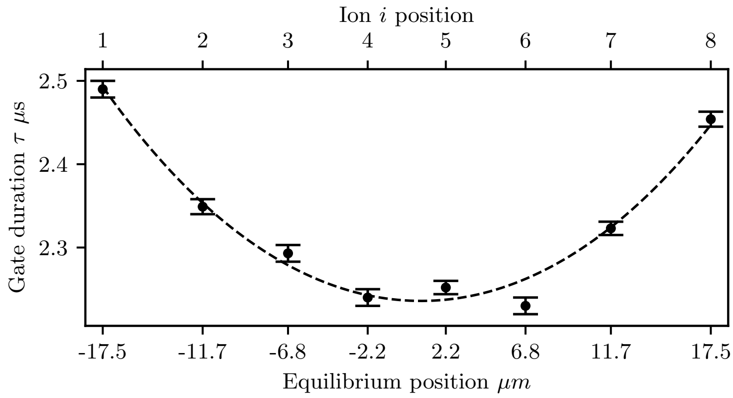



**Figure 5.12:** Variation of $\pi$-pulse gate duration along eight ion chain due to the Gaussian waist of the global addressing beam. Ion labels, $i$, are given on the upper axis, and equilibrium position given on the lower.

as described in section 5.6.4. This reduces the number of samples required to accurately estimate the correlated and uncorrelated error contributions, which is of practical importance in collecting data in reasonable time (for reference, this data was collected over $\sim 3$ hours). It can be seen that $\varepsilon_{\text{corr}}$ increases only when injecting *common noise*, and remains constant when local errors are injected. Dashed lines are simple linear fits given by $\varepsilon_{\text{fit}} = \varepsilon_0 + \varepsilon_{\text{inj}}$, where $\varepsilon_{\text{fit}}$ are the extracted errors, $\varepsilon_0$ is some residual error baseline. These straight line fits show good agreement between measured and injected error when accounting for some fit baseline error rates.

### 5.6.5 Scaling to a multi-ion register

Pairwise *k-copy* RB extends simply to $> 2$ qubits: correlated populations are fit to equation 5.36 independently for each qubit pair. For a register of $k$ qubits, all $k(k-1)/2$ pairwise outcomes are extracted from the same shot record — no additional trials on the quantum register are required, although classical post-processing scales with the number of pair combinations. The protocol itself requires only synchronised application of identical single-qubit Clifford sequences and spatially resolved readout.

This scaling is demonstrated on an eight-ion chain, with parallel single-qubit rotations driven by an elliptical 729-nm beam. We expect to observe structured

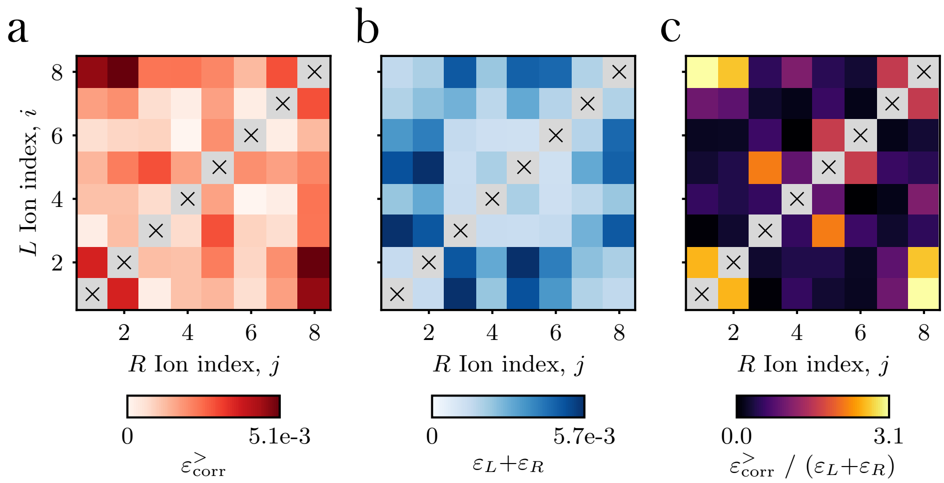



**Figure 5.13:** Correlated and local error per Cliffords measured for all qubit pairs after applying *k-copy* RB to an 8 ion chain. Noise is dominated by position dependent duration of $R_x(\pi/2)$ gates consistent with inhomogeneous illumination by the global drive. Panel (a) displays the extracted excess correlated errors, $\varepsilon^{(i,j)}_{\mathrm{corr}}$ for each qubit pair, $(i, j)$. Panel (b) displays the sum of both independent depolarising errors, $(\varepsilon_L + \varepsilon_R)^{(i,j)}$. Panel (c) displays the ratio of excess correlated noise to the sum of local errors given in (b). This plot highlights the outer qubits are dominated by spatially correlated errors.

errors due to the Gaussian intensity profile of the beam. To verify the expected gate duration dependence due to the finite beam, we run parallel gate duration calibration experiments (see section 4.2.4 for details). The gate duration, $\tau$, as a function of qubit position in the beam is expected to follow,

$$\tau(r) \propto \frac{1}{\sqrt{I(r)}}, \tag{5.78}$$

where $I(r)$ is a Gaussian distribution, $I = a\exp\left(\frac{-(r-r_0)^2}{2c^2}\right)$. Figure 5.12 shows the measured $\pi$-pulse gate durations for each ion, with the fit given by the dashed line. From this, we find an expected major-axis waist radius of 27.7(9)µm aligned to the ion chain. The qubit spacing, on the lower axis of the calibration plot, is inferred through the axial confinement of 0.5 MHz using equilibrium positions of a linear chain given in [85]. At this confinement, the chain extent is 35 µm, giving an expected Rabi-frequency difference of 10.3(6)% between the outer and inner ions. We calibrate global single-qubit gates using the inner ion, $i = 4$, leading to an expected per Clifford error of (using fidelity definition from appendix ??),

$$\varepsilon_{c,max} = \frac{2}{3}\kappa\left(\alpha\frac{\pi}{4}\right)^2 = 5.08(2)\times 10^{-3}, \tag{5.79}$$

where, here the Clifford decomposition used gives $\kappa = 1.167$, and the fractional Rabi frequency difference is given by the above calibration curve, $\alpha = 0.103(6)$.

As with the single qubit Stark-shift dataset, we ensure that correlated errors dominate to improve the fitting sensitivity. This is done through a fixed phase noise error corresponding to $\varepsilon_{corr,0} = 1.3\times 10^{-2}$. In the following data, we subtract this baseline offset to highlight the excess correlated noise due to the beam structure, $\varepsilon^{>}_{corr} = \varepsilon_{corr} - \varepsilon_{corr,0}$.

Figure 5.13 shows the extracted pairwise correlations. The excess correlated error is shown in (a), with the largest values seen for the outer ion pairs, agreeing with the expected maximum error of $\varepsilon^{>}_{corr,max} = \varepsilon_{c,max}$. Panel (b) shows the sum of the local depolarisation error per Cliffords on both qubits, with the largest values now corresponding to pairs consisting of one central ion, and one outer ion. We plot the ratio of excess correlated error and depolarisation errors in panel (c) to further highlight the error distribution due to the beam shape.

## 5.7 Outlook

**Distill what learnt.** *k-copy* RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. The symmetry enforced by twirling over the diagonal single-qubit Clifford representation $Cl_1^{\otimes k}$ (section 5.3.2), causes structured errors to drive the decay of correlated populations into invariant subspaces. Observing transient populations in these invariant subspaces allows for error characterisation without full process reconstruction. Given a specified error model, here focussing on common unitary errors due to fluctuating global fields, these correlated decays can be used to directly quantify the magnitude of the individual error channels contributing to the total error per Clifford that is typically measured in standard RB.

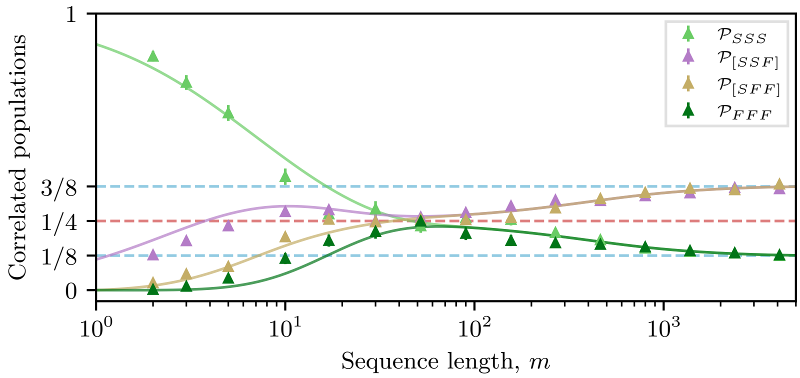



**Figure 5.14:** Experimental data and analytic fit for 3-qubits subject to correlated random-unitary single-qubit noise injected via random phase jitter ($\sigma_\phi = 0.06$ turns). $\mathcal{P}_{[SSF]}$ and $\mathcal{P}_{[SFF]}$ represent the sum of the permutations of single- and two-qubit failures respectively. Explicitly, $\mathcal{P}_{[SSF]} = \mathcal{P}_{SSF} + \mathcal{P}_{SFS} + \mathcal{P}_{FSS}$.

In this chapter we proposed and experimentally demonstrated this method applied on on up to an eight-qubit register through pairwise comparisons (section 5.6.5). This approach permits highly efficient parallel data collection: the time required on the device is independent of qubit number, and the protocol requires exclusively local operations followed by classical post-processing of correlated survival rates.

**reiterate relevance.**

### 5.7.1 Limitations

**add parameter regime that is hard to fit. add the need for explicit channels?** This chapter has analysed structured errors on two qubits, extending to more qubits only through pairwise comparisons. The current pairwise protocol described here does not determine correlated errors between $>$ 2-qubits. These higher weight errors may be particularly detrimental to error correction circuits where the distance of the code, $d$, is less than the correlated error channel weight [**<empty citation>**]**citation needed**, therefore these errors would be advantageous to characterise. We expect the symmetry resolved dynamics revealed by pairwise *k-copy* RB to generalise to higher weight errors, however, without prior work on the sub-representation structure of the single-qubit Clifford group applied diagonally to $> 2$ qubits, extending the analysis for *k-copy* RB is challenging. The number of relevant irreps does not increase, however, the multiplicity does increase dramatically. Some headway can be made if we consider the *common noise* error mode discussed throughout this chapter. A simple relation can be found for the asymptotic states, namely, under *common noise* the state is driven to the maximally mixed state on the symmetric subspace, as this is the SU(2)-invariant state on the symmetric irrep. This state is given by,

$$\rho_{\text{sym}} = \frac{1}{k+1} \sum_{l=0}^{k} |D_l^k\rangle \langle D_l^k| \,, \tag{5.80}$$

where $|D_l^k\rangle$ is the Dicke state for $l$ excitations (here successful recoveries of target state), out of $k$-qubits. From this expression we see that an asymptote of $\mathcal{P}_{SS\ldots S}|_{m\to\infty} = 1/(k+1)$ is expected under *common noise*, while the maximally mixed state under local depolarisation leads to the familiar asymptote of $\mathcal{P}_{SS\ldots S}|_{m\to\infty} = 1/(2^k)$. This result suggests that the presence of *common noise* affecting many qubits may be simply observed, however, without the explicit representation structure, it cannot be easily quantified.

We confirm this asymptote experimentally with data shown in figure 5.14. For three-qubits, we leverage symbolic computation to diagonalise the resulting PTM representation of *common noise* and recover the analytic decay used in this fit. This direct diagonalisation approach is possible for this small system, and with the highly structured *common noise* error channel, however, for a full treatment of $k$-body error under $Cl_1^{\otimes k}$ twirling, we suspect the sub-representation structure will need to be found.

### 5.7.2 Future work

One avenue for future work is captured by the above limitations section: scaling the analysis to multi-body errors through studying the sub-representation structure of the single qubit Clifford group applied diagonally to $k$-qubits. We propose two further dimensions in which to extend the *k-copy* analysis:

First, the subsystem on which the local twirling is applied could be extended to

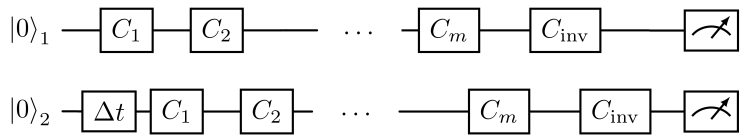



**Figure 5.15:** Circuit for probing time-dependent environmental errors by introducing a delay between random sequence execution on parallel copies in *k-copy* RB.

$>$ 1-qubit. As an example, the two-qubit Clifford group [86] or the SLeRB [73] protocol could be applied in parallel to $k$-copies of 2-qubit subsystems. This would reveal correlated errors and leakage that reduce the fidelity of two-qubit entangling gates.

Second, we may utilise this method to probe correlations in time. As the protocol uses exclusively single-qubit rotations, with no explicit coupling made between qubits by the ideal circuit, we could apply some time delay, $\Delta t$, between the parallel gate sequences. A proposed circuit is shown in figure 5.15 for the two-copy case. In the case that $\Delta t$ is much longer than the random sequence length, we are effectively probing the stability of the circuit over time (albeit on parallel subsystems) — specifically we are looking for deviations in circuit fidelity due to the external context in which the circuit is executed. This method is similar to the context dependent tests studied in reference [69]. However, as we execute parallel sequences on two copies, we can also set $\Delta t$ to be significantly shorter than the random sequence length. In this regime, we could, for example, probe the frequency and coherence of noise on the common drive field used to execute the parallel gates. This would provide evidence and quantify violations of the Markovian QIP model, without full process tomography, as discussed in section 5.1.

# 6

# Towards Hybrid System Control and Quantum Error Mitigation

## Contents



**Keep in mind: Want to make clear that this is future work, and not specifically QEM on CV!** Suggested rough intro: Having established and characterised the experimental platform and demonstrated its application to *k-copy* randomised benchmarking, this chapter considers future research goals utilising the multi-qubit array, and control over both the discrete qubits subsystems, as well as the continuous bosonic motional modes. In broad terms, we are interested in including the motional modes as computational resources, following recent progress in spin-conditioned bosonic operations demonstrated within our group [**bazavan_thesis**, **saner_thesis**, **shinbrough_applications_2026**, 4, 5]. In section 6.1 we estabilish a gateset required for continuous-variable control of the ion motion, and demonstrate the generation of a squeezed state on the new apparatus.

The second half of this chapter considers an error mitigation technique applicable to both discrete and continuous systems. Simulation and computation utilising both discrete and continuous resources has been proposed to offer advantages over fully discrete systems due to resource scaling... [**<empty citation>**]. Demonstrations include [**feynman**, 7]. However, to extract faithful results, we must reduce the impact of errors. We explore in section 6.2 [**virtual_cooling**, **balint**, **huggins**] the error mitigation technique virtual distillation, which has applications on both discrete [**google**, 87] and continuous systems [**<empty citation>**]**virtual cooling**[1]. This section begins with the theory behind virtual distillation, and two protocols for its application. We then provide initial demonstrations of this protocol applied to a discrete-variable only system, and explore the performance under both local depolarising and under the correlated noise due to a common fluctuating control field, explored in detail in chapter 5. We end this chapter with an outlook of how VD will be applied to larger system sizes, including the ultimate goal of application to hybrid systems.

## 6.1 Continuous-Variable Control in Trapped Ions

**Short summary of section** One paragraph on why this apparatus is well placed for this work. **this section develops the CV gateset that is a prerequisite for hybrid QEM, without yet performing error mitigation on it**

### 6.1.1 Motivation

**One paragraphs on what is CV, and why it is interesting. One paragraph on its conception in ITQC. One paragraph on why error mitigation in this framework is important.**

[1] In fact, the intial proposal of virtual cooling was on a continuous system

| Gate | $H_{\text{osc}}$ | Gaussian | modes num. | $\sigma$ | $m$ |
|---|---|---|---|---|---|
| Displacement | $a+a^\dagger$ | yes | 1 | $\sigma_x$ | 0 |
| Squeezing | $a+a^\dagger$ | yes | 1 | $\sigma_z$ | 1 |
| two-mode squeezing | $a+a^\dagger$ | yes | 2 | $\sigma_z$ | 1 |
| dispersive | $a+a^\dagger$ | yes | 1 | $\sigma_z$ | 1 |
| beam splitter | $a+a^\dagger$ | yes | 2 | $\sigma_z$ | 1 |
| 3rd order squeezing | $a+a^\dagger$ | no | 1 | $\sigma_x$ | 2 |

**Table 6.1:** **Gateset, need to check factors and phases for all gates. Check if Hermitian!**

### 6.1.2 Theory

#### Computing on a Hybrid System

How information is stored in oscillator and spin, and retrieved via spin?

Summary from a couple of reviews (Liu?)

#### Gate Implementation

Two-SDF toolkit from TS+RS

Could also mention SW and higher order S.B. approaches.

Magnus expansion

**point to gateset table of how to generate interaction using TS+RS method**

#### Readout

Define state for CV.

Define characteristic and Wigner function.

$$\mathcal{P}_a = \frac{1 + Re(\xi(\beta))}{2}, \tag{6.1}$$

$$\mathcal{P}_a = \frac{1 + Im(\xi(\beta))}{2}, \tag{6.2}$$

### 6.1.3 Experimental Demonstration

**Short description of what we will show**

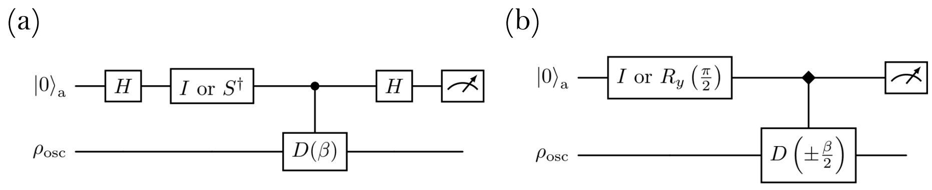



**Figure 6.1:** **Figure: circuit diagram for tomography.**

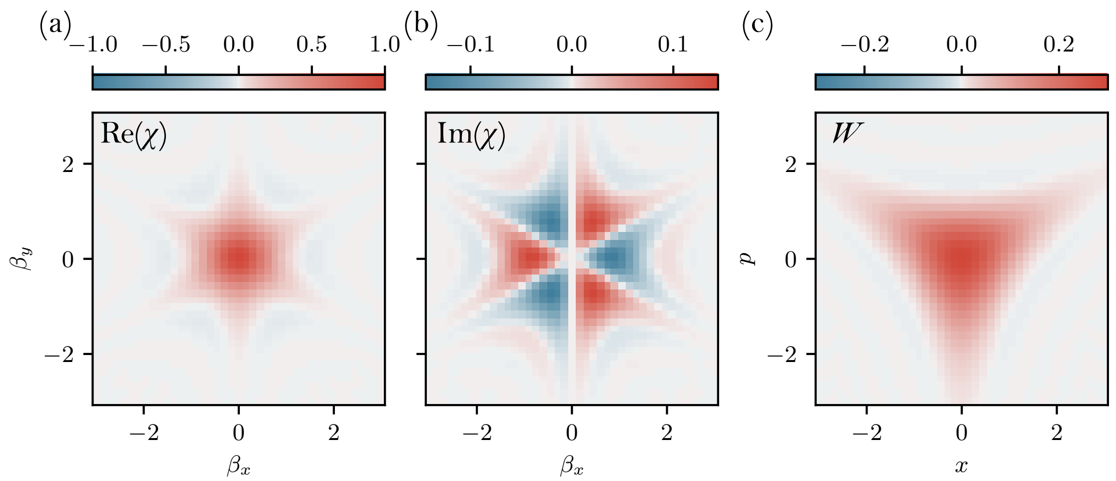



**Figure 6.2:** **Figure: characteristic function of tri-sqz state Fock state. TODO: verify axis convention with GC**

### Two Spin Dependent Forces

Experimental sequence, hardware involved, thermal drifts

Four tones on one AOM.

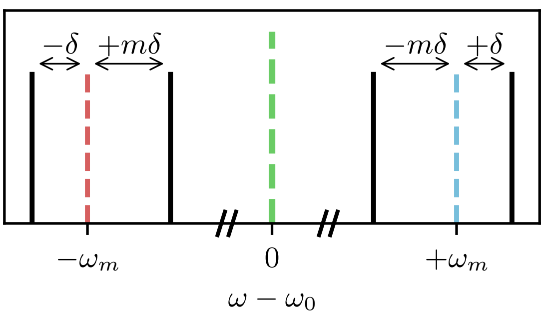



**Figure 6.3:** **Figure: detuning for four tones, Z measurement with ramp**

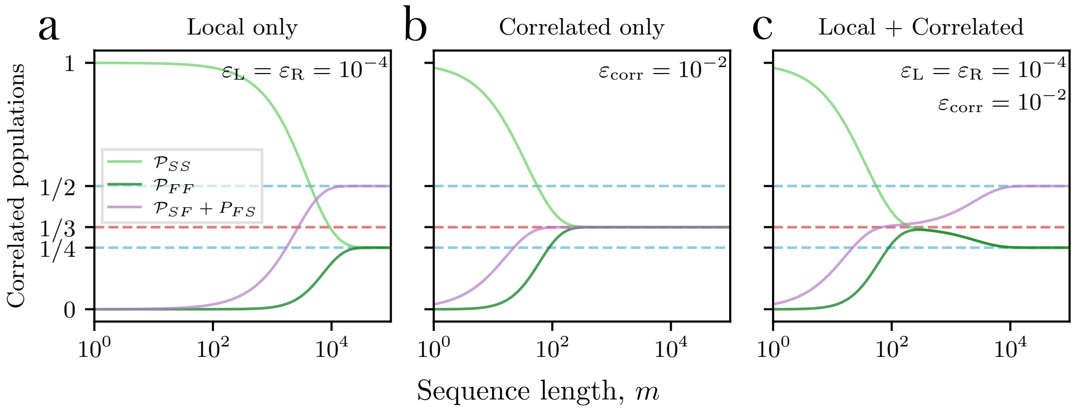



**Figure 6.4:** **Figure: Sqz characteristic function with varying duration**

**Squeezing**

Experimental sequence, dephasing dominated, improved by grounding

### 6.1.4 Outlook

Some next steps for demonstrating full CV control

## 6.2 Virtual Distillation

**Short summary of section** Here we develop experimental tools for demonstrating VD on the discrete variable portion of the full hybrid system.

### 6.2.1 Error Mitigation

Noise coupled to a quantum system results in unintended coherent, and incoherent evolution. Deviations of the realised state from the target state are known as errors, and, if left unchecked, lead to the failure of quantum algorithms. This failure can either be due to a reduction of signal to noise ratio, or a bias in the measured expectation value, resulting in misleading conclusions. Ultimately, the goal of quantum error correction, QEC, is to detect and correct these errors through embedding information in a larger Hilbert space [**qec_review**]. Full

QEC will allow for the execution of arbitrarily long algorithms ensuring errors (of up to a certain distance) do not accumulate over time. However, current QEC methods require large overheads in both qubit number and circuit depth, and are therefore not feasible on current quantum hardware. Furthermore, QEC is formulated for digital "gate-based" quantum computers, which although promise to be universal, pay for that privilege in often inefficient decompositions of target unitaries into a discrete set of gates. This poses an opportunity for near term devices to show quantum advantage using either short depth, uncorrected circuits, or analog quantum simulators, which although not universal, can provide meaningful results for specific problems [**nisq_review**]. Particularly, phrasing the problem in terms of expectation values of an observable on a physical final state, rather than a sequence of measurements on a set of logical state, allows for a more efficient use of resources. As a tactile example, consider the problem of finding the ground state energy of a Hamiltonian. This can be done by preparing a trial state, and directly measuring the expectation value of the Hamiltonian on that state, building up statistics over many samples [**variational_qc_review**]. Increasing the number of samples will reduce the statistical error, however, if the trial state is not prepared perfectly due to either environmental noise, or to imperfect application of unitary evolutions, the measured expectation value will be biased. To this end, quantum error mitigation (QEM) techniques have been developed to reduce the bias of measured expectation value of an observable at the expense of requiring more samples. Reference [41] defines QEM as "... algorithmic schemes that reduce the noise-induced bias in the expectation value by postprocessing outputs from an ensemble of circuit runs, using circuits at the same noise level as the original unmitigated circuit or above." This highlights the crux of how information on the system is collected by a QEM implementation, which I shall explore with the following thought experiment on the "perfect" QEM algorithm. Consider the problem of measuring the expectation value of an observable $\hat{O}$ on the target state $\rho_{id}$. The state is prepared by a quantum circuit, which due to noise, instead prepares the mixed state $\rho = (1-\lambda)\rho_{id} + \lambda\rho_{err}$, where $\rho_{err}$ is the error state. Each measurement of the observable $\hat{O}$ on the noisy

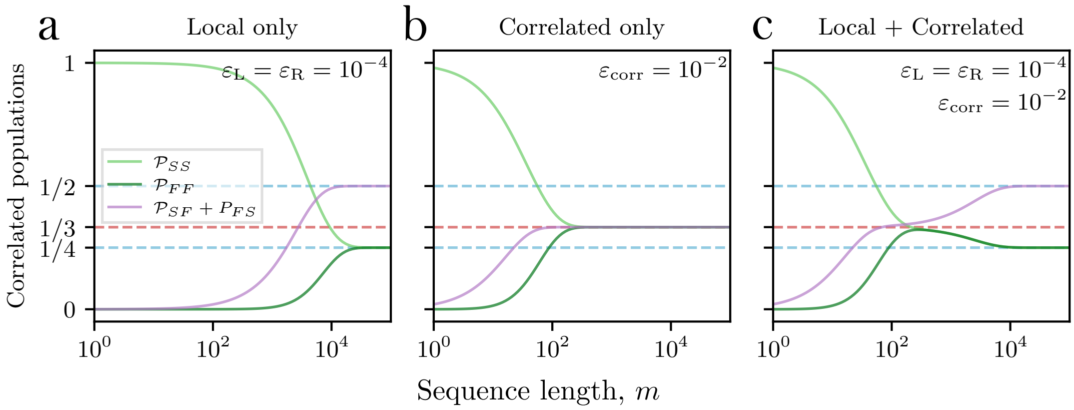



**Figure 6.5:** **Figure: unmitigated vs mitigated expectation value, showing how bias is reduced at the expense of more samples.**

state $\rho$ will return a value $o_i$, and the average of these measurements will converge to the expectation value $\langle \hat{O} \rangle = \mathrm{Tr}[\hat{O}\rho]$. Given $S$ total samples, the final result will be,

$$\langle \hat{O} \rangle = \frac{1}{S}\left(\sum_{i=1}^{S_{id}} o_i + \sum_{j=1}^{S_{err}} o_j\right),$$

where $S_{id}$ and $S_{err}$ are the number of samples that return a measurement outcome from the ideal and error states respectively, with $S = S_{id} + S_{err}$. The bias in the measured expectation value from the ideal is then given by contributions from the samples $S_{err}$. An ideal QEM algorithm would flag these error samples, and remove them from the final average, resulting in a bias free expectation value using the restricted samples $S_{id}$.

- Error mitigation motivation. Mainly on contrasting to quantum error correction.
- QEM is framed as finding expectation value of observable, rather than maintaining coherence for a long computation.
- How QEM reduces bias (at the expense of required samples). Look at Cai paper for structured argument.

### 6.2.2 Virtual Distillation Protocol

#### Purification

Estimate expectation value of observable on a state without ever physically preparing the state. Purification here refers to approaching the dominant pure state of a mixed state. The $k$th degree purified state is given by,

$$\rho^{(k)}_{\text{corrected}} = \frac{\rho^k}{\text{Tr}[\rho^k]} = \sum_{i=1}^{d} \frac{\lambda_i}{\sum_{i=1}^{d} \lambda_i^k} |\psi_i\rangle \langle\psi_i| , \tag{6.3}$$

where $d$ is the dimension of the Hilbert space, $\lambda_i$ are the eigenvalues of $\rho$, and $|\psi_i\rangle$ are the corresponding eigenvectors. If we assume that one eigenvalue is dominant, $\lambda_1 \gg \lambda_{i>1}$, then the normalised purified state exponentially approaches the dominant eigenvector, $\rho^{(k)}_{\text{corrected}} \to |\psi_1\rangle \langle\psi_1|$, as $k$ increases. The hope, which we shall verify in section XX on coherent mismatch, is that the dominant eigenvector of the noisy state closely approximates the ideal state, and therefore effective meaurements on the purified state yield accurate expectation values of the ideal state. The estimated expectation value of measuring an observable, $\hat{O}$, on this purified state is then,

$$\langle\hat{O}\rangle_{\text{corrected}} = \text{Tr}[\hat{O}\rho^{(k)}_{\text{corrected}}] = \frac{\text{Tr}[\hat{O}\rho^k]}{\text{Tr}[\rho^k]}. \tag{6.4}$$

Next, how to access this corrected state on which to make these measurements? Kozcor and Huggins show that this can be accessed without explicit preparation of the purified state. Instead the numerator and denominator may be found through preparing $k$ copies of the noisy state, applying some derangement operation to entangle all copies together, and measuring the desired observable on one of the copies. This can be understood through the following identity,

$$\text{Tr}[\hat{O}\rho^k] = \text{Tr}[\hat{O}^i S^{(k)} \rho^{\otimes k}], \tag{6.5}$$

where $S^{(k)}$ is the derangement operator, which permutes the $k$ copies of the state, and $\hat{O}^i$ is the observable acting on the $i$th copy. We shall define the derangement operator explicitly as the cyclic permutation of $k$ objects,

$$S^{(k)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes ... \otimes |\psi_k\rangle = |\psi_2\rangle |\psi_3\rangle \otimes ... \otimes |\psi_k\rangle \otimes |\psi_1\rangle . \tag{6.6}$$

Huggins provides a proof of this identity, which I reproduce here for the simple example of $k = 2$. The derangement operator for two copies is simply the swap operator, $S^{(2)} = \text{SWAP}$, and we will assume the observable acts only on the first copy, giving $\hat{O}^1 = \hat{O} \otimes \mathbb{I}$. The identity is then,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \tag{6.7}$$

$$= \sum_{\substack{i_1,i_2\\j_1,j_2\\k_1,k_2}} \langle i_1| \hat{O} |k_1\rangle \, \delta_{i_2,k_2} \langle k_1, k_2| S^{(2)} |j_1, j_2\rangle \langle j_1| \rho |i_1\rangle \langle j_2| \rho |i_2\rangle \tag{6.8}$$

$$= \sum_{\substack{i_1,i_2\\j_1,j_2\\k_1,k_2}} \hat{O}_{i_1,k_1} \delta_{i_2,k_2} \delta_{k_1,j_2} \delta_{k_2,j_1} \rho_{j_1,i_1} \rho_{j_2,i_2} \tag{6.9}$$

and contracting the Kronecker deltas, we find,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \tag{6.10}$$

$$= \sum_{i_1,j_1,j_1} \hat{O}_{i_1,j_2} \rho_{j_1,i_1} \rho_{j_2,j_1} \tag{6.11}$$

$$= \text{Tr}[\hat{O}\rho^2], \tag{6.12}$$

with the last equality following from the definition of matrix multiplication.

The denominator of the corrected expectation value is found by replacing the observable with the identity and measuring the expectation value of the derangement operator on $k$ copies of the noisy state.

Access to this corrected expectation value is then possible if we can measure the multi-qubit observable $\hat{O}^i S^{(k)}$ on $k$ copies of the noisy state. We will discuss two methods of implementing this measurement. First, a Hadamard test, given in Fig. **??**, which uses an ancilla qubit to control the derangement and desired observable operation, and then subsequent measurement of the ancilla only. The second method, given in Fig. **??**, uses a B-matrix to implement the derangement operation, and then measures the observable on one of the copies. The Hadamard test is more general, as it can be used to measure any observable, however, it requires an additional ancilla qubit and controlled operations. The B-matrix method has better qubit number and operation depth efficiency, however, requires simultaneous

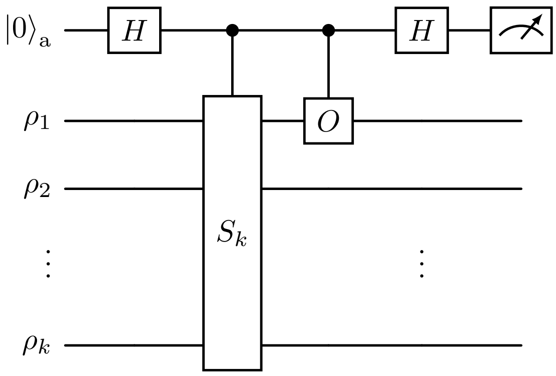



**Figure 6.6:** **Figure: circuit diagram for ancilla based virtual distillation, Hadamard test, derangement operator by CSwap**

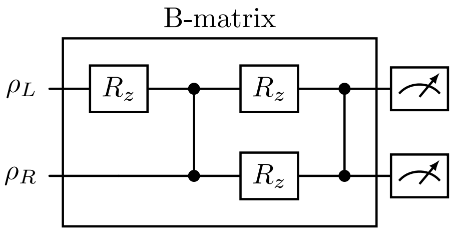



**Figure 6.7:** **Figure: circuit diagram for B matrix**

diagonalisation of $S^{(k)}$ and $\hat{O}^i S^{(k)}$, which is not always possible for an arbitrary observable and derangement operator.

Next, need to derive scaling of bias with copy number. And finally, discuss the variance of measurement with shot number (and how this scales with copy number?).

### Circuit

Ancilla method

B-matrix protocol

If we can diagonalise both S and SO. Can apply transversally

On hybrid systems

It can be used for both spin and bosonic objects

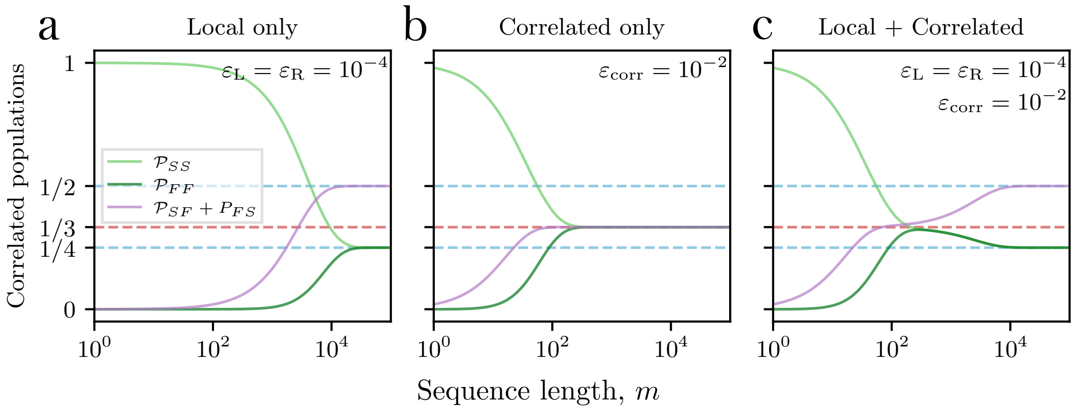



**Figure 6.8:** **Figure: B matrix decomposition.**

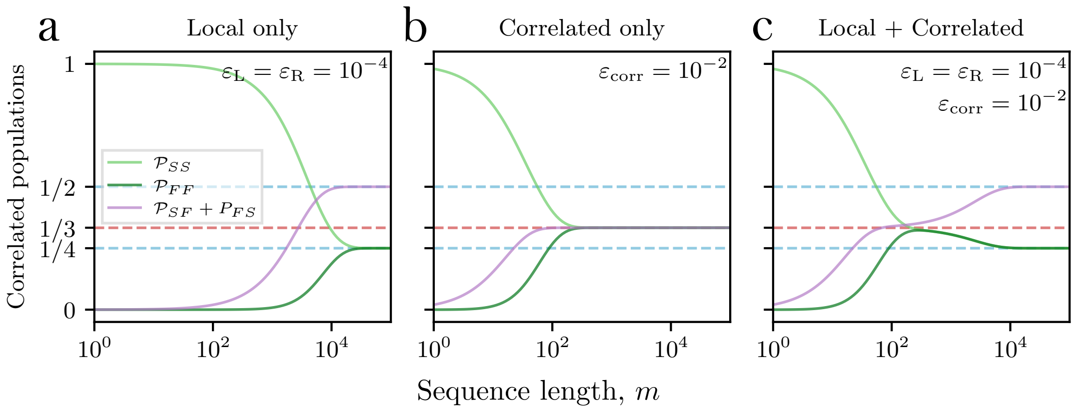



**Figure 6.9:** **Figure: pi-4 MS gate calibration**. **Figure: Heuristic signal of B-matrix working**. **Tomography would be best!**

### 6.2.3 Experimental Demonstration

**Short description of following results.**

#### Implementation

Protocol, circuit diagram, hardware involved.

#### B-matrix Calibration

pi-4 gates, single addressing

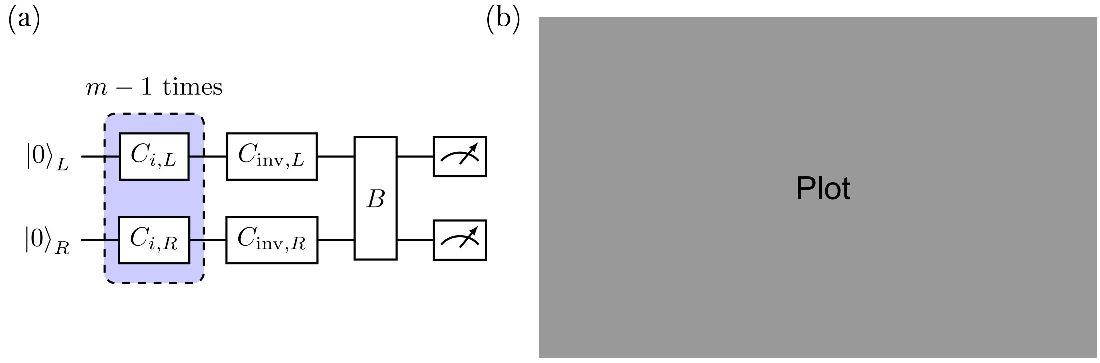



**Figure 6.10: Figure: Cicruit diagram of RB + VD, exp data for noise floor for two copy B-matrix, with and without spatial correlations**

#### Injecting Spatial Correlations

two copy, B matrix demonstration

Analysis of this data.

## 6.3 Proposed experiments and technical requirements

How CVDV and VD combine.

#### Extending to $>2$ copies

plan for c-swap implementation.

What is difficult, what numbers we need (gate fidelities, shot number)

#### Virtual Distillation on Hybrid Systems

Cite Quantum virtual cooling

Why important to develop

What errors does this mitigate?

How it looks on a thermal state and on various quantum states (sqz, coherent, fock)

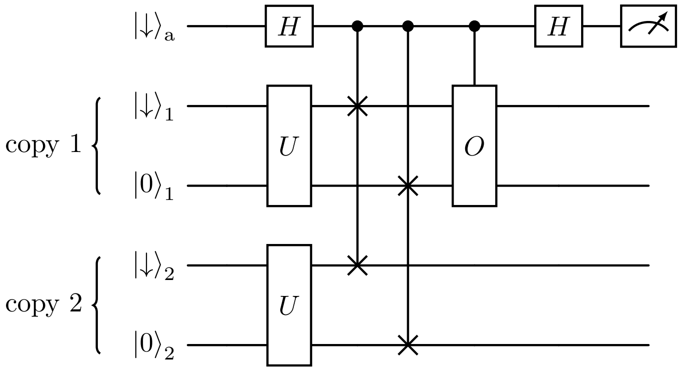



**Figure 6.11:** **Figure: Fredkin gate on oscillator, and how to compose derangement transversally.**

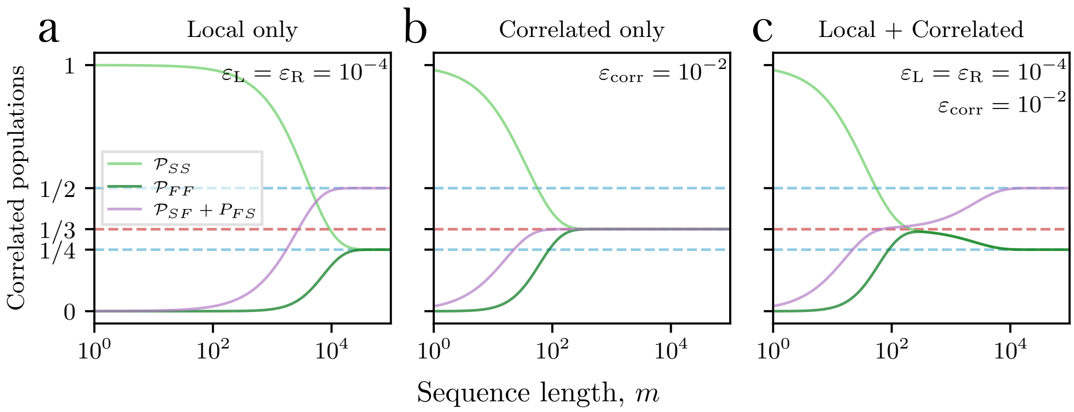



**Figure 6.12:** **Figure: Simulation of tri-sqz state with VD to show negativity!**

What is difficult, what numbers we need (gate fidelities, shot number)

## 6.4 Outlook

**Broader perspective of these QEM research directions, and also how we can play with our larger CVDV system Can generate sqz using SW**

# 7
# Conclusion

# Appendices

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